MAGNETOM

Startup System

Start-up / Magnet System

Aera
Skyra

Document Version

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Verifying the electronic low pressure switch is part of the filling procedure.

If the magnet does not require a helium fill, the test must be performed as a stand-alone procedure.

> If the magnet does not require a helium fill at start-up, the Low Pressure Switch must be tested.
>
> Refer to (→ Verifying the electronic low pressure switch / Page 44) in the Filling Procedure

2.1 Quench tube check

- Perform a visual inspection of the quench tube.
  - » Refer to (→ Quench tube / M7-010.833)
- Check if the Installation protocol is signed by the project manager and Quench line manufacturer.

> Ensure the Quench Line installation is complete before performing the first filling procedure.
>
> The Quench Line Acceptance Protocol must be checked before the first ramp.
>
> Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer.
>
> The acceptance protocol can be found under:
>
> **Product Information Home > Planning > Planning Guides > MR Systems > Quench Line Design and Calculation Tool**

2.2 Filling the static system

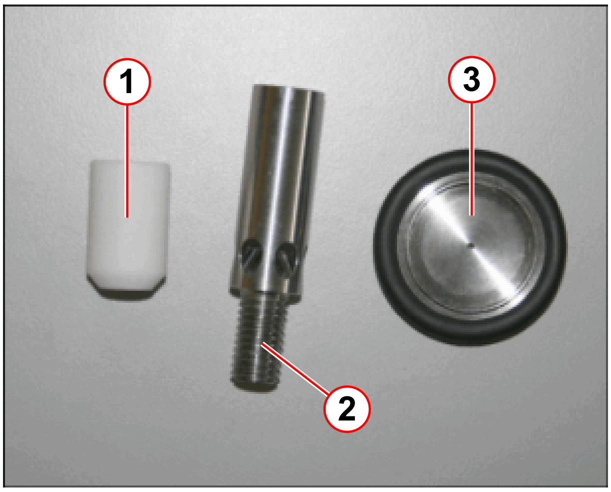
2.2.1 General

2.2.1.1 Static systems

The magnet design does not include a permanently fitted syphon. A short magnet service syphon must be fitted when filling with liquid helium. The service syphon is tooling only and must be removed immediately after filling has been completed.

This procedure can be performed on magnets at field and magnets at zero field.

Fig. 1: Upgrade kit 10129438 - De-ice tooling



(1) PTFE tip
(2) De-ice helium fill diffuser
(3) NW25 flow restrictor

The Magnet service syphon kit is now upgraded to include **de-ice tooling.**

The de-ice tooling is used for **helium top fill**, and **de-icing of the syphon port.**

- Use for **helium top filling**:
  - PTFE tip
  - 6 hole metal de-ice helium fill diffuser
- Use for **de-icing of the syphon port**:
  - PTFE tip
  - 6 hole metal de-ice helium fill diffuser
  - NW25 flow resistor

The new 6 hole diffuser **is more efficient,** and must be used **in place of** the 60 mm PTFE diffuser.

Only Siemens trained and accredited CSEs are allowed to perform this work.

During filling, do not exceed 95%.

System Type	Recommended helium level for the first ramp to field	Minimum helium level for future ramping	Minimum helium level for customer hand over
Aera OR98	60%	40%	50%
Skyra OR99	65%	60%	50%
Skyra / Spectra OR103	65%	40%	50%

To recover a potentially warm magnet, refer to magnet temperatures in the Bottom filling procedure (➔ Bottom filling the magnet / Page 50). Contact Magnet HSC for advice.

It is recommended that fills are performed during service visits and before ramping up.

The minimum level for filling at field is 40%

Wait at least 2 hours after filling before ramping the magnet.

If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

2.2.1.2 Venting system

CAUTION

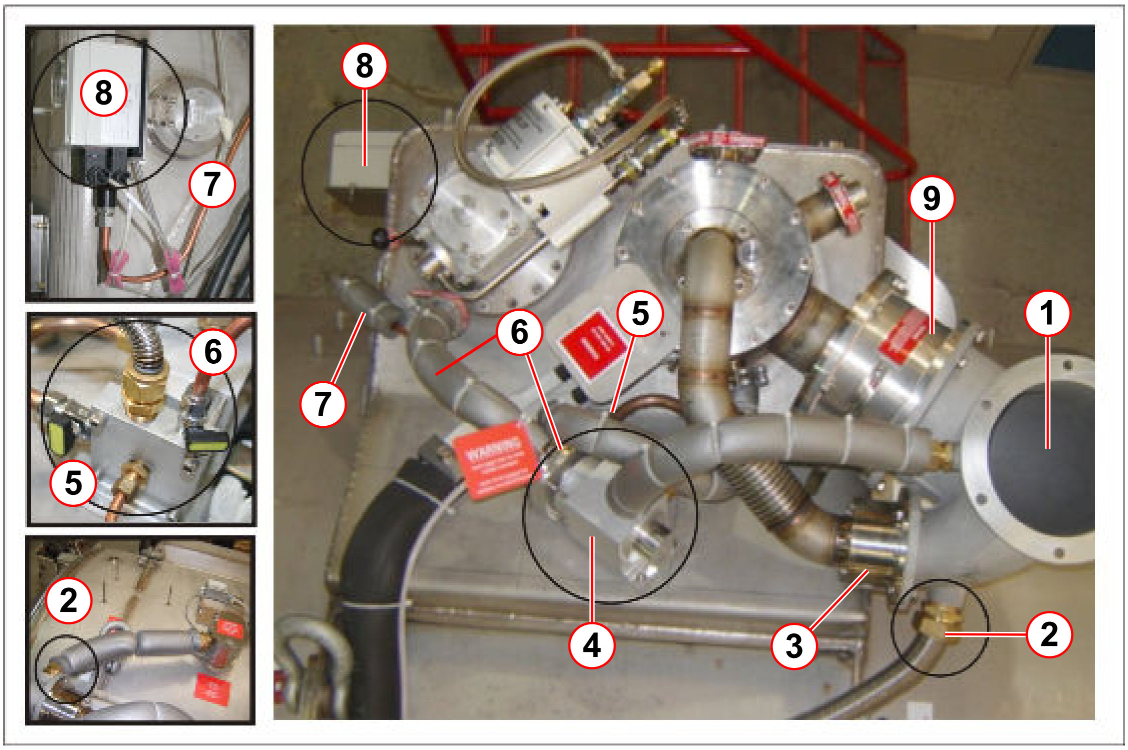
Cold head failure.

The cold head seals can fail, and leak to atmoshphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

This document describes the filling procedure for **OR98, OR99 and OR103** magnets.

Fig. 2: Normal venting configuration



(1) Quench valve elbow
(2) Flexible connection from OVC vent port
(3) Auxiliary burst disk (1.38 bar)
(4) Venting valve assembly
(5) Magnet system bypass line
(6) Cold head vent line and bypass valve
(7) Pressure display & control switch line
(8) Pressure display & control switch
(9) Quench valve with integral 15 psi (G) burst disk

2.2.2 Safety

WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » **Only non-magnetic equipment may be taken into the magnet room.**
- » **Never transport the GHe steel cylinder into the magnet room.**

Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

» Do not vent helium gas into the magnet room during helium transfer. As an exception, helium gas may be vented to cool the syphon.

» Ensure that the turret venting is connected to quench pipe.

» Oxygen monitors must be activated at all times.

CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

» Protective clothing, gloves and eye protection must be worn.

» Filling must be performed by trained personnel only.

CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

» Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.

CAUTION

Risk of quench during filling

If the magnet quenches during helium transfer

» Stop the helium transfer if possible and safe to do so.

» Follow Post Quench procedures

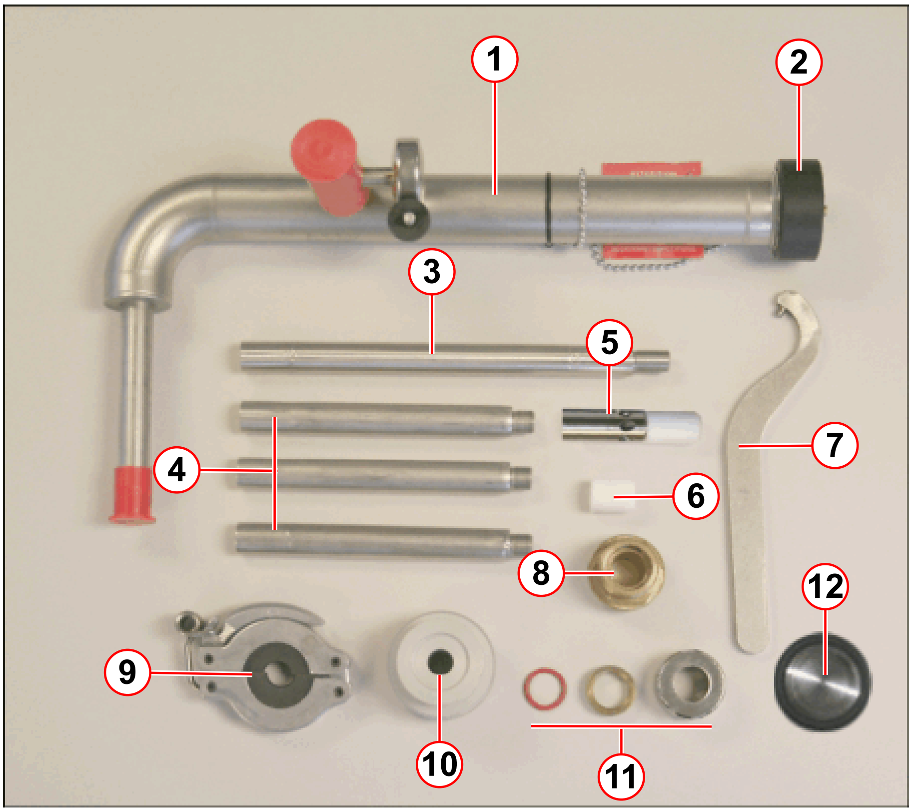
NOTICE

Mishandling of the dewar.

» The dewar-relevant safety regulations of the manufacturer must be observed.

2.2.2.1 Equipment required

Fig. 3: Magnet service syphon kit



(1) Service syphon assembly
(2) Syphon end cap
(3) 200 mm extension leg (1 off)
(4) 140 mm extension legs (3 off)
(5) 6 hole helium fill diffuser tip (top fill)
(6) 22mm PTFE seal (bottom fill)
(7) Syphon C-spanner/wrench
(8) M28 brass syphon adaptor bush (OR105, OR122)
(9) Syphon leg support clamp (low ceiling)
(10) Nitrogen fill adaptor and O-ring seal
(11) Syphon O-ring seal, brass syphon flange, syphon ring nut
(12) NW25 flow restricter

- Magnet service syphon kit (part no **10126651 (Revision 01)**). (→ Fig. 3 Page 11)
- Edwards hand held leak checker, (part number **10614850**)
- Dewar syphon.
- Dewar neck adaptor, clamps, seals, and hose.
- Heat gun.
- Paper towelling.
- Helium gas supply (at least 99.9% pure).
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes.
- Replacement copper seal for syphon port on the turret.

- Service platform.
- Gas bottle wrench.
- A room or personal O_2 Monitor.

2.2.3 Site preparation

- Ensure warning signs are correctly positioned.
- Ensure there is a clear path to exit the imaging suite.

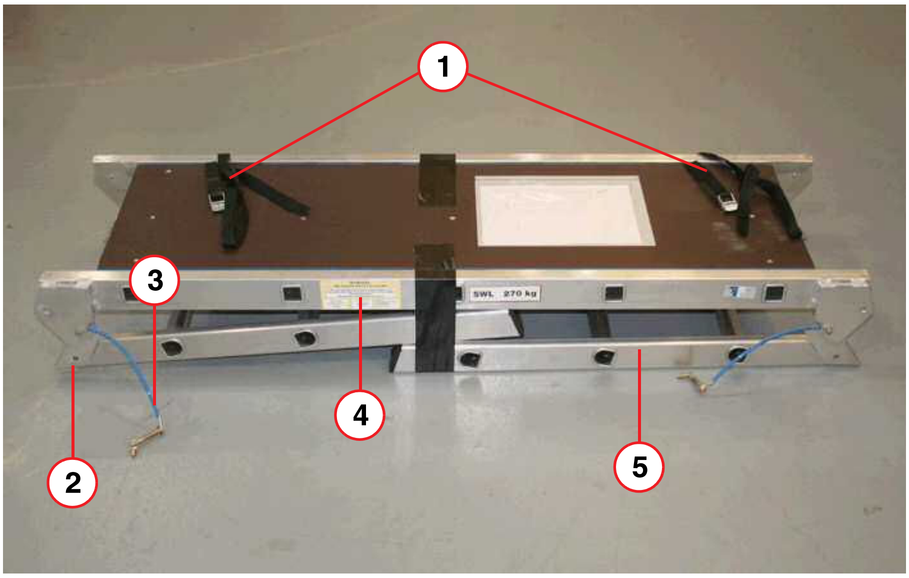
2.2.4 System preparation

Certain operations must be performed without delay to ensure efficient transfer:-

- Ensure that there is **open and clear access** to the magnet for **helium dewar(s).**
- Ensure that there is **open and clear access** to the **syphon**.
- **Position** the **Service platform** next to the magnet. .
  - The platform is secured to the magnet, using **2 long straps** which are passed **through the magnet bore**.
- During installation, if the **MRC (MRAWP) Host** is not available, take a helium sample using the **MSUP**. (→ Helium level check at the MSUP / Page 31)
- **Remove** the appropriate **covers**.

2.2.4.1 Service platform installation

Fig. 4: Service platform - as delivered



(1) Safety straps (2 off)
(2) Hinged locking pin location (each end)
(3) Retaining wire and hinged locking pin (each end)
(4) Warning label
(5) Step legs (each end)

Service platform safety

Before using the service platform **read the following safety cautions** and perform the **required safety checks**.

⚠ CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » **Always check the platform and hinge locking pins for damage.**
- » **Always check the hinge locking pins are secured.**
- » **Always secure the platform to the magnet frame or with the two long straps through the magnet bore.**
- » **Always ascend and descend (Walk UP or DOWN) the steps facing the platform.**
- » **Where possible use the frame or covers for additional hand support.**
- » **Never climb onto the platform from the side.**
- » **Never jump from the steps or top of the platform.**

Fig. 5: *Correct method - Up to platform*

Fig. 6: *Incorrect method - Down from platform*

CAUTION

Risk of personal injury

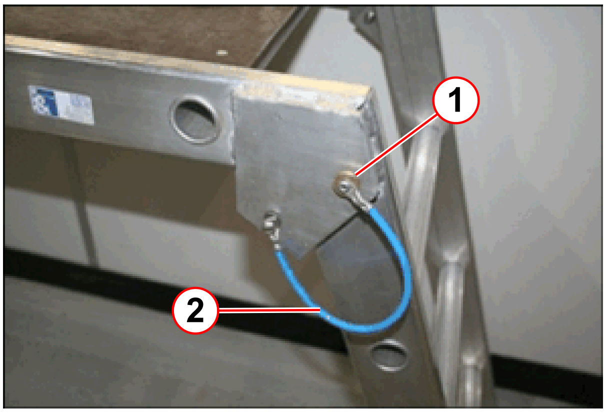
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- **DO NOT carry heavy components up or down the platform steps.**
- **Place heavy objects (i.e. cold head) onto the platform first.**
- **Only lift the component when standing safely on the platform.**

Platform assembly

- Remove any packaging.
- Unfold both step legs.

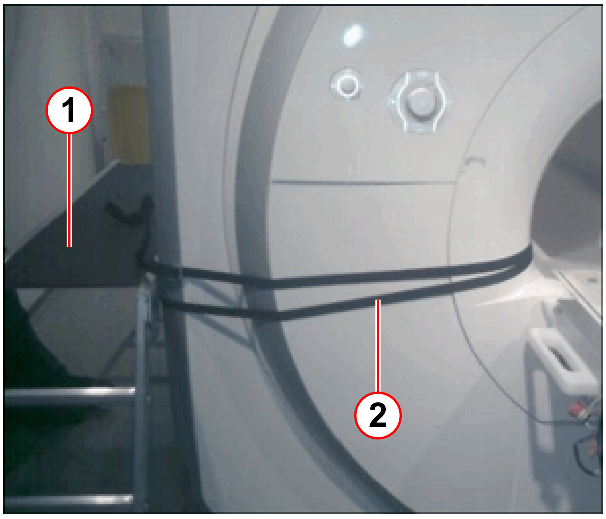
Fig. 7: Service platform - lock legs in position



(1) Hinged locking pin
(2) Retaining wire

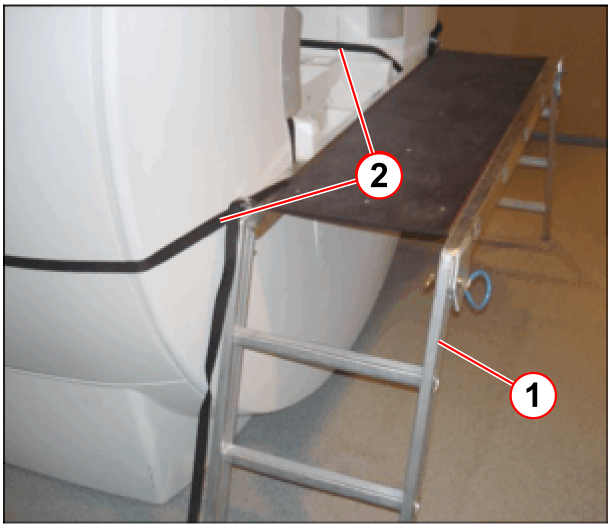
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 8: Securing the service platform (example)



(1) Service platform
(2) Long straps (fitted throught the magnet bore)

Fig. 9: Securing the service platform (example)



(1) Service platform
(2) Long straps (fitted through the magnet bore)

Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.

The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

2.2.4.2 Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

The procedure covers all dewars that are externally pressurized with helium gas.

Refer to the dewar manufacturer's documentation when using side fill and/or self-pressurizing dewars.

Refer to the dewar manufacturer's safety guidelines and check the dewar valve functions before commencing the LHe fill.

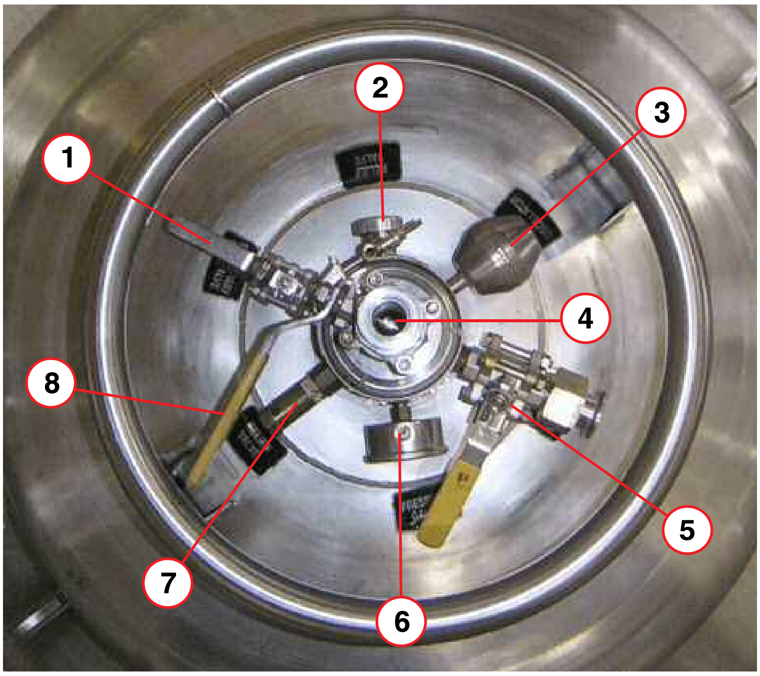
- If the dewar is **highly pressurized**, (greater than 0.3bar (4.4psi)), release the pressure by slowly opening the **primary ball valve** (→ 5/Fig. 10 Page 16) to minimize the flash losses of helium.
  - This must be done in a **well ventilated area - NOT in the imaging suite.**

Check for the cause of the high pressure, i.e. was the low pressure relief valve closed?

Inform the supplier if the cause cannot be found.

- Fit the **dewar syphon adaptor** (→ 3/Fig. 12 Page 18) to the top of the dewar.

Fig. 10: Dewar Ports (typical)

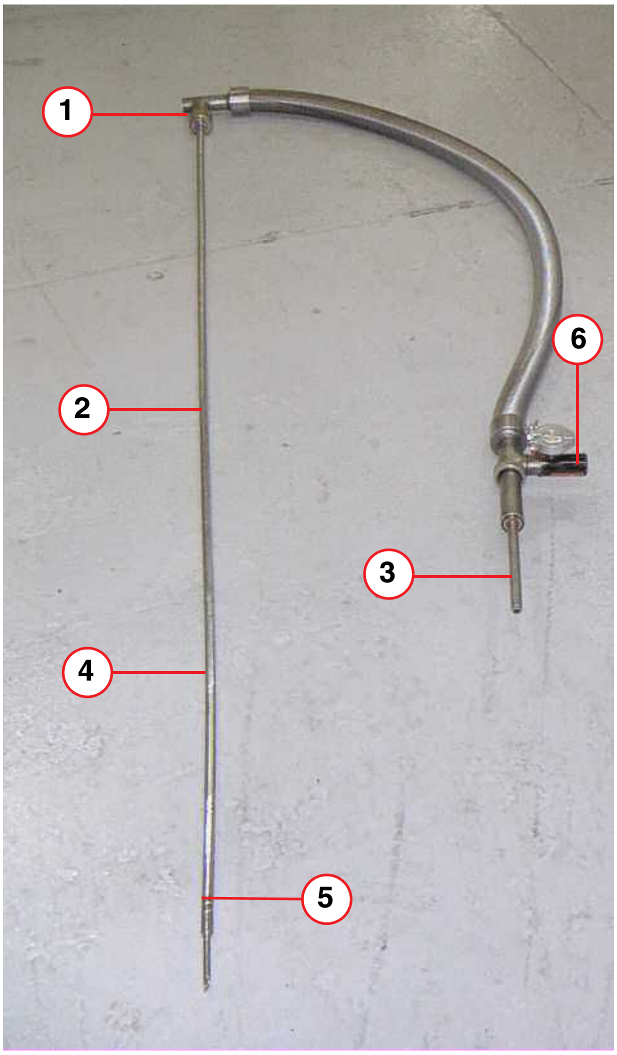


(1) Low pressure relief valve (shown open)
(2) Relief valve
(3) Anti-oscillator
(4) Syphon entry port (shown closed)
(5) Primary ball valve (shown closed)
(6) Pressure gauge
(7) Relief valve
(8) Syphon entry valve

- **Close** the **primary ball valve** (→ 5/Fig. 10 Page 16) and **low pressure relief valve**. (→ 1/Fig. 10 Page 16)

Dewar syphon preparation

Fig. 11: Dewar syphon



(1) Syphon flow valve
(2) Syphon leg
(3) Bayonet connection (male)
(4) Extension leg
(5) Sludge inhibitor
(6) Syphon vacuum valve

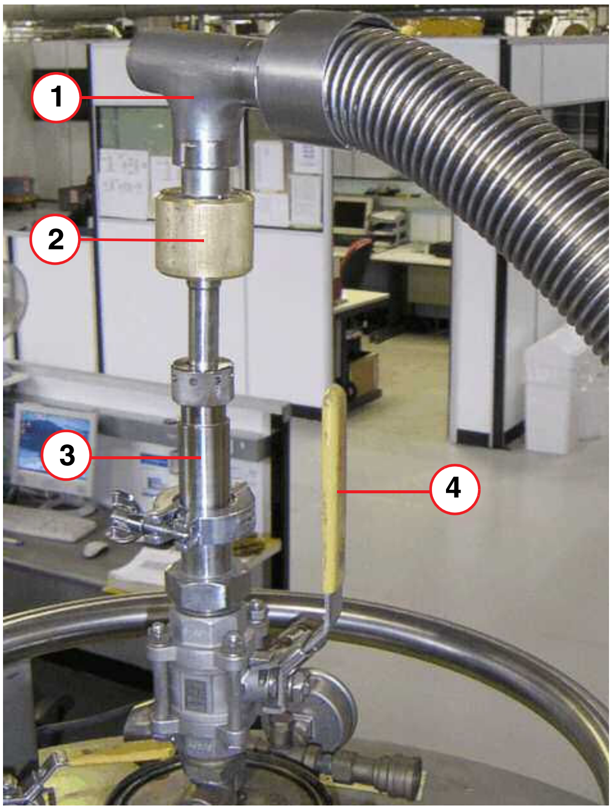
- Fit the extension leg (s) and sludge inhibitor, to the dewar **syphon leg**.

Do not open the syphon vacuum valve (6).

- Fit the **helium regulator** to the helium **gas bottle**.

Precool the dewar syphon

Fig. 12: Dewar syphon fitment



(1) Syphon
(2) Syphon flow valve
(3) Dewar syphon adaptor
(4) Syphon entry valve

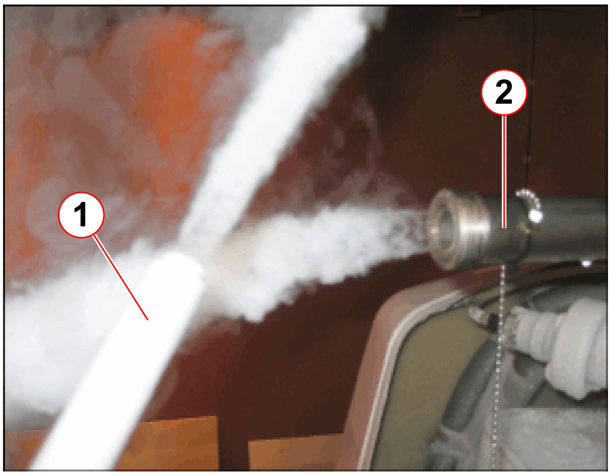
- Open the **syphon flow valve** (pos 2) two turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.

  (Do not fully tighten the ring nut at this stage).
- Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.

- Push the syphon **fully down** into the dewar, (very slowly to minimize pressure build-up and flash losses), until the sludge inhibitor touches the bottom.

- Fully tighten the **ring nut** and **seal.**

Fig. 13: Pre-cooling the syphons

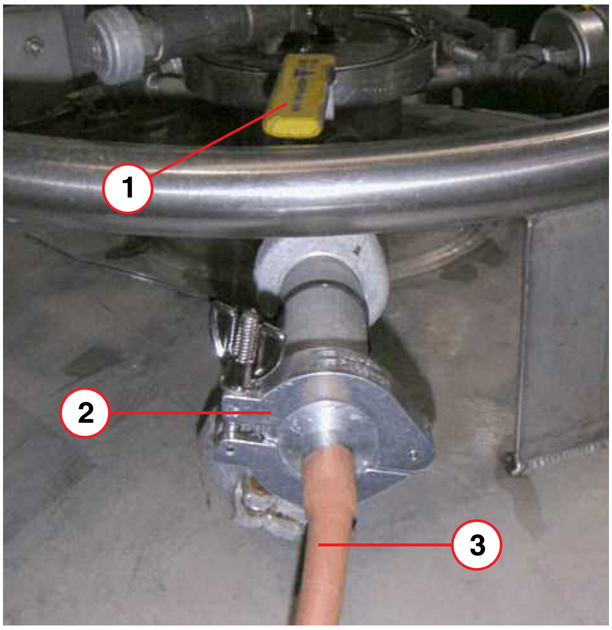


(1) Dewar syphon
(2) Service syphon

- Allow the syphon to cool until a dense white “cascading plume” appears.
- **Do not** allow high pressure to build in the dewar, only enough to cool the syphon.

- When the syphon is cold, close the **syphon flow valve** (→ 2/Fig. 12 Page 18) and move the dewar so that the syphon can be connected without excessive bending.

Fig. 14: Dewar gas connection



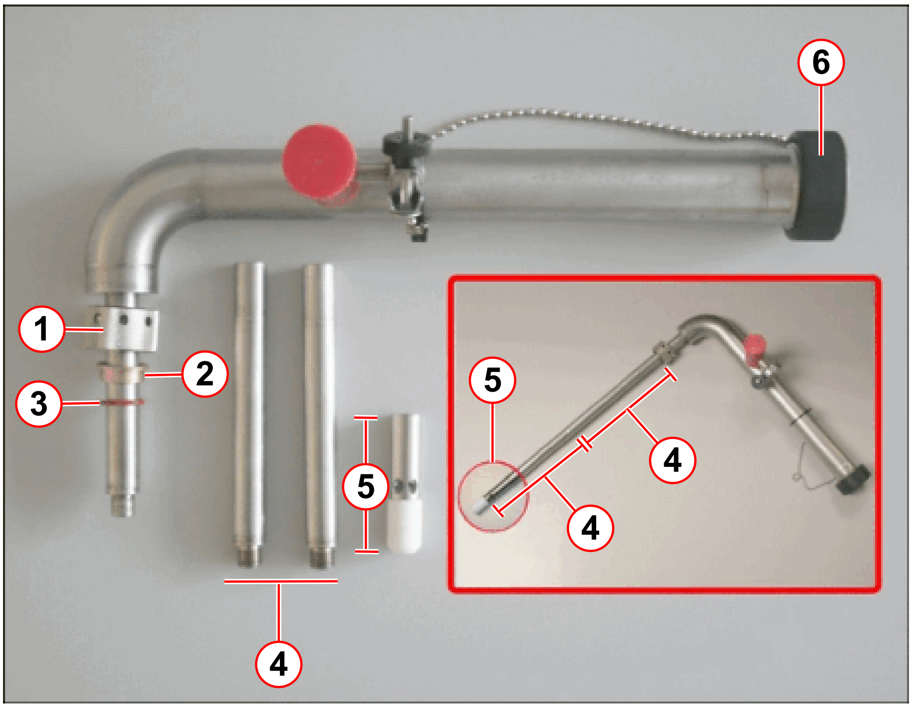
(1) Primary ball valve
(2) NW25 clamp and seal
(3) Gas supply line

- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet if the magnet is at field.
- **Purge** the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an **NW25** clamp and seal (pos 2).

- **Open** the **primary ball valve** (→ 5/Fig. 10 Page 16) and pressurize the dewar to **0.27bar (4 psi (G))**.

2.2.5 Assemble the service syphon

Fig. 15: Magnet service syphon assembly - top filling



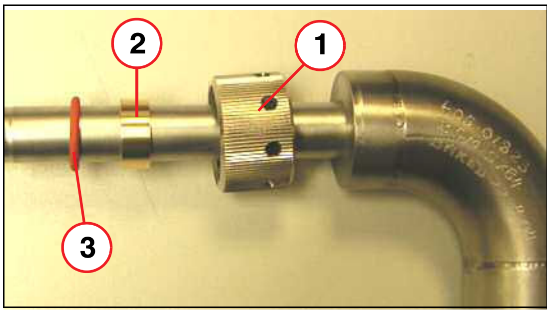
(1) Ring nut
(2) Brass flange
(3) O-ring seal
(4) 140 mm syphon legs (2 off)
(5) De-ice and helium fill diffuser and PTFE tip
(6) Syphon end cap

Tab. 1 OR98 / OR99 / OR103 TOP filling syphon extension legs

Magnet type	140 mm leg	60 mm De-ice & helium fill diffuser & PTFE tip
Aera (OR98) Skyra (OR99/OR103)	2	1
The **syphon diffuser** should optimally be 50-80 mm above the syphon cone for **top filling**. If damaged, the **PTFE diffuser tip** must be replaced.		

- Check the **ceiling height.** (→ Low ceiling height / Page 41)
  - Assemble the **syphon** (only if it can be fitted), or
  - Assemble the **diffuser**, bottom **extension** and support **clamp**.

Fig. 16: Assembling the Ring Nut Seal



(1) Ring nut
(2) Brass spacer
(3) Silicon O-ring

- Fit the ring nut, brass spacer, and the O-ring to the **service syphon leg.**

Refer to table above (➔ Tab. 1 Page 20)

- Connect the **extension legs** and diffuser:
  - Screw the (2 off) **140 mm legs** together.
  - Connect the leg to the **syphon**.
  - Fit the 60 mm **diffuser** and PTFE tip.

2.2.6 Fit the magnet service syphon

CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmoshphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

CAUTION

Pressurized helium vessel.

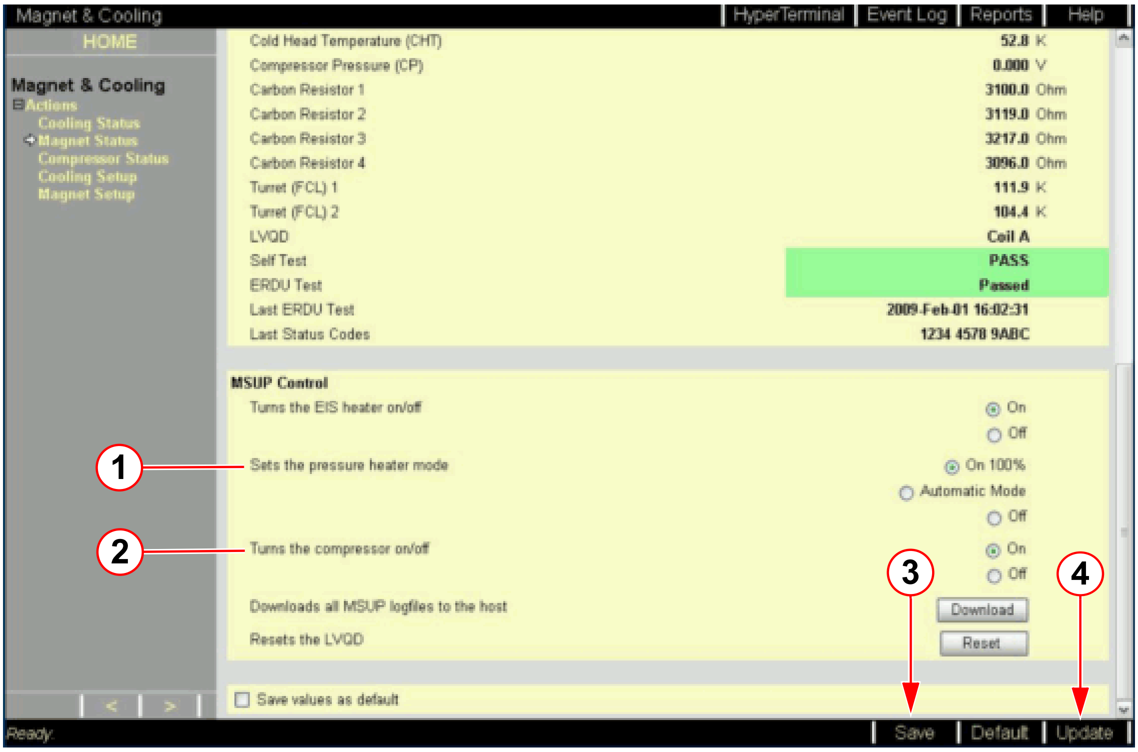
Failure to observe the following safety information may result in physical injury.

» **Do not disconnect any of the fittings without first depressurizing the helium vessel.**

2.2.6.1 Deactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **OFF** button (➔ 1/Fig. 17 Page 22).
- Select **SAVE** (➔ 3/Fig. 17 Page 22).

Fig. 17: Service screen Magnet & Cooling - Magnet status - MSUP Control



(1) Sets the pressure heater mode
(2) Sets the the compressor on/off
(3) Save button
(4) Update button

> **i** The pressure heater will reactivate within 60 min.

2.2.6.2 Depressurize the helium vessel

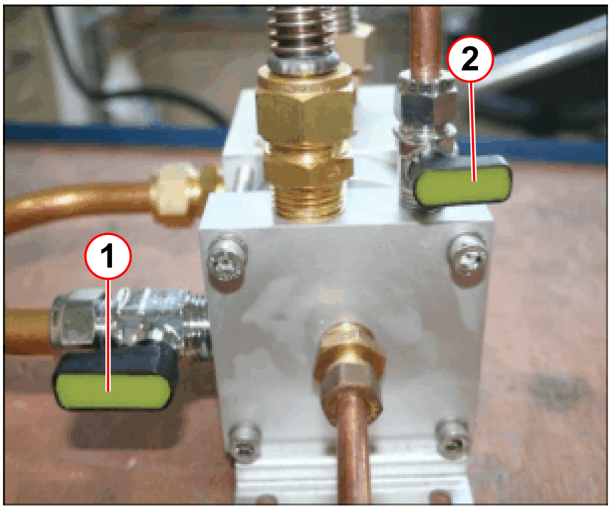
⚠ CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

» **To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.**

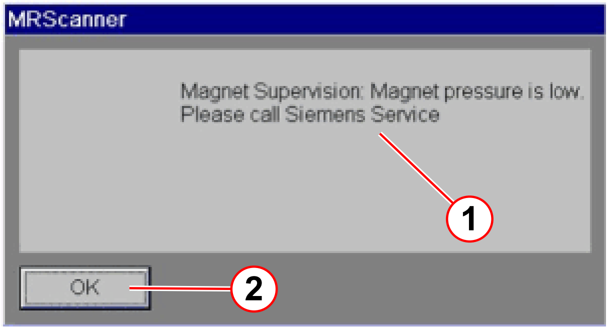
Fig. 18: Bypass valve positions for depressurization



(1) Magnet bypass valve OPEN
(2) Cold head bypass valve CLOSED

- **Fully open** the **magnet bypass valve** to **depressurize** the helium vessel.
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch** will activate.
  - » the **cold head** will turn **off**.
- **Close** the magnet bypass valve to prevent **air ingress**.
- **Warm up** the vent valve assembly.
  - » **Take care** not to damage any components when using a **heat gun**.

Fig. 19: MSUP - Magnet pressure is low



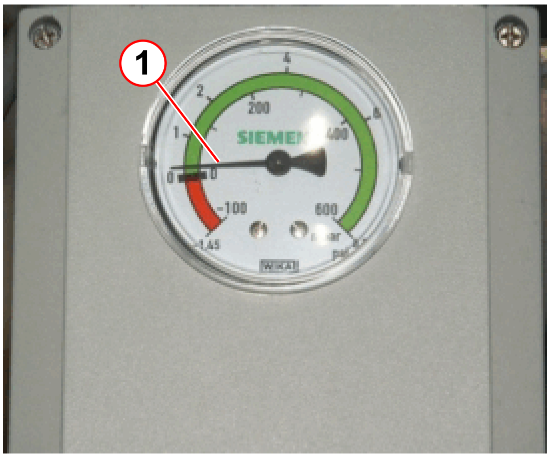
(1) Pop-up message
(2) Click OK to clear

On the **MR Scanner:**

- A pop-up **alarm message** is displayed.
  - » **Magnet pressure is low**.
  - » Click **OK** to clear the alarm.
- The **alarm box** will sound when **pressure falls** to **alarm limits.**
  - » The yellow **WARNING LED** lights on the alarm box.
  - » The alarm **buzzer** is **activated**
  - » Press **AUDIO ALARM OFF** to silence the alarm.

- Check the **pressure** is positive when the **compressor** switches **OFF**.

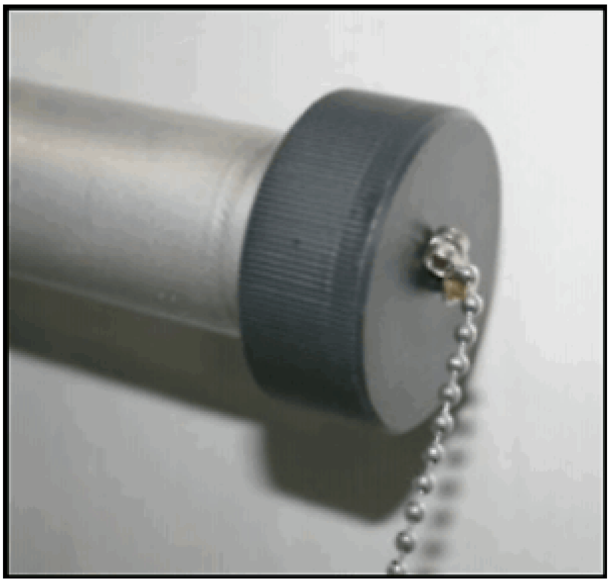
Fig. 20: WIKKA Pressure switch - with a small positive pressure



- **Check** the helium vessel has a **small positive pressure.**

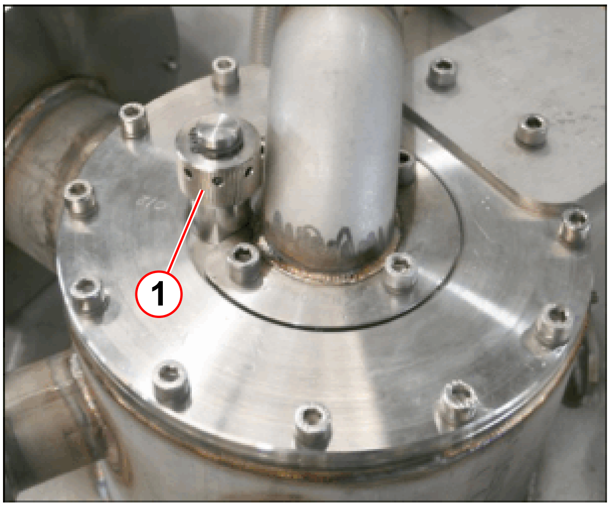
- Warm and dry the **syphon blanking plug** using a heat gun to ease removal. (➔ 1/Fig. 22 Page 24)

Fig. 21: Syphon end cap



- **Remove** the syphon **end cap.**

Fig. 22: Fit/remove the syphon blanking plug



- **Remove** the **syphon blanking plug** using the syphon ‘C’ spanner.

⚠ CAUTION

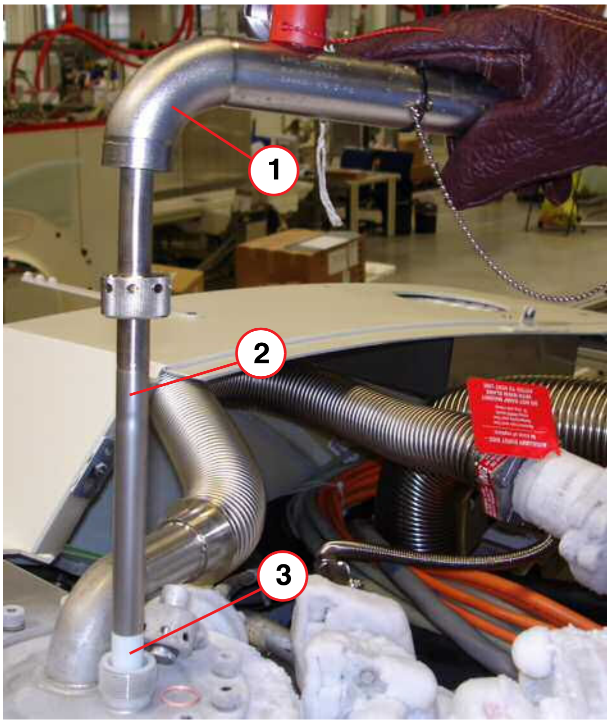
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

Fig. 23: Helium syphon



(1) Service Syphon
(2) Two Extension Legs
(3) PTFE Syphon Diffuser

- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon fully into the magnet.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.

There is a risk of quenching the magnet if the syphon is inserted too quickly.

- **Re-fit** the syphon **end cap.** (➔ Fig. 21 Page 24)

If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

2.2.7 Precool the service syphon

The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use, (Refer to (➔ Dewar preparation / Page 15)).

If the temperature of the magnet is above 4.2K, a dense cascading plume will not form on cooling the service syphon.

CAUTION

Risk of quench

Warm helium gas can cause a quench.

» **If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.**

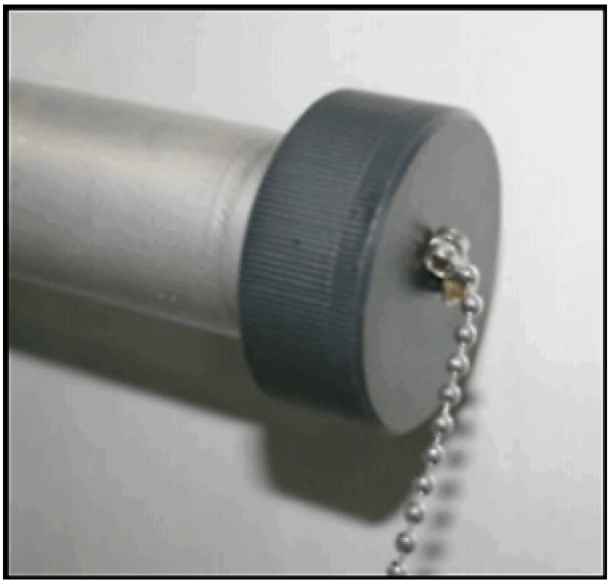
» **Ensure syphons are correctly pre-cooled before joining together.**

» **Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.**

» **Frost on the syphon coupling is acceptable.**

Cool down the **service syphon** as follows:-

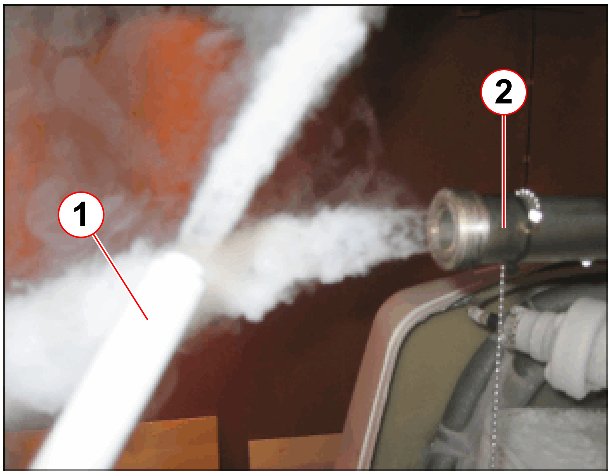
Fig. 24: Syphon end cap



- Ensure the **syphon end cap** is fitted and the **bypass valve** is **closed**.

- **At the MRC, reactivate** the **pressure heater** to achieve a positive pressure of about **0.3 psi (G)** in the magnet. (➔ Reactivate the pressure heater / Page 34)
- **Deactivate** the pressure heater (➔ Deactivate the pressure heater / Page 21).

Fig. 25: Pre-cooling the syphons



(1) Dewar syphon
(2) Service syphon

- **Remove** the **syphon end cap**.
- Allow the syphon to vent until **fully cold**.
  - a “cascading plume” of helium gas appears.

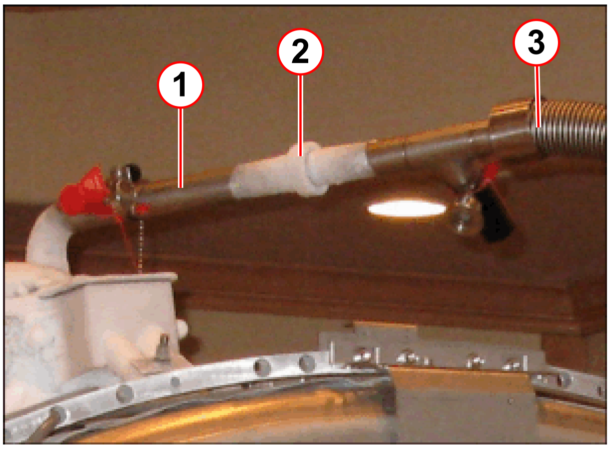
> **CAUTION**
>
> **Risk of cold burns.**
>
> **Cold gas will exhaust from the syphon port on opening.**
>
> » **Stand clear when opening.**
>
> » **Wear protective gloves and safety glasses.**

2.2.8 Liquid helium transfer - top fill

Before helium transfer, refer to (➔ Helium pressures at altitude / Page 42)

- Check the **dewar** gas supply pressure is set at **0.27bar (4psi (G)).**
- As the **service syphon** nears the point of being fully cold:
  - open the **dewar syphon flow valve** half way (2 turns). (➔ Fig. 27 Page 29)
  - **precool the syphon** again until a dense white “cascading plume” appears.

Fig. 26: Dewar and service syphon coupling



(1) Service syphon
(2) Lock nut
(3) Dewar syphon

- When **both** syphons are giving a dense white “cascading plume” of helium gas, connect them together and secure the lock nut.

CAUTION

Risk of quench

Warm helium gas can cause a quench.

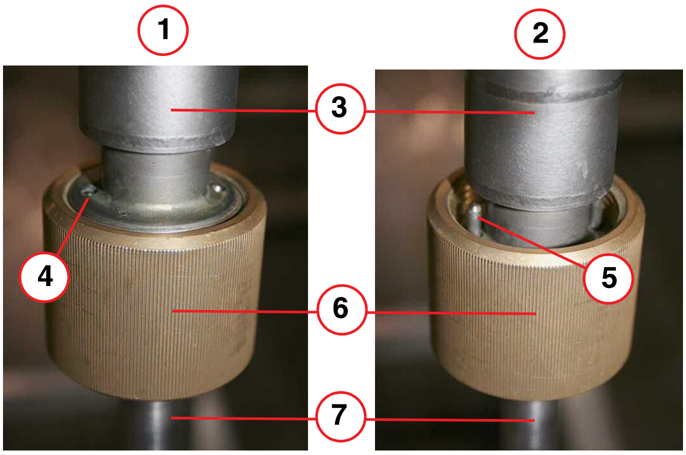
» **If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.**

» **Ensure syphons are correctly pre-cooled before joining together.**

» **Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.**

» **Frost on the syphon coupling is acceptable.**

Fig. 27: Dewar syphon flow valves - open and closed positions



(1) FULLY OPEN position
(2) FULLY CLOSED position
(3) Stainless steel body
(4) Guide pin flush with body
(5) Guide pin
(6) Brass nut
(7) Syphon leg

Fig. 28: Helium filling

- Fully opening the **dewar syphon flow valve**.
  - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body (➔ 4/Fig. 27 Page 29).
  - » From fully closed to fully open, is **4 turns**.
- Ensure the transfer dewar pressure is **4 psi (A)** at normal altitude. Refer to (➔ Helium pressures at altitude / Page 42).

2.2.8.1 Perform a helium level check

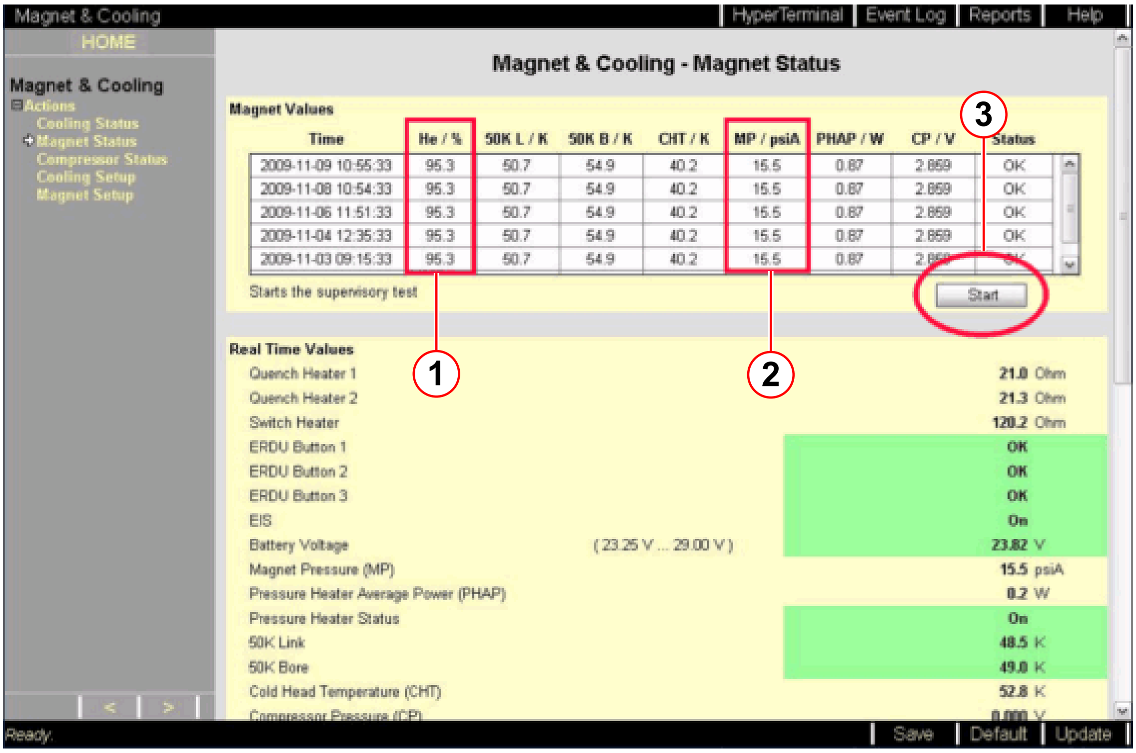
> **i** During installation, if the MRC is not available:
>
> 1.Take a helium sample using the MSUP. (➔ Helium level check at the MSUP / Page 31)

Helium level check at the MRC

- Check the liquid helium level by using Service Software on MRC (MRAWP) Host.
  - Check the liquid helium level using the MRC.
  - Access the Service Software at the MRC.
  - Select the menu item Magnet & Cooling - Magnet Status.

- Press the Start button (→ 3/Fig. 29 Page 30) to initiate the helium measurement. (→ 1/Fig. 29 Page 30)

Fig. 29: Checking liquid helium and magnet pressure levels



(1) Liquid helium level

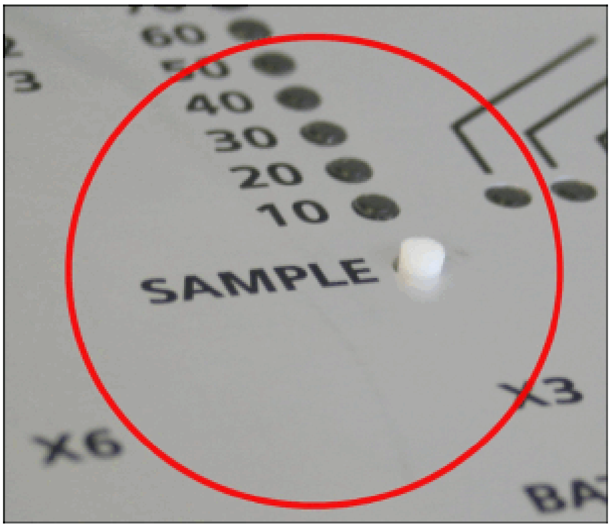
(2) Magnet pressure

(3) Start button

Data is only accurate from the last refresh of the page, or from when the page was opened last.

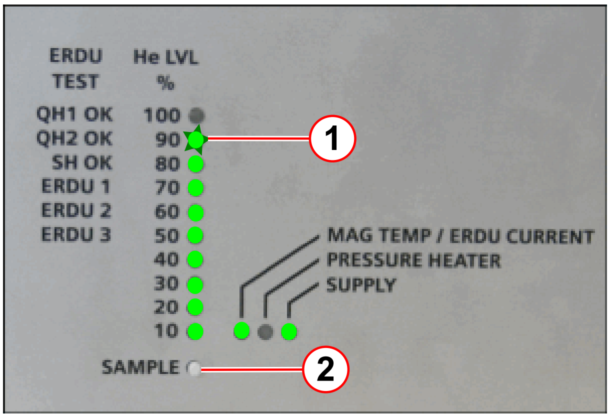
Helium level check at the MSUP

Fig. 30: MSUP 2306 - depress SAMPLE to activate



- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.

Fig. 31: MSUP - Helium level sample



(1) Sample switch
(2) Helium level

- The green LEDs illuminate and display the actual liquid helium level percentage.
- If the 90% LED is flashing, STOP the transfer. Refer to (→ Tab. 2 Page 31).

Tab. 2 MSUP Helium display

MSUP LEDS	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue to fill as required.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATLEY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED.

* The LEDs return to ON after approximately 30 seconds.

To avoid overfilling, the helium transfer must be stopped as soon as the 90% LED illuminates.

- Repeat the measurement every 5 minutes. An increase in level should be seen within 20 minutes.
  » If not, stop the transfer and investigate the reason for poor transfer efficiency.

Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.

Maintain a pressure of about 0.27 bar (4 psi (G) at the helium dewar.

Too much or too little pressure will result in inefficiency.

At 4 psi (G), the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum.**Stop the transfer immediately - the syphon may need to be replaced.**

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

» **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

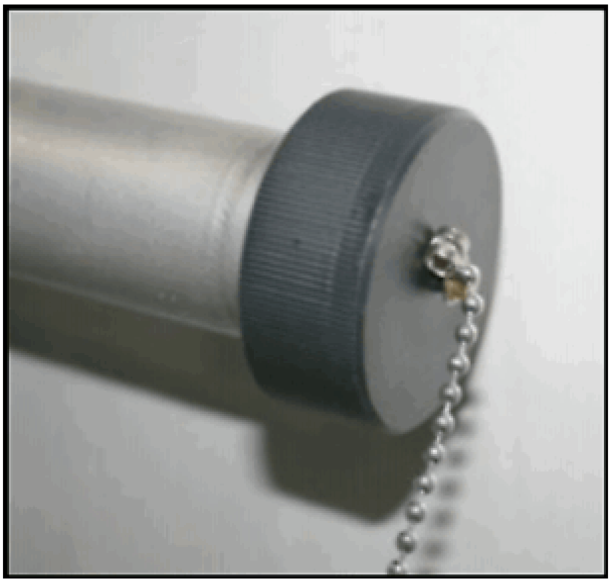
2.2.9 Removing the service syphon

- **Depressurize** the helium vessel. (➔ Depressurize the helium vessel / Page 22)

Fig. 32: Warm up the syphon joint with heat gun.

- **Warm** the connecting **syphon joint** to room temperature (using a **heat gun**) to remove ice.
- **Warm** the syphon **entry valve** to room temperature and dry.

Fig. 33: Syphon end cap



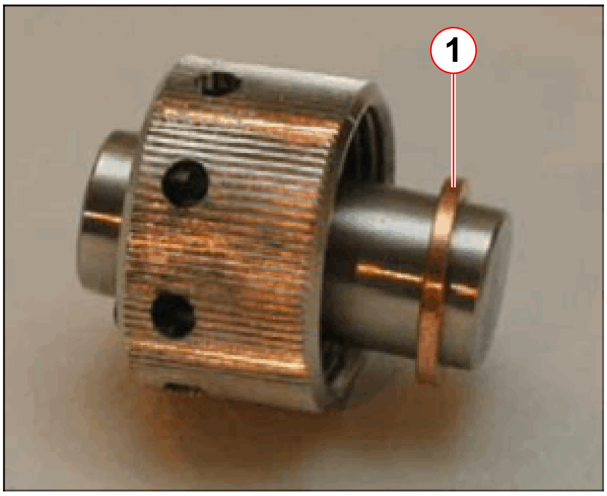
- **Warm** the syphon **end cap** ready for fitment onto the syphon.
- **Disconnect** the syphons.

> **⚠ CAUTION**
>
> **Risk of cold burns.**
>
> **Cold gas will exhaust from the syphon port on opening.**
>
> » **Stand clear when opening.**
>
> » **Wear protective gloves and safety glasses.**

- **Close off** the end of the service syphon with the **syphon end cap.** (➔ Fig. 33 Page 33)

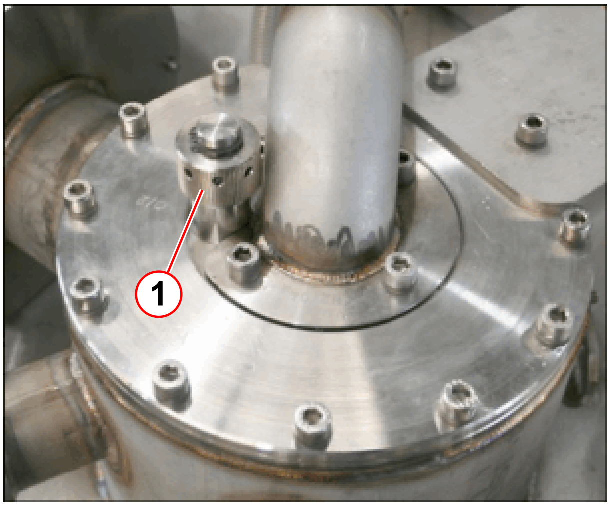
- **Immediately remove** the service **syphon** from the magnet.

Fig. 34: Syphon blanking plug



(1) Copper seal

Fig. 35: Fit/remove the syphon blanking plug



- **Immediately** fit the **syphon blanking plug** using a new copper seal. (➔ Fig. 34 Page 34)
- **Warm up the threads** on the plug and **carefully** tighten with the syphon 'C' spanner.
  - There might be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.

⚠ CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **Pressurize** the helium vessel to **15.5 psi (A).**

2.2.9.1 Reactivate the pressure heater

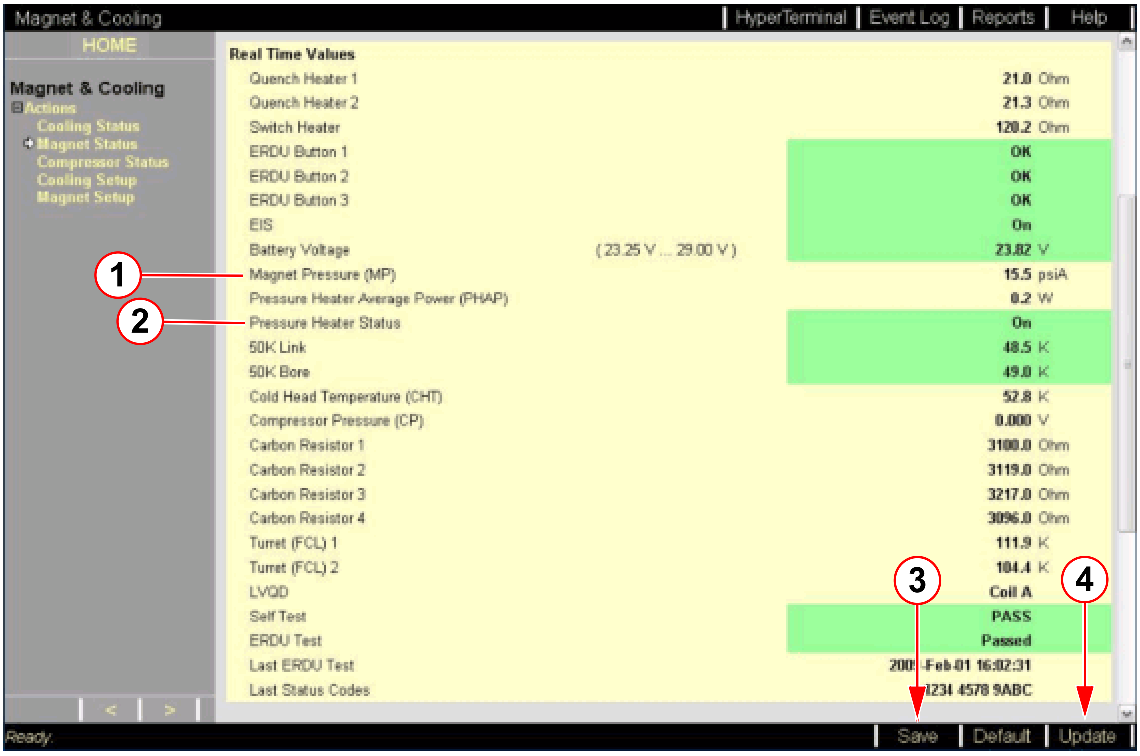
- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status.**
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button. (➔ 1/Fig. 17 Page 22)
- Select **SAVE.** (➔ 3/Fig. 17 Page 22)

- **At Real-Time Values** - Monitor the **magnet pressure.**
  - Select **Update** to refresh data until pressure reaches ≥ **15.5 psi (A).**

> Data is only accurate from the last refresh of the page, or from when the page was opened last.

> During pressurization, the cold head will switch on automatically.
> If the cold head was turned off at MRC, select (→ 2/Fig. 17 Page 22)

Fig. 36: Check magnet pressure level



(1) Magnet pressure (MP)
(2) Pressure heater status
(3) Save button
(4) Update button

- At **MSUP Control - Sets the pressure heater mode** select the **Automatic mode** button. (→ 1/Fig. 17 Page 22)
- Select **SAVE.** (→ 3/Fig. 36 Page 35)

2.2.9.2 Leak check

- After helium fill has been completed, **leak check** the **turret.**

> Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- **Move the dewar** out of the imaging suite into a **well ventilated area**.

- Complete **depressurizing** the dewar by opening the **primary ball valve.** (➔ 5/Fig. 10 Page 16)
- When at the gas pressure is at **zero**, **close** the **primary ball valve** on the dewar.
- **Remove** the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar. (➔ 4/Fig. 10 Page 16)
- **Open** the **low-pressure dewar relief valve.** (➔ 1/Fig. 10 Page 16)
- **Warm** both **syphons** with the heat gun **until dry**, and store safely.

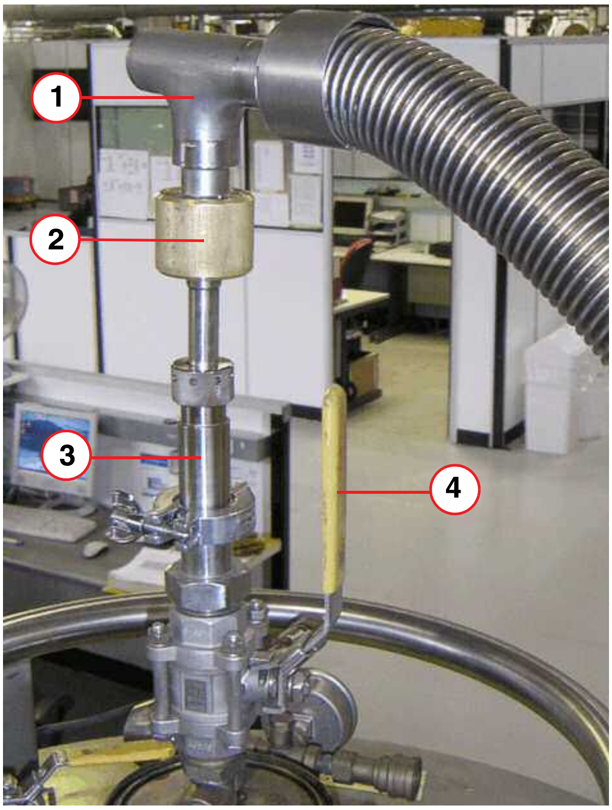
2.2.10 Changing supply dewars during transfer

> Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

Fig. 37: Dewar syphon fitment



(1) Syphon
(2) Syphon flow valve
(3) Dewar syphon adaptor
(4) Syphon entry valve

- When the dewar empties, stop the transfer **immediately** by closing the **dewar syphon flow valve** (pos 2).

CAUTION

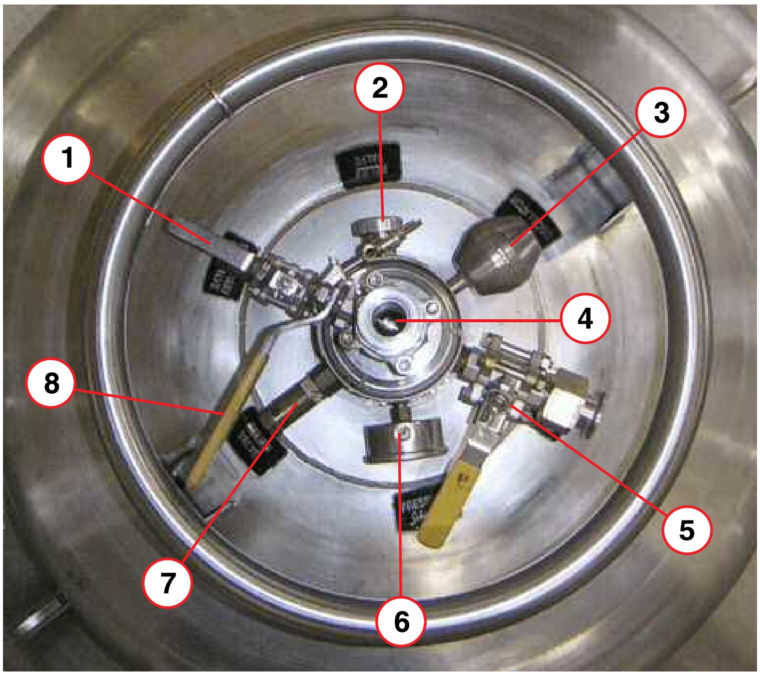
Risk of quench.

Failure to close the valve immediately, when the magnet is at field, will result in a quench.

» **Close the valve immediately.**

- Turn **off** the dewar gas supply:
  - remove the **gas connection**.
  - vent the dewar down to atmospheric pressure.
  - **Ensure the area is well ventilated**.
- **Close** the **primary dewar ball valve**. (→ 5/Fig. 38 Page 37)

Fig. 38: Dewar Ports (typical)



(1) Low pressure relief valve (shown open)
(2) Relief valve
(3) Anti-oscillator
(4) Syphon entry port (shown closed)
(5) Primary ball valve (shown closed)
(6) Pressure gauge
(7) Relief valve
(8) Syphon entry valve

- Using a heat gun, **warm** the **syphon connecting joint** to room temperature.

- **Disconnect** the syphons.

> **CAUTION**
>
> **Risk of cold burns.**
>
> **A “cascading plume” of cold gas will exhaust from the syphons when they are disconnected.**
>
> » **Stand clear of the syphon joint when disconnecting.**
>
> » **Wear protective gloves and safety glasses.**

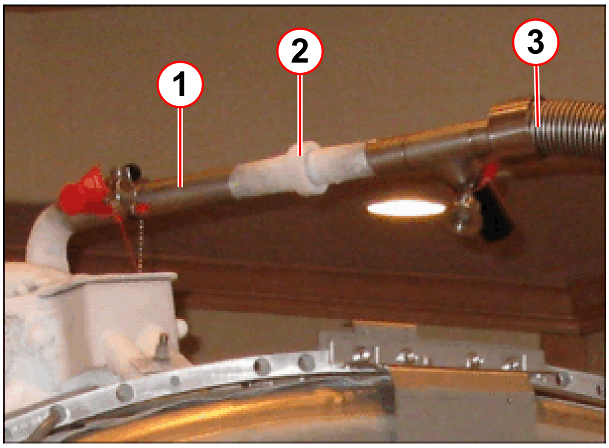
- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap.** (→ Fig. 33 Page 33)

> **CAUTION**
>
> **Danger of air ingress.**
>
> **Internal ice blockage could cause excessive pressure.**
>
> » **Ensure the syphons are capped and plugged immediately and securely.**

- Move the empty dewar close to the full dewar.
- Vent both storage dewars down to **atmospheric pressure**.
  » **This must be done in a well ventilated area - NOT in the imaging suite.**
- **Open** the **low pressure relief valve** on the **empty** dewar. (→ 1/Fig. 38 Page 37)
- **Remove** the dewar **syphon** and **adaptor** from the empty dewar and **immediately** Insert the syphon leg into the **full dewar**.
- **Close** the **syphon entry valve** (→ 4/Fig. 38 Page 37), of the **empty** dewar. and **open the low pressure relief valve**.(→ 1/Fig. 38 Page 37)
- **Close** the **low pressure relief valve** of the **full** dewar.
- **Purge** the gas supply line.
- **Connect** to the full dewar and **open** the dewar **primary ball valve**.
- Set the **gas pressure** to 0.27bar (4psi (G)).
- **Precool** the **dewar syphon**.(→ Precool the dewar syphon / Page 18)
- Move the dewar next to the magnet system.

- **Precool** the **service syphon.**

Fig. 39: Dewar and service syphon coupling



(1) Service syphon
(2) Lock nut
(3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, **connect** them together and **secure** the lock nut.

- **Continue** with the transfer.

2.2.11 Stopping the helium transfer

Fig. 40: Dewar manometer

- As soon as the helium level reads the required level or at a maximum **95%**, or the dewar is empty, **stop the transfer** by closing the flow valve at the dewar syphon. **Do not fill above 95%.** (→ Helium level check at the MRC / Page 29)
  - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/Fig. 27 Page 29)
- Set the dewar gas pressure regulator to **zero**.
- Close the dewar **primary ball valve.** (→ 1/Fig. 14 Page 19)
- **Remove** the **gas line** from the **dewar**.
- Refer to (→ Changing supply dewars during transfer / Page 36) if the magnet is not full.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

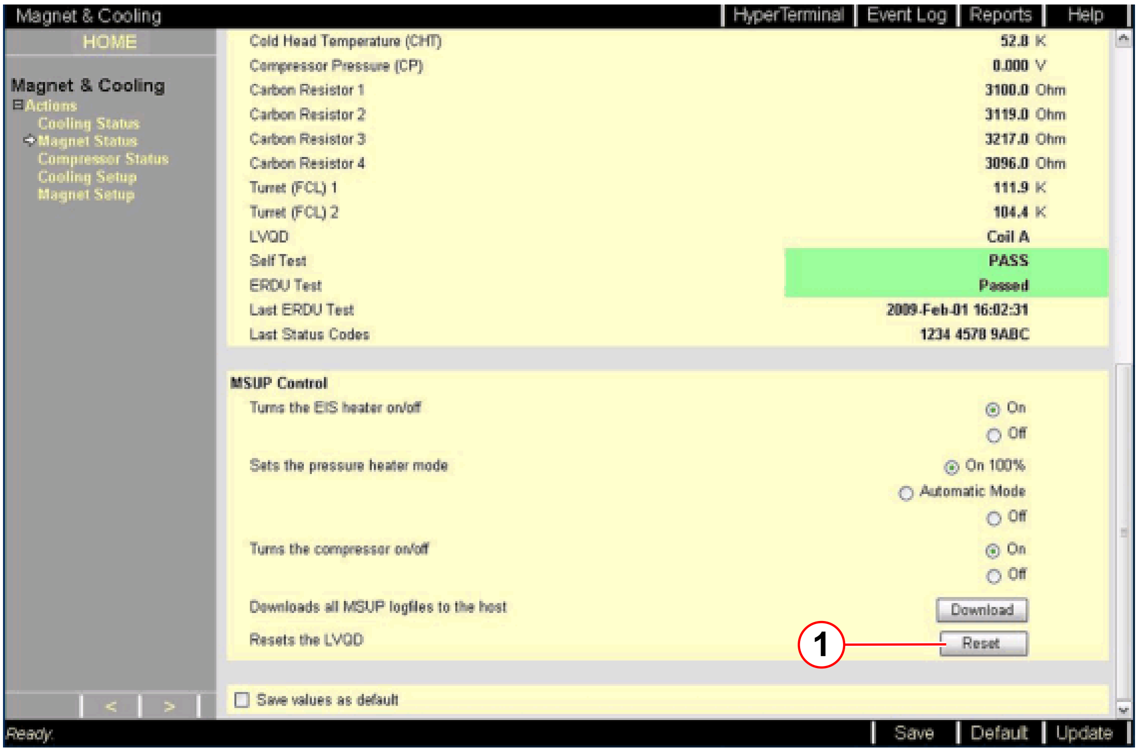
» **If the magnet is still at field, stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure drops.**

2.2.12 Final work steps

After transfer, check:-

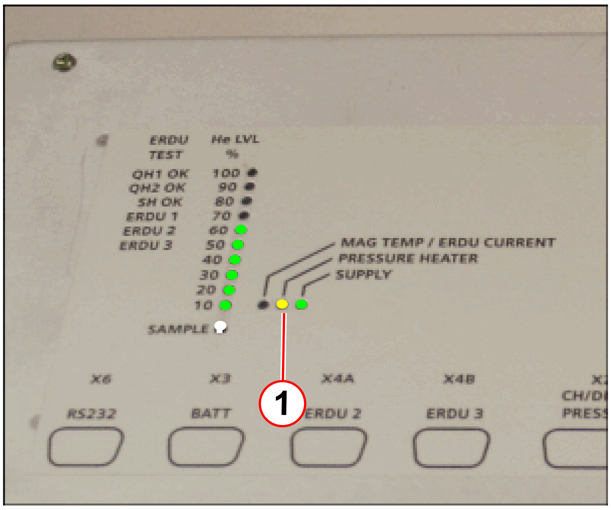
- Press the **LVQD reset button** to ensure the LVQD shows **waiting trigger**.

Fig. 41: LVQD reset button



- The **magnet bypass valve** is **closed.**
- The **pressure heater** is operating correctly.

Fig. 42: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

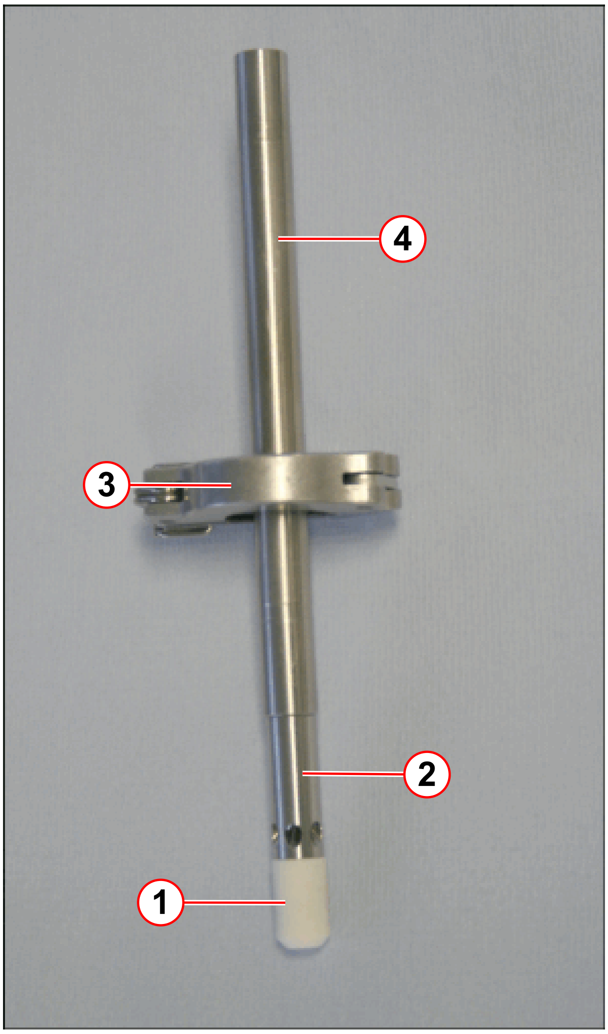
- The pressure heater **LED** on the **MSUP** is illuminated, or switching on and off.
- **No alarm** signals are evident at the **MSUP** or the **alarm box.**
- The **compressor** is switched **on** and **running**.
- The **magnet pressure** is stable at **15.5psi (A).**
- The **turret** and **venting** system are **warm** and **dry**.
- The filling **port** and **venting** system have **no leaks.** (Refer to **Venting Leak Check procedure).**
- The 4K **cold head flange** has **no leaks**.
- The used **dewars** are correctly **closed**, with their low pressure relief valves **open**.

2.2.13 Low ceiling height

Where magnets have been installed into rooms with a **low ceiling height**, it can be difficult to fit the assembled service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** must be used (➔ 3/Fig. 43 Page 42).

- Check the **ceiling height**:
  - Assemble the syphon (only if it can be fitted), or
  - Assemble the diffuser, bottom extension and support clamp

Fig. 43: Syphon configuration for low ceiling height



(1) PTFE extension tip (top fill)
(2) Helium fill diffuser tip (top fill)
(3) Syphon leg support clamp
(4) Extension leg

Refer to Magnet Service syphon kit

- Remove the **syphon end cap.**
- Ensure the **ring nut**, **brass spacer and O-ring** are fitted correctly.
- Fit the **diffuser** to the extension leg.
- Fit **support clamp** to the extension leg.
- **Warm** the syphon **blanking plug** to ease removal.
- **Remove** the syphon **blanking plug** using the C-spanner.
- **Lower** the assembly into the **neck**.
- **Screw** on the second **extension leg**.
- **Move** the syphon **clamp** up the extension leg and lower extensions into the entry port, to allow room for the syphon body.
- **Screw** the **diffuser leg** to the syphon **body**.
- **Remove** the syphon **clamp** and slowly **lower the syphon fully** into the syphon entry.
- **Screw** the **syphon port ring** nut into place and **tighten** to make a gastight **seal**.
- **Re-fit** the syphon **end cap**.

Use the support clamp for removing the syphon.

2.2.14 Helium pressures at altitude

2.2.14.1 Helium transfer with magnet bypass valve closed

As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

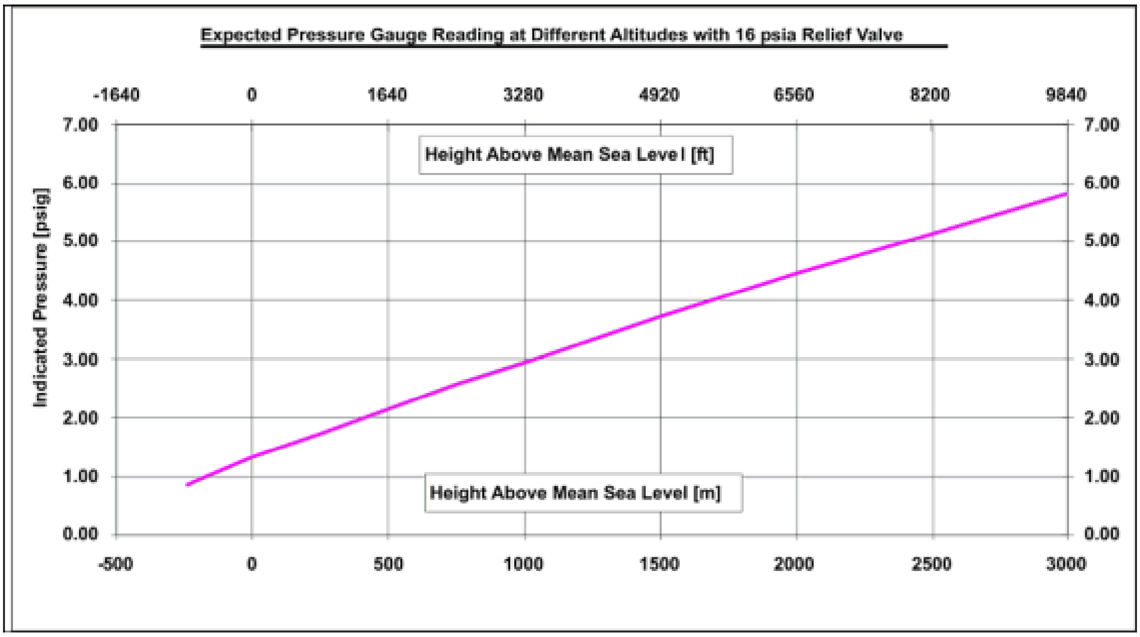
The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi.To calculate this:

- Determine the **magnet's altitude**

- Determine the **magnet gauge pressure** from the graphs below.

2.2.14.2 OR98 magnet - 16 psi (A) valve

Fig. 44: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 3 Dewar transfer pressures (typical)

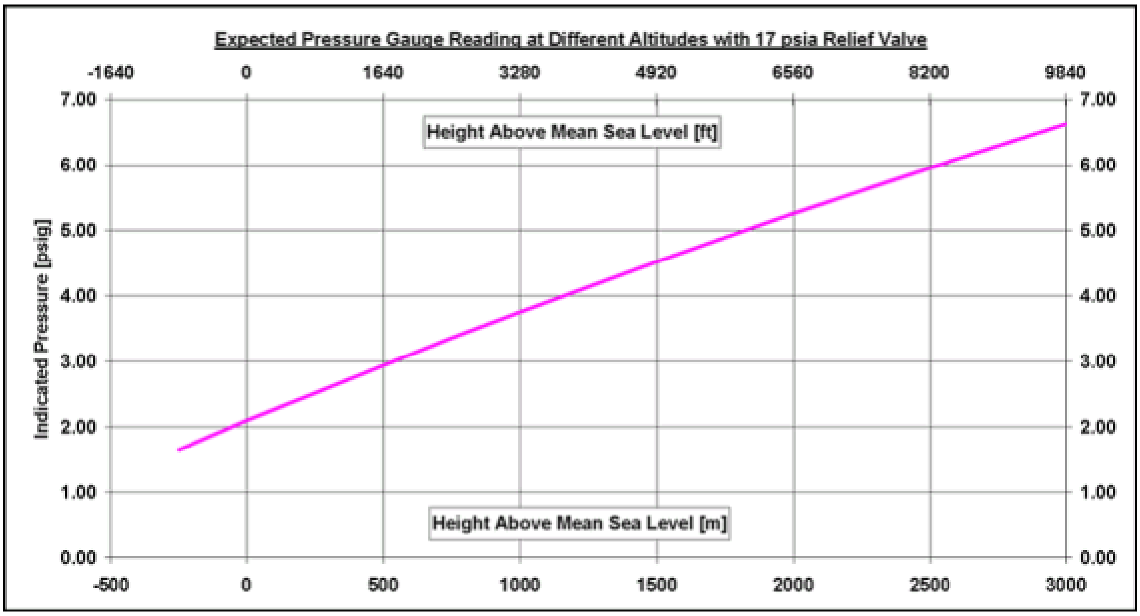
Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- **Add 4 psi** to this number to calculate the **required transfer pressure**
  - » **Example of 16 psi (A) vent valve assembly**
    Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

Maximum dewar pressure is 10.5 psi (G).

2.2.14.3 OR99/OR103 magnet 17 psi (A) valve

Fig. 45: Effect of a change in altitude on gauge pressure (17psi (A))



Tab. 4 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	6	7.75	9.25	10.5

- **Add 4 psi** to this number to calculate the **required transfer pressure**
  - » Example of 17 psi (A) vent valve assembly
    Altitude (1000m) = 3.75 psi (G) + 4 psi = 7.75 psi (G)

Maximum dewar pressure is 10.5 psi (G).

2.2.15 Verifying the electronic low pressure switch

This procedure may be carried out in conjunction with any other process involving de-pressurization of the helium vessel, i.e. fitting the service syphon, cold head replacement.

Pressure control of the helium vessel is managed by the **MSUP**, via the **Pressure heater control unit.** The low pressure switch must be checked **whenever the magnet is dep-ressurized.**

When pressure in the helium vessel drops to **within 0.2 psi (G)** above atmospheric pres-sure, the **electrical pressure switch** will open and turn **off** the compressor.

Atmospheric pressure may differ, depending on the altitude and local weather conditions.

As the helium vessel is **pressurized,** the switch closes and the **compressor** will **re-start**.

The pressure switch is factory calibrated to operate at 0.2 psi (G) above atmospheric pressure.

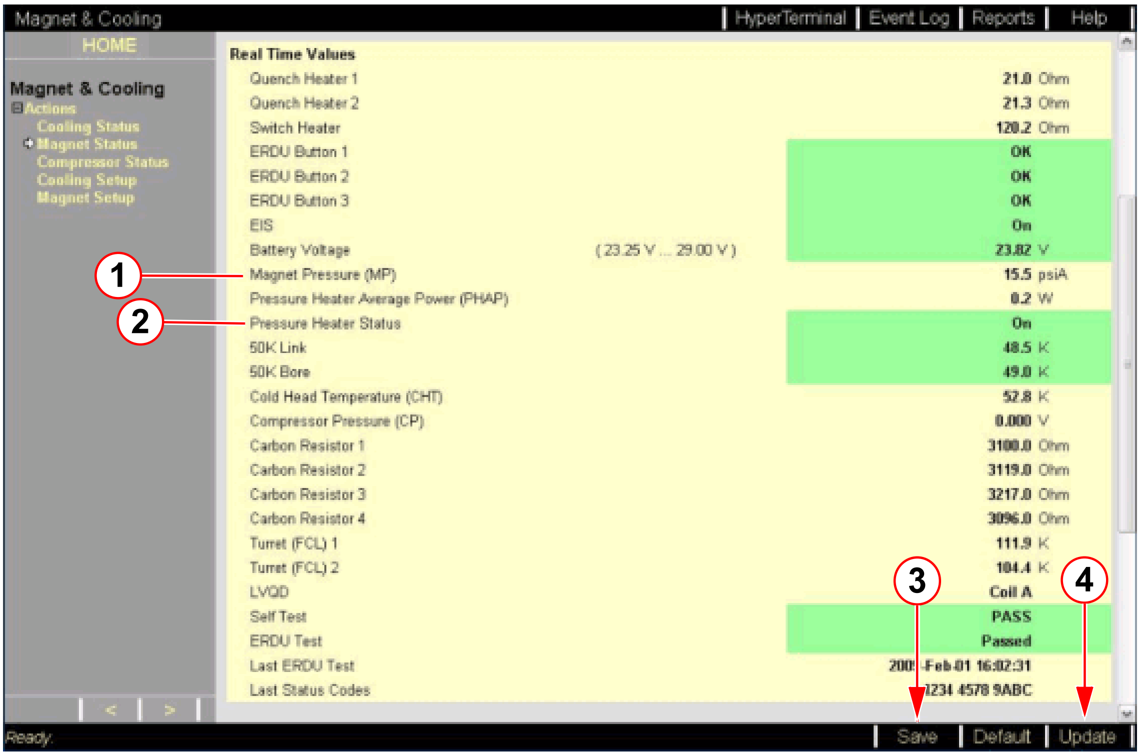
Perform this check after helium filling when the magnet syphon is *removed.

The filling port is closed and the refrigerator switched on.

Note: *Mobile systems have a fixed syphon.

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- Check **magnet pressure** is normal (**15.5 psi (A)**).

Fig. 46: Check magnet pressure level



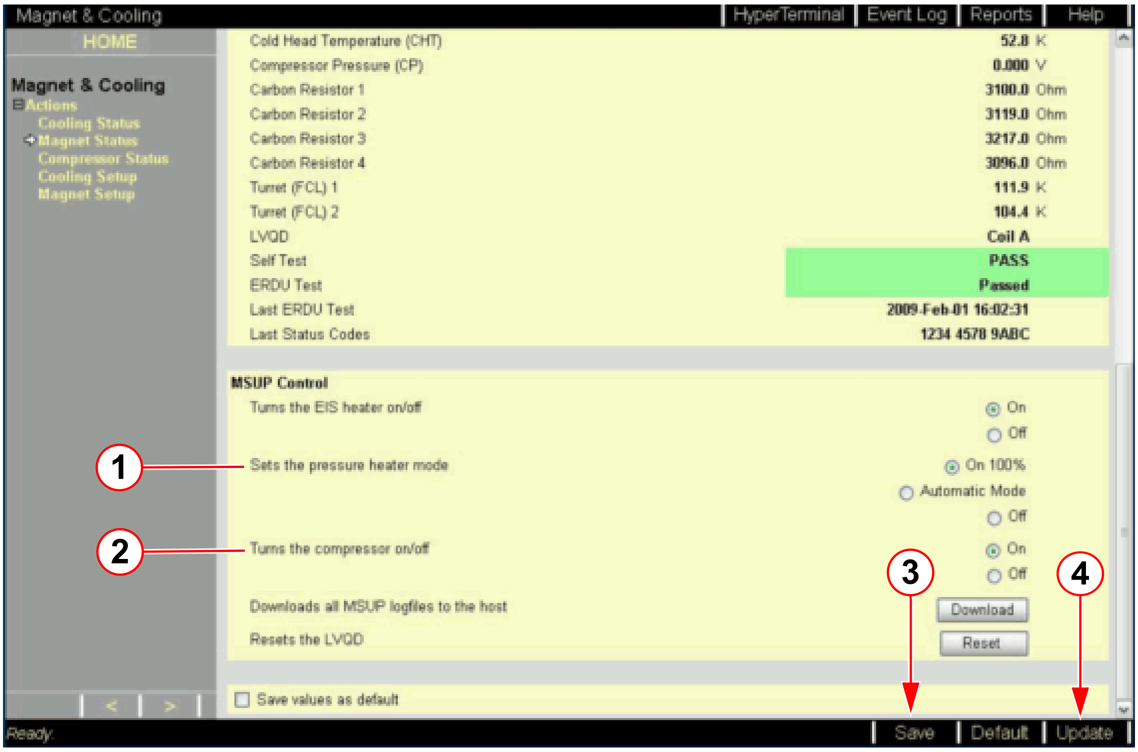
(1) Magnet pressure (MP)
(2) Pressure heater status
(3) Save button
(4) Update button

- Check the **compressor** is switched **on** and **running**.

Deactivate the pressure heater

- At **MSUP Control - Sets the pressure heater mode** select the **OFF** button. (➔ 1/Fig. 47 Page 46)

Fig. 47: Service screen Magnet & Cooling - Magnet status - MSUP Control

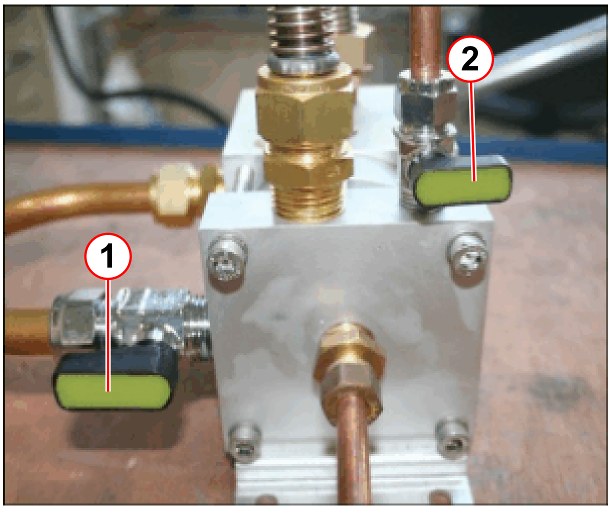


(1) Sets the pressure heater mode
(2) Sets the the compressor on/off
(3) Save button
(4) Update button

2.2.15.1 Depressurize the helium vessel

- **Depressurize** to within **0.2 psi (A)** of atmospheric pressure.

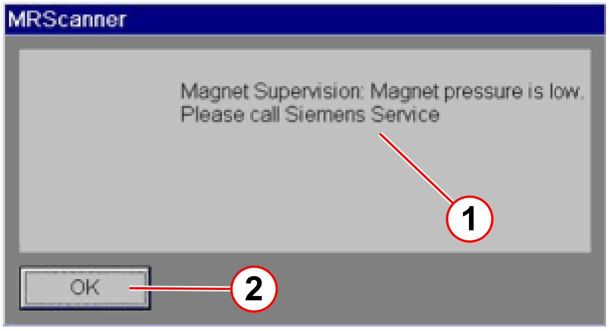
Fig. 48: Bypass valve positions for depressurization



(1) Magnet bypass valve OPEN
(2) Cold head bypass valve CLOSED

- **Fully open** the **magnet bypass valve** to **depressurize** the helium vessel.
- The **pressure switch** will activate when system pressure is within **0.2 psi (A)** of atmospheric pressure.
  - » this turns the **cold head OFF**.
- **Close** the magnet **bypass valve** to prevent **air ingress**.
- **Warm up** the vent **valve assembly.**
  - » **Take care** not to damage any components when using a **heat gun**.

Fig. 49: MSUP - Magnet pressure is low



(1) Pop-up message
(2) Click OK to clear

On the **MR Scanner:**

- A pop-up **alarm message** is displayed.
  - » **Magnet pressure is low**.
  - » Click **OK** to clear the alarm.
- The **alarm box** will sound when **pressure falls** to **alarm limits.**
  - » The yellow **WARNING LED** lights on the **alarm box**.
  - » The alarm **buzzer** is **activated**
  - » Press **AUDIO ALARM OFF** to silence the alarm.

- Check the **pressure is positive** when the **compressor** switches **OFF**.
- **Repressurize** the helium vessel.[1]

2.2.15.2 Reactivate the pressure heater

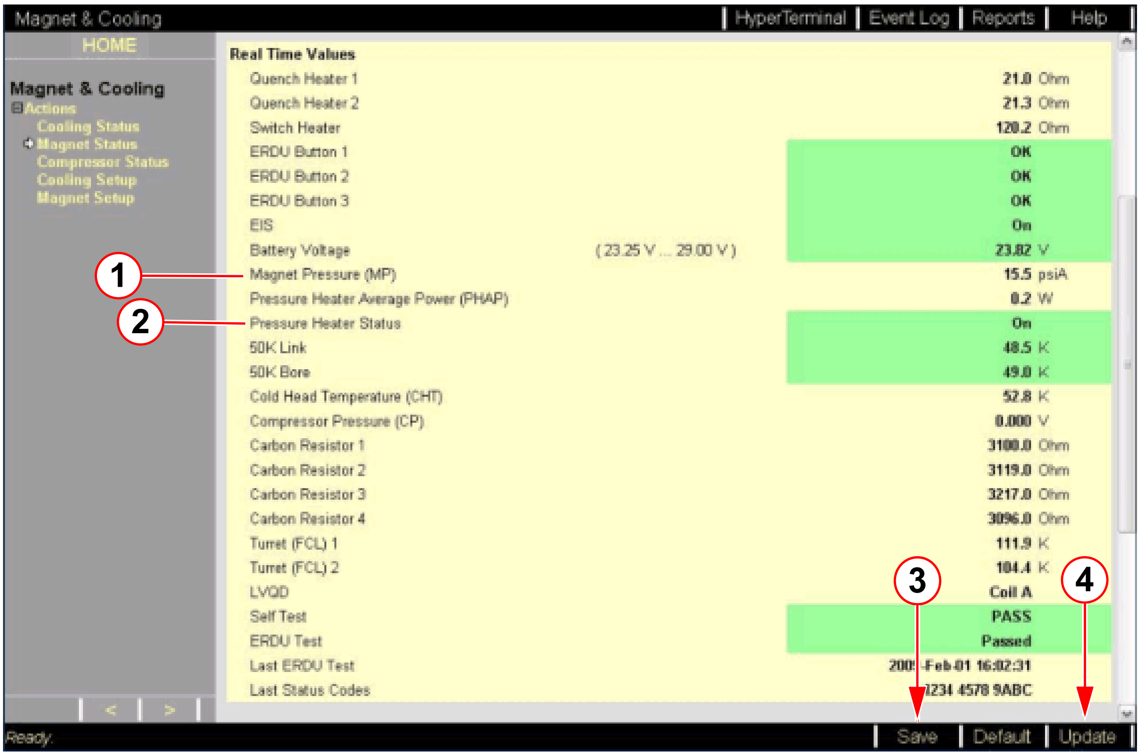
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button. (→ 1/Fig. 47 Page 46)
- Click **UPDATE**.(→ 4/Fig. 47 Page 46)
- Check that the **compressor** activates **automatically** as the pressure rises above **0.2 psi (A)** atmospheric.

1. If the verification is being carried out with another process (i.e. fitting the Magnet Service Syphon (except for Mobile systems), or replacing the Cold head), follow the **re-pressurization** part of this procedure when those processes are complete.

- **At Real-Time Values** - Monitor the **magnet pressure**.
  - Select **Update** to refresh data until pressure reaches ≥ **15.6 psi (A)**.

> Data is only accurate from the last refresh of the page, or from when the page was opened last.

Fig. 50: Check magnet pressure level



(1) Magnet pressure (MP)
(2) Pressure heater status
(3) Save button
(4) Update button

- At **MSUP Control - Sets the pressure heater mode** select the **Automatic mode** button (→ 1/Fig. 47 Page 46)
- Click **UPDATE**. (→ 4/Fig. 47 Page 46)
- Cancel any **alarms**.

> Check that the bypass valve is locked securely in the closed position.

> Contact the Magnet HSC if the test fails.

As the cold head recondenses the helium gas, the system pressure is returned to 15.5 psi (A).

2.3 Bottom filling the magnet

CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmoshphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

The bottom fill procedure must ONLY be performed when the magnet is at ZERO FIELD.

Bottom filling of the magnet will be required if:-

- The liquid helium level -
  - Falls to **zero** percent,
  - **and**
  - The temperatures measured from the ceramic carbon resistors **(CCR)** and coil diodes, indicates temperatures **no warmer than** 110K. Refer to (→ Fig. 51 Page 51), (→ Fig. 52 Page 52), (→ Fig. 53 Page 53).

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.

Inform Magnet HSC immediately if the helium level is zero. Magnet coil temperatures must be confirmed before continuing with the helium fill.

Possible causes for this include:-

- Post magnet quench, (ramping or from persistence).
- Faulty helium level probes.
- Alarm box buttons still depressed.
- Pressure heater is left on.
- Long transportation and storage times.
- Cold head or compressor failure.

2.3.1 Checking magnet temperatures

Fig. 51: OR98 Magnet Acceptance Document (example)

SIEMENS Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21236477 **SMT Reference** 21245226 05.10

System Description

1.495T Actively Shielded Superconducting Twin Turret Magnet System OR98

Measured Magnet Centre (Offset)	+1 (P)	mm
Gradient Coil Position	39	mm
Overall System Length	1409	mm

This system requires 12 Z2 rings and 2.38 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Measured value	
Operating Current @ 1.495T	510.83	AMPS
Field/Current Ratio	12.46	kHz/0.1 Amp

Magnet Type: OR98--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC Outer (D-E	926		2951
CCR3 Top Inner (C-E	1060		2812
CCR2 Middle Inner (B-E)	1035		2977
CCR1 Bottom Inner (A-E	1093		3138
TDC Calibration Values:	882.4	1181.8	3075.3

Helium Level Probe Resistance **(When using a 250 mA supply)**

	Probe 1	Probe 2	
0% Resistance	107.0	107.0	Ohms
0 % Calibration	448	448	
100 % Resistance	17.9	17.9	Ohms
100 % Calibration	75	75	
Diode Table No	-2		
Pressure Heater Res	22.1		Ohms

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
56	55	41	K	782 Milliwatts

Checked by K Suraina Signature

Date 28/05/2010

Fig. 52: OR99 Magnet Acceptance Document (example)

SIEMENS

Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21240447 **SMT Reference** 21243544 05.10

System Description

2.894T Actively Shielded Superconducting Twin Turret Magnet System OR99

Measured Magnet Centre (Offset)	-3 (S)	mm
Gradient Coil Position	43	mm
Overall System Length	1669	mm

This system requires 12 Z2 rings and 1.44 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Specification value	Measured value	
Operating Current @ 2.894T	< 500 AMPS	465.97	AMPS
Field/Current Ratio	N/A	26.44	kHz/0.1 Amp

Magnet Type: OR99--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC (D-E)	1072		2863
CCR3 Top (C-E)	872		3114
CCR2 Middle (B-E)	1027		3182
CCR1 Bottom (A-E)	1036		2963
TDC Calibration Values:	1022.6	1322.1	2991.4

Helium Level Probe Resistance **(When using a 250 mA supply)**

	Probe 1	Probe 2	
0% Resistance	80.3	80.3	Ohms
0 % Calibration	336	336	
100 % Resistance	0.5	0.5	Ohms
100 % Calibration	002	002	
Diode Table No	-2		
Pressure Heater Res	20.5	Ohms	

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
54	50	40	K	874 Milliwatts

Checked by K Suraina Signature

Date 10/05/2010

Page 1 of 1 25-AD EPS 000 01

Fig. 53: OR103 Magnet Acceptance Document

SIEMENS

Magnet Acceptance Documentation
Installation data
MR Magnet Technology Quality Regulations

Cryostat number 33002284 **SMT Shipped System** 33002281 07.11

System description 2.894T Actively Shielded Superconducting Twin Turret Magnet System OR103

	SKYRA	MONET	
Measured magnet centre (Offset)	-4 (S)	-4 (S)	mm
Gradient Coil Position	174	174	mm
Overall Magnet Length	1670	1670	mm
This system requires	0	0	Z2 Rings
This system requires	2.63	2.63	kg of iron to shim

(This is a theoretical calculation that may change due to site conditions)

Operating parameters

Description	**Specification value**	**Measured value**
Operating current @ 2.894T	< 500 Amps	487.35 Amps
Field/Current Ratio	N/A	25.28 kHz/0.1 Amp

Magnet type: OR103-00

Carbon resistor values

Position	**Room temperature**	**Nitrogen temperature**	**Helium temperature**
CCR4 d - e top	1071 Ohms		2573 Ohms
CCR3 c - e	991 Ohms		2612 Ohms
CCR2 b - e	929 Ohms		2797 Ohms
CCR1 a - e bottom	1007 Ohms		2841 Ohms
CCR4 calibration values	1031 Ohms	1304 Ohms	2621 Ohms

Helium level probe resistance (when using a 250mA supply)

	Probe 1	**Probe 2**
0% resistance	107 Ohms	107 Ohms
0% calibration	448	448
100% resistance	10.8 Ohms	10.8 Ohms
100% calibration	45	45
Diode table no.	2	
Pressure heater resistance	20.5 Ohms	

Final temperature sensor recon values

Shield bore	Shield Link	Coldhead	Recon margin
55K	48K	39K	853mW

Checked by K. Baker **Signed** KmBaker

Date 11/07/2011

Page 1 of 1 25-AD EPS 000 01

To **recover** a magnet with a **bottom helium fill**, the following must be checked:-

Access the Service software at the **MRC (MRAWP) Host** to measure:

- **Helium level** values.
- **CCR** values.

Tab. 5 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

Carbon 4/CCR4 is for measuring helium gas temperatures during ramping and is not applicable to filling.

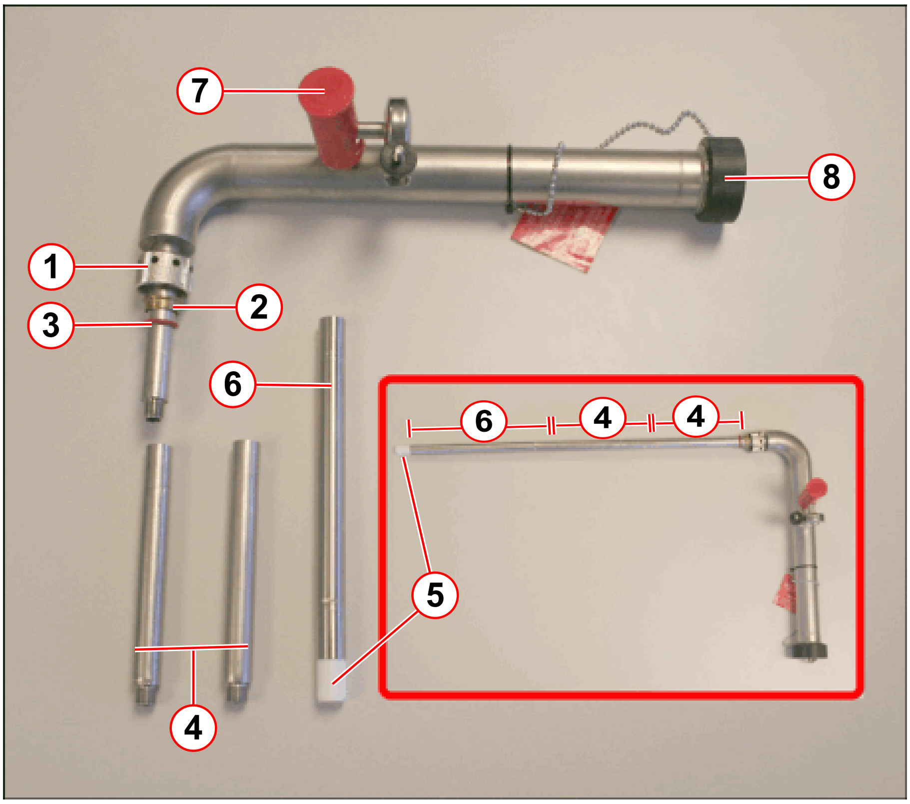
2.3.2 Bottom fill procedure

The bottom fill procedure must ONLY be performed when the magnet is at ZERO FIELD.

- Prepare the **dewar** and **dewar syphon.** (→ Dewar preparation / Page 15), (→ Dewar syphon preparation / Page 17)
- Check the **ceiling height.** (→ Low ceiling height / Page 41)
  - Assemble the **syphon** (only if it can be fitted), or
  - Assemble the **diffuser**, bottom **extension** and support **clamp**.

2.3.2.1 Assemble the service syphon

Fig. 54: Magnet service syphon assembly - bottom filling

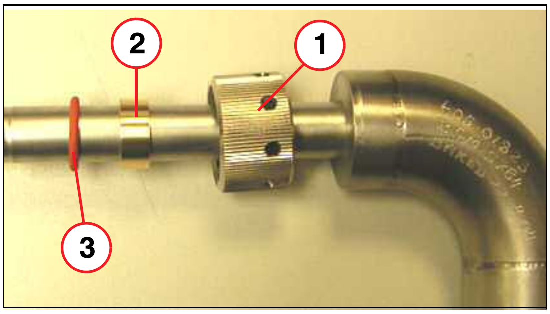


(1) Ring nut
(2) Brass spacer
(3) O-ring seal
(4) 140 mm syphon legs (2 off)
(5) 22 mm PTFE seal (bottom fill)
(6) 200 mm syphon leg (1 off)
(7) Vacuum valve
(8) Syphon end cap

Tab. 6 OR98 / OR99 / OR103 BOTTOM filling syphon extension legs

Magnet type	140 mm leg	200 mm leg	22 mm PTFE seal
OR98 (Aera) OR99/103 (Skra)	2	1	1

Fig. 55: *Assembling the Ring Nut Seal*



(1) Ring nut
(2) Brass spacer
(3) Silicon 0-ring

- Fit the ring nut, brass spacer and the O-ring to the **service syphon leg.**

Refer to table above (➔ Tab. 6 Page 55)

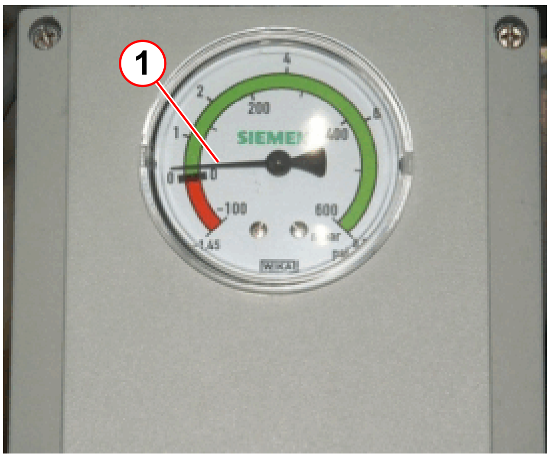
- Connect the **extension legs** and the 22 mm **PTFE seal:**
  - Screw the 2 x **140 mm legs** together, and connect to the **syphon**.
  - Fit the 1 x **200 mm leg.**
  - Fit the **22 mm PTFE seal** to the 200mm leg.

> The minimum length required to reach the syphon cone is:
>
> OR98 - 554 mm (+/- 5 mm)
>
> OR99 - 540 mm (+/- 5 mm)

2.3.2.2 Depressurize the helium vessel

- **De-activate** the **pressure heater.** (➔ Deactivate the pressure heater / Page 21)
- **Depressurize** the helium vessel (➔ Depressurize the helium vessel / Page 22)
- Note the **pressure** (using the **MRC**, or **Laptop**) when the **compressor** switches **OFF**.

Fig. 56: *WIKKA Pressure switch - with a small positive pressure*



- **Check** the helium vessel has a **small positive pressure.**
- **Warm** and dry the **syphon blanking plug** using a heat gun to ease removal.

- **Dry** the syphon **port**.
- **Remove** the syphon **end cap.** (→ Fig. 33 Page 33)
- **Remove** the syphon **blanking plug.**

CAUTION

Risk of cold burns.

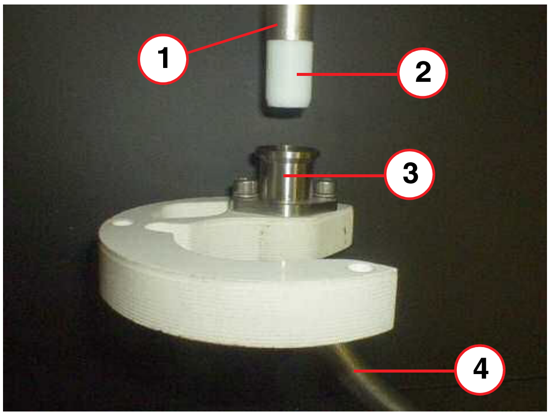
Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

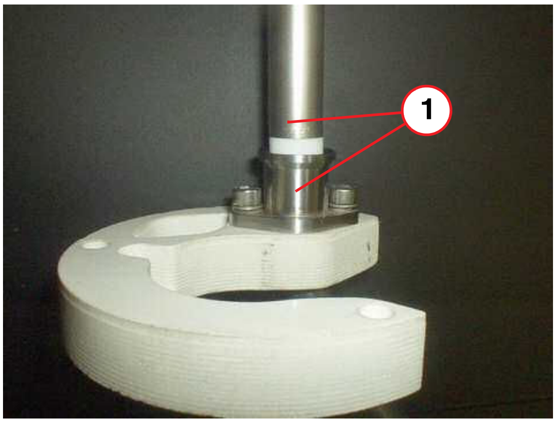
- **Slowly** lower the **service syphon** fully into the neck.
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 57: Bottom fill syphon cone (pictorial view)



(1) Syphon leg
(2) PTFE Seal (Bottom Fill)
(3) Syphon cone
(4) Syphon tube

Fig. 58: Bottom fill - syphon leg fully engaged (pictorial view)



(1) Syphon leg - fully engaged

Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the **syphon port ring nut** (→ 1/Fig. 55 Page 56) by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap.**

2.3.3 Liquid helium transfer - bottom fill

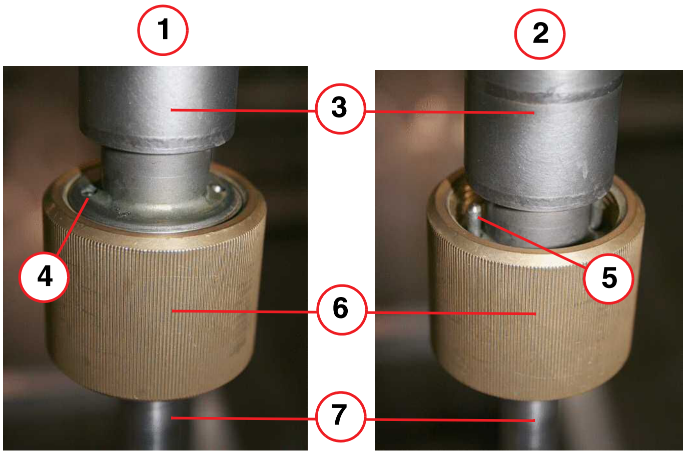
Before helium transfer, refer to (→ Helium pressures at altitude / Page 42)

The helium dewar must be positioned close to the magnet, with the dewar syphon fitted.

For bottom filling a magnet at zero, there is no need to pre-cool both syphons.

- **Check** the **dewar** gas supply pressure is set at **0.27bar (4 psi (G)).**
- **Open** the **dewar syphon flow valve** half way (2 turns) to **purge** the syphon of air.

Fig. 59: Dewar syphon flow valves - open and closed positions



(1) FULLY OPEN position
(2) FULLY CLOSED position
(3) Stainless steel body
(4) Guide pin flush with body
(5) Guide pin
(6) Brass nut
(7) Syphon leg

Fig. 60: Helium filling

- **Close** the dewar syphon **flow valve.**
- **Remove** the service syphon end cap and **couple the two syphons together** and secure the **locknut**.
- **Open** the dewar syphon **flow valve.**
- Maximize the transfer by fully opening the **dewar syphon flow valve**.
  - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/Fig. 59 Page 58)
  - » From **fully closed** to **fully open**, is **4 turns**.
- Ensure the transfer dewar pressure is **4 psi (A)** at normal altitude. Refer to (→ Helium pressures at altitude / Page 42).

2.3.3.1 Monitor the CCRs

- Monitor the **CCRs** every **15 minutes**, to measure the **temperature drop**.
  » Refer to the **acceptance documentation** (Installation Protocol) for required values.
  » If the CCRs **do not** get cooler, **contact Magnet HSC.**
- When the required value is reached, **monitor** the **helium level** at the **Host** or **MSUP** .

2.3.3.2 Perform a helium level check

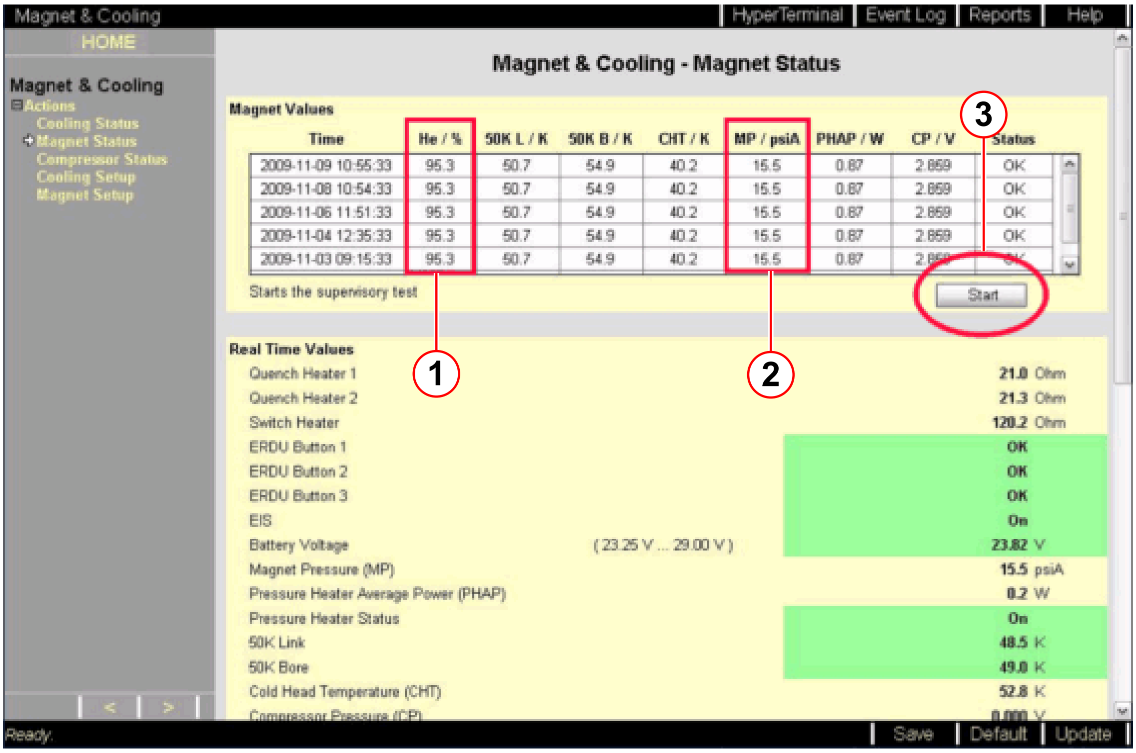
> During installation, if the MRC is not available:
> 1.Take a helium sample using the MSUP. (→ Helium level check at the MSUP / Page 61)

Helium level check at the MRC

- Check the liquid helium level by using Service Software on MRC (MRAWP) Host.
  - Check the liquid helium level using the MRC.
  - Access the Service Software at the MRC.
  - Select the menu item Magnet & Cooling - Magnet Status.

- Press the Start button (→ 3/Fig. 61 Page 60) to initiate the helium measurement. (→ 1/Fig. 61 Page 60)

Fig. 61: Checking liquid helium and magnet pressure levels

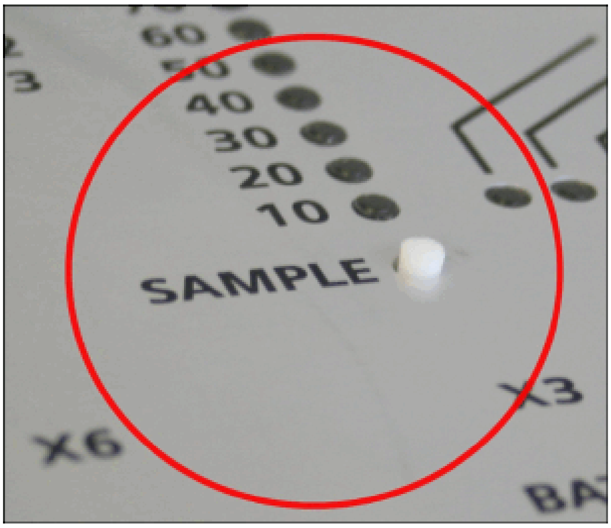


(1) Liquid helium level
(2) Magnet pressure
(3) Start button

> Data is only accurate from the last refresh of the page, or from when the page was opened last.

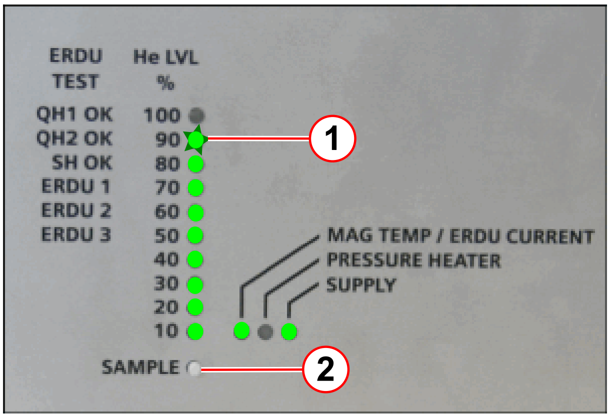
Helium level check at the MSUP

Fig. 62: MSUP 2306 - depress SAMPLE to activate



- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.

Fig. 63: MSUP - Helium level sample



(1) Sample switch
(2) Helium level

- The green LEDs illuminate and display the actual liquid helium level percentage.
- If the 90% LED is flashing, STOP the transfer. Refer to (→ Tab. 7 Page 61).

Tab. 7 MSUP Helium display

MSUP LEDS	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue to fill as required.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATLEY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED.

* The LEDs return to ON after approximately 30 seconds.

To avoid overfilling, the helium transfer must be stopped as soon as the 90% LED illuminates.

- Repeat the measurement every 5 minutes. An increase in level should be seen within 20 minutes.
  » If not, stop the transfer and investigate the reason for poor transfer efficiency.

Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.

Maintain a pressure of about 0.27 bar (4 psi (G) at the helium dewar.

Too much or too little pressure will result in inefficiency.

At 4 psi (G), the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum.**Stop the transfer immediately - the syphon may need to be replaced.**

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

» **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

2.3.4 Stopping the helium transfer

Fig. 64: *Dewar manometer*

- As soon as the required helium level is obtained or a maximum of level of **95%**, or the dewar is empty, stop the transfer by closing the **flow valve** at the dewar syphon. **Do not fill above 95%**. (→ Helium level check at the MRC / Page 59)
  - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties. (→ 6/Fig. 59 Page 58)
- Set the **dewar** gas pressure regulator to **zero**.
- Close the dewar **primary ball valve**. (→ 1/Fig. 14 Page 19)
- **Remove** the **gas line** from the **dewar**.
- Refer to (→ Changing supply dewars during transfer / Page 36) if the magnet is not full.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

» **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

2.3.5 Removing the service syphon

- **Deactivate** the pressure heater. (→ Deactivate the pressure heater / Page 21)
- **Depressurize** the helium vessel. (→ Depressurize the helium vessel / Page 22)

CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **Warm** the **syphon joint** to room temperature (using a heat gun).
- **Warm** the syphon **entry valve** to room temperature and **dry**.
- **Warm** the syphon **end cap** ready for fitment onto the syphon.
- **Disconnect** the syphons.

CAUTION

Risk of cold burns.

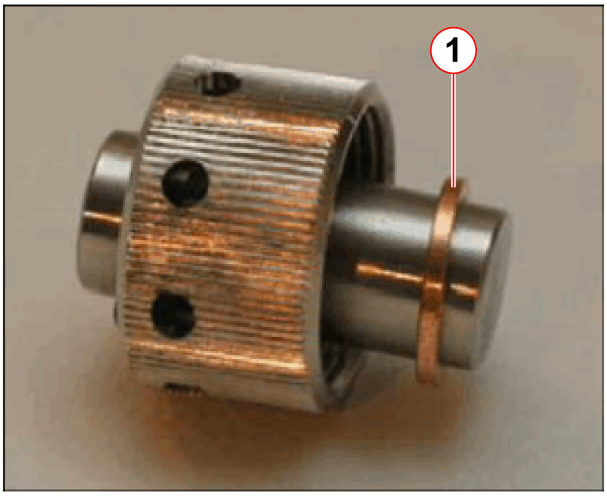
Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

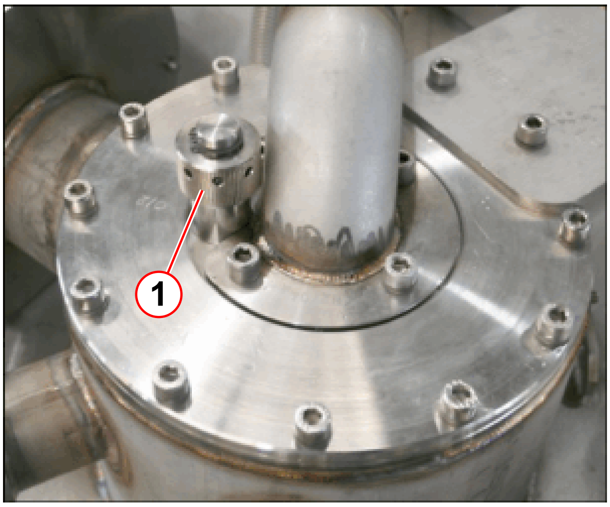
- **Immediately** close off the **end** of the service syphon with the **syphon end cap**.
- **Immediately remove** the service **syphon** from the magnet.

Fig. 65: Syphon blanking plug



(1) Copper seal

Fig. 66: Fit/remove the syphon blanking plug



- **Immediately** fit the **syphon blanking plug** using a new copper seal (➔ Fig. 65 Page 64).
- **Warm up the threads** on the plug and **carefully** tighten with the syphon C-spanner.
  - There might be **ice** on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.

- **Warm** up the magnet bypass **valve assembly** and check the **bypass valve** is **closed**.

CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **Pressurize** the **helium vessel** to **15.5 psi (A).**
  - **Reactivate** the pressure heater.(→ Reactivate the pressure heater / Page 47)
  - Check that all **pressure alarms** are deactivated.
  - Check the **compressor** is **running**.
- **Warm** the **extension legs** using a heat gun.
- Remove the **bottom fill extension leg** and **PTFE seal.**
- **Warm** the **turret and venting valves** to room temperature with a heat gun.
  - **remove** any **surplus water** with paper towelling.
- **Move the dewar** out of the imaging suite into a **well ventilated area**.
- Complete **depressurizing** the dewar by opening the **primary ball valve.** (→ 5/Fig. 38 Page 37)
- When at the gas pressure is at **zero, close** the **primary ball valve** on the dewar.
- **Remove** the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar. (→ 4/Fig. 38 Page 37)
- **Open** the **low-pressure dewar relief valve.** (→ 1/Fig. 38 Page 37)
- **Warm** both **syphons** with the heat gun **until dry**, and store safely.
- **Leak check** the turret and venting.

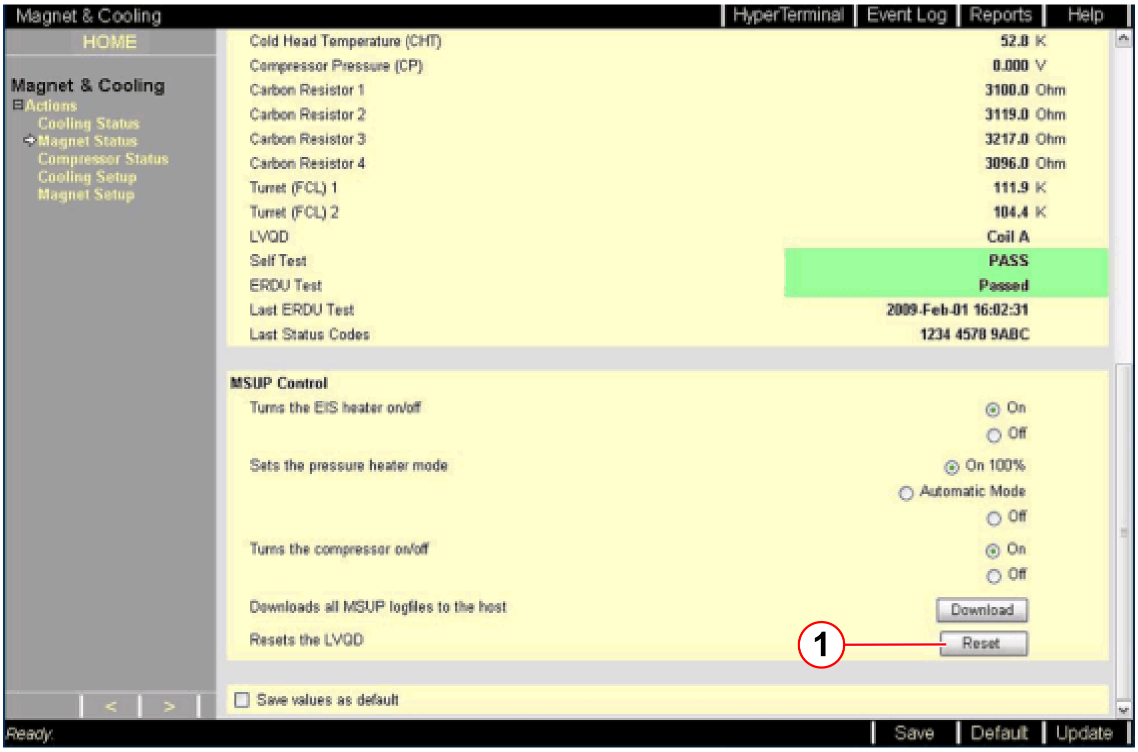
After a bottom fill, it is recommended that the first magnet energization is performed the next day. This will allow the magnet coils to thermally stabilize.

2.3.6 Final work steps

After transfer, check:-

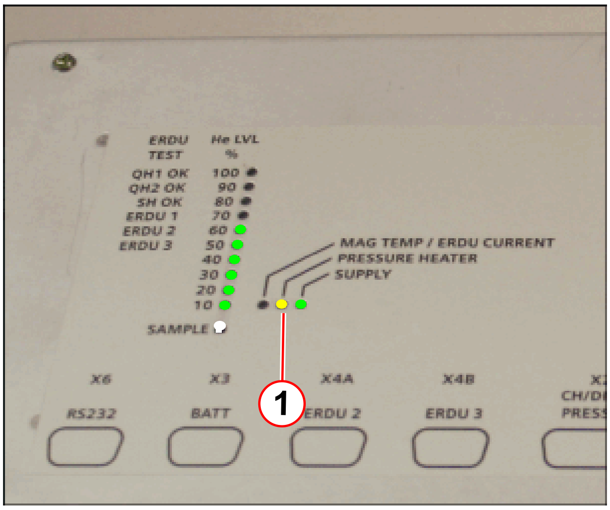
- Press the **LVQD reset button** to ensure the functionality of the **LVQD circuit**.

Fig. 67: LVQD reset button



- The **magnet bypass valve** is **closed.**
- The **pressure heater** is operating correctly.

Fig. 68: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

- The pressure heater **LED** on the **MSUP** is illuminated, or switching on and off.

- **No alarm** signals are evident at the **MSUP** or the **alarm box.**
- The **compressor** is switched **on** and **running**.
- The **magnet pressure** is stable at **15.5psi (A).**

- The **turret** and **venting** system are **warm** and **dry**.
- The filling **port** and **venting** system have **no leaks.** (Refer to **Venting Leak Check procedure).**
- The 4K **cold head flange** has **no leaks**.
- The used **dewars** are correctly **closed**, with their low pressure relief valves **open**.

3.1 Quench tube check

- Perform a visual inspection of the quench tube.
  » Refer to (→ Quench tube / M7-010.833)
- Check if the Installation protocol is signed by the Quench line manufacturer.

> Ensure the Quench Line installation is complete before performing the first ramping procedure.
>
> The Quench Line Acceptance Protocol must be checked before the first ramp.
>
> Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer.
>
> Acceptance Protocol can be found under:
>
> **Product Information Home > Planning > Planning Guides > MR Systems > Quench Line Design and Calculation Tool**

3.2 Identification of the controlled access area (0.5 mT area)

- Check that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.

  » Refer to: (→ Identification of the controlled access area (0.5 mT area) / M7-010.833)

WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

» **Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.**

» **Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.**

» **The sign must be posted in a way that it is visible even when the door to the controlled area is opened.**

3.3 Magnet stop button test

- Perform the Magnet stop button test
  » Refer to (➔ ERDU / M7-010.833)

3.4 Venting leak check

3.4.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1 x 10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as **1 x 10^{-4} mbar l/s.**

The magnet must be checked after any **service operations** have been performed, such as:

- During installation.
- After helium filling.
- After a cold head exchange.
- After any exchange or disturbance of a part on the high pressure side of the magnet.

CAUTION

Risk of icing

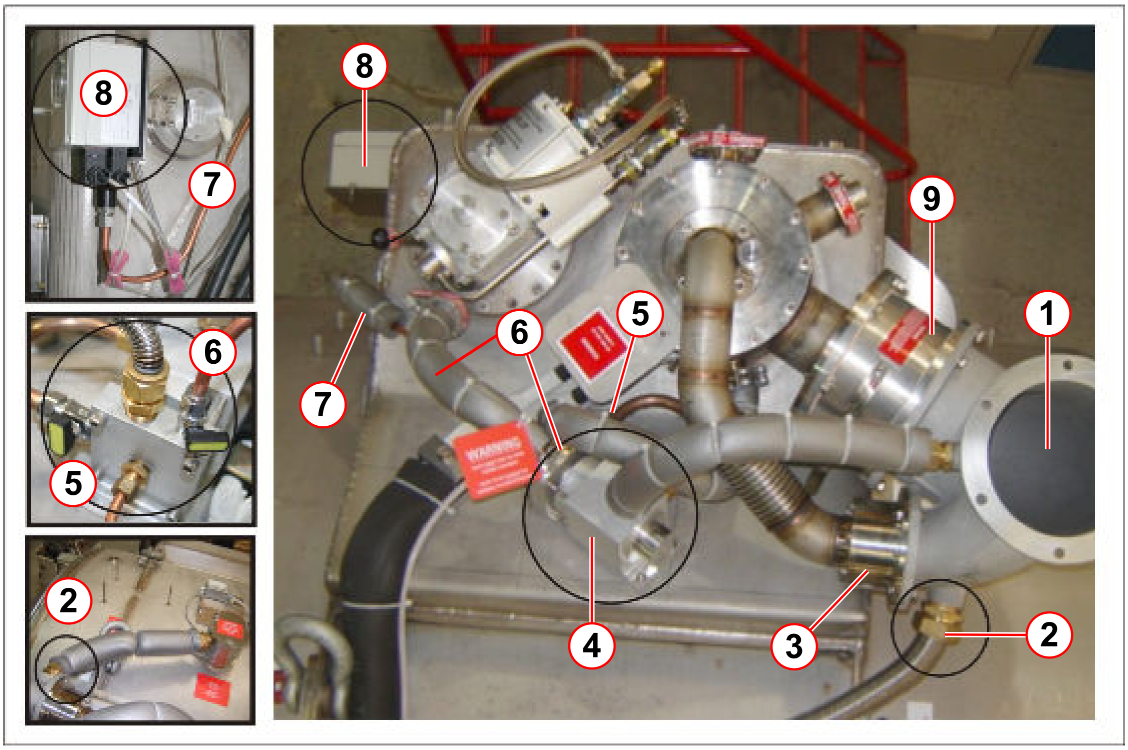
Fine leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

» **Leak check with a suitable leak check device.**

3.4.2 Venting system

The document describes the procedure for **OR98**, **OR99** and **OR103** venting systems.

Fig. 69: Normal venting configuration



(1) Quench valve elbow
(2) Flexible connection from OVC vent port
(3) Auxiliary burst disk (1.38 bar)
(4) Venting valve assembly
(5) Magnet system bypass line
(6) Cold head vent line and bypass valve
(7) Pressure display & control switch line
(8) Pressure display & control switch
(9) Quench valve with integral 15 psi (G) burst disk

3.4.3 Safety

CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

» **Fittings on the high-pressure side must not be disconnected without first de-pressurizing the helium vessel.**

WARNING

Magnet at field.

Personal injury or damage to equipment may occur.

» **Do not use magnetic tools for repairing or tightening seals when the magnet is at field.**

» **Only non-magnetic equipment may be taken into the magnet room.**

It may also be necessary to disable the pressurization heater.

3.4.4 Equipment required

- Heat gun.
- Paper towelling.
- **Accepted** leak detection equipment.
  - **Helium mass spectrometer**.

Fig. 70: Edwards Leak detector

- Use the Edwards hand-held leak detector (part no **10614850**).

  This is the **recommended equipment** for leaks **greater than 1 x 10^{-4} mbar l/s.**

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10^{-4} mbar l/s **=** 1 x 10^{-4} ml/sec at normal pressure and temperature.

The functionality of the Edwards leak detector can be compromised by magnetic fields.

The mass spectrometer is used for the detection of ultra fine leaks less than 1 x 10^{-5} mbar l/s. This equipment must only be used when trouble-shooting is required.

- Check that the equipment is detecting helium.
  - Check the equipment using helium from a known gas supply (i.e. regulated gas bottle).
  - Refer to the equipment manufacturer's operating instructions.

- The calibration certificate must be valid.

3.4.5 Pre-requisites for leak checking

- A suitable **electronic leak detector** is available.
- A suitably long **remote sensor** must be used when magnets are at field.
- The helium vessel must at least be at **normal operating pressure (15.5 psi (A))** prior to leak checking.
  - Activate the **pressure heater** (if required) to raise pressure.
- All joints are **dry** and at **room temperature** (use a heat gun).

3.4.6 Leak Checks at installation/service

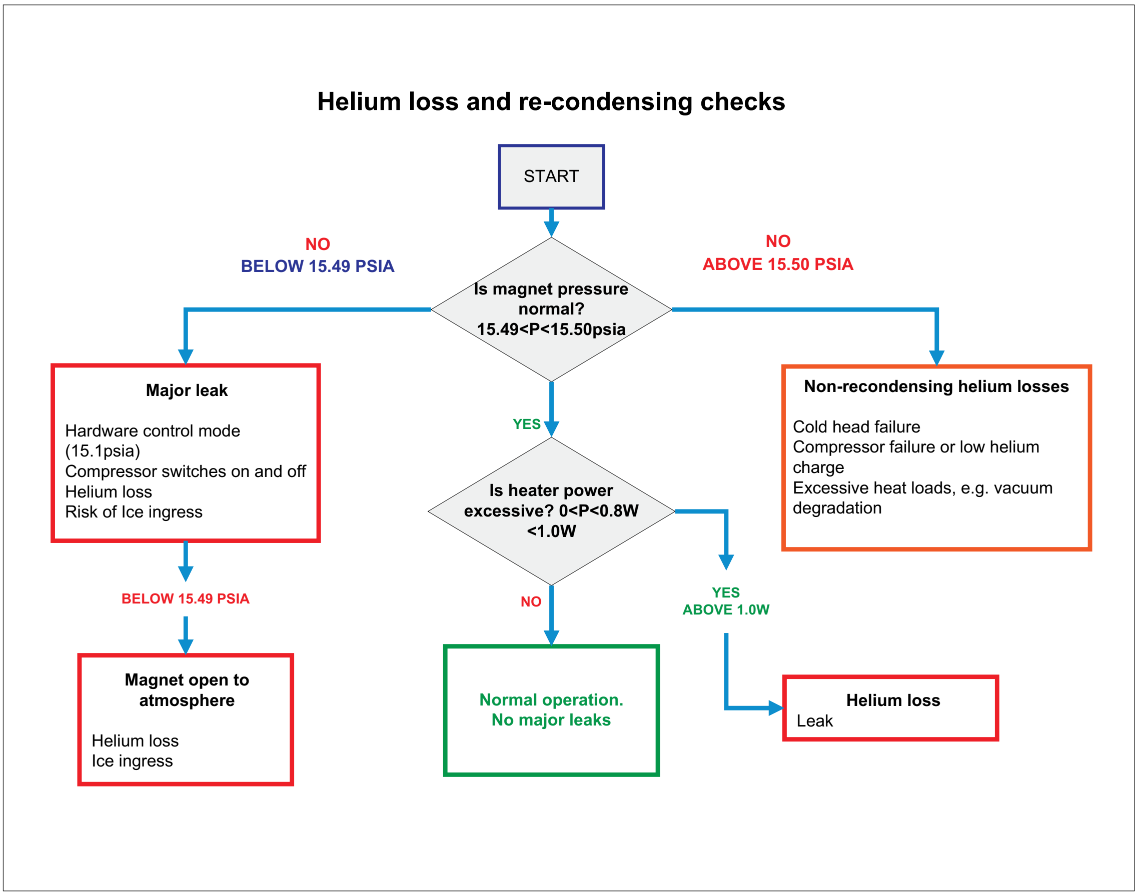
Installation

- Check the **system pressure** on arrival at the installation. If the system has **low**, or **zero** pressure then a **gross leak** could have occurred during shipment.
  - All seals, burst disks and fittings on the **high pressure side** of the vent valve must be checked **before** the cold head is turned on and the magnet run to field.
  - Any **joints** which are broken and re-made during installation must be checked.
  - Leaks can occur through the **internal venting system** (valves) as well as to atmosphere.

Service

- Leak check all seals that have been subjected to **prolonged low temperature** and the subsequent build up of frost. For example, those which have been cooled to low temperature by **helium filling**, **depressurization**, or magnet **ramping**.
  - Any **joints** which are broken and re-made during servicing must be checked and replaced as required.
  - The turret, venting valve assembly and quench valve must be **warmed up to room temperature** after any service procedures have been performed, **prior** to leak check.

Fig. 71: 4K- Helium and re-condensing checks



The illustration above shows the condition in which the magnet should return, after a number of hours from use, and has not been affected by external heat loads.

3.4.7 Procedure

The leak detector must be set for maximum sensitivity.

- Switch on the leak detector and allow it to warm up.
- Check the equipment is working correctly (detecting helium). This should be done both before and after use.

Ensure that the atmospheric environment is not saturated with helium gas (i.e. do not leak check soon after a helium fill, or depressurization).

- Determine the background level in the magnet room.

The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).

This level will be undetectable by the hand-held device.

- Note response time of the detector.

During leak checking, move the instrument slowly over the leak point to give the detector time to respond.

Start at the **lowest** point:

- Hold the detector at the suspected leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is still indicated.

The following points must **always** be observed:-

- Joints on the **'HIGH' pressure** side of the venting system will generally be fitted with **metal seals.** Before dismantling a joint, ensure a replacement seal is available.
- **DO NOT** use rubber seals as a substitute for metal seals.
- **DO NOT** re-use metal seals, always replace with a new seal.
- **DO NOT** use vacuum grease.

Do not use any form of vacuum grease on O-rings and seals.

It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

- After repair, leak check the entire venting system, not just the joints which have been repaired.

3.4.8 External leak checks

3.4.8.1 External leaks

The venting and components on the **high pressure** side of the magnet system require leak checking to ensure that the re-condensing performance is achieved. This includes:

- Cold head (➔ Cold head checks / Page 77).
- Turret top plate, including helium syphon port and venting. (➔ Turret and venting / Page 80)
- Vent valve assembly (➔ Fig. 79 Page 83)
  - **16** psi (A) = **OR98**
  - **17** psi (A) = **OR99/OR103**
- Pressure gauge assembly. (➔ Fig. 80 Page 84)

- Connecting pipe couplings.

If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.

The magnet must be RUN TO ZERO before leak checking inside the quench valve or auxiliary vent.

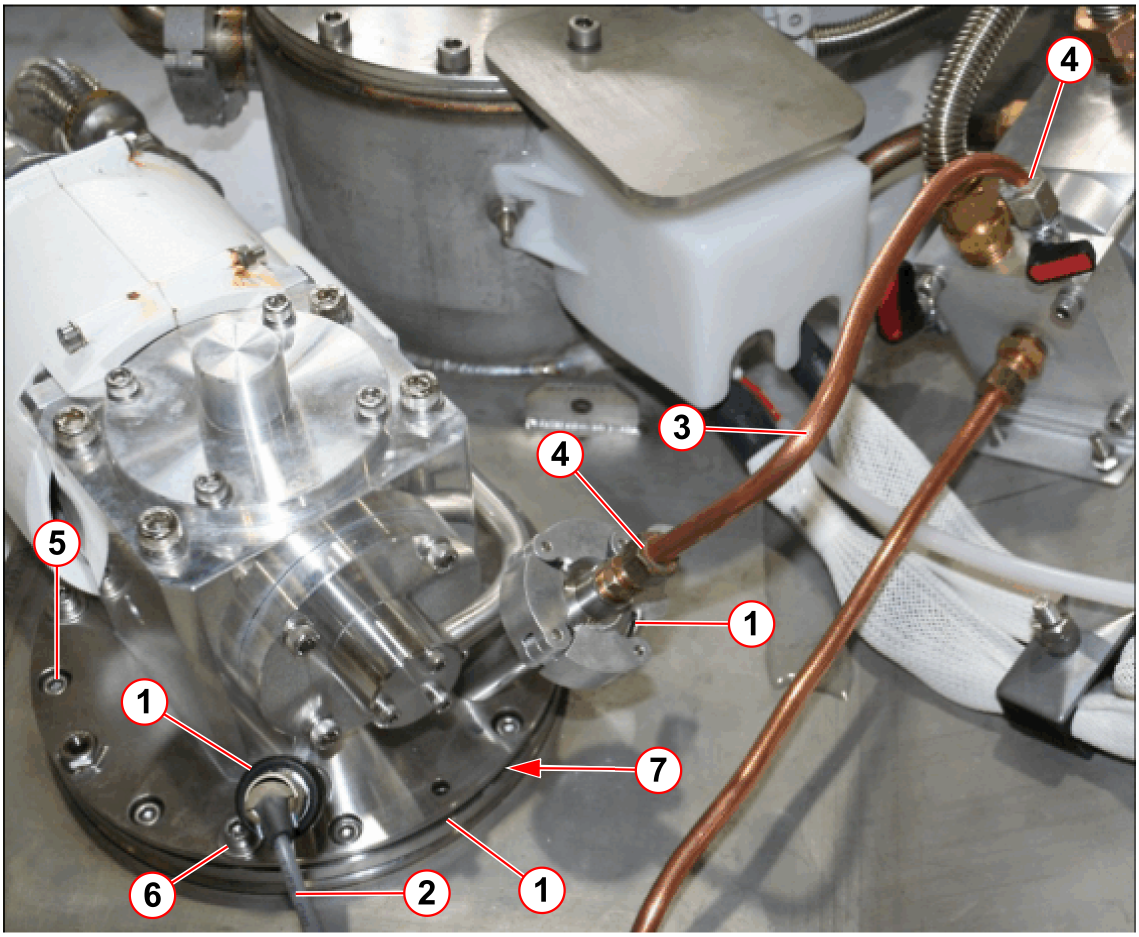
If leaks are found at **vent line connections**, first tighten the clamp, or securing screws and re-check. **DO NOT OVERTIGHTEN**.

If this does not cure the leak, replace the **metal seal**.

3.4.8.2 Cold head checks

The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

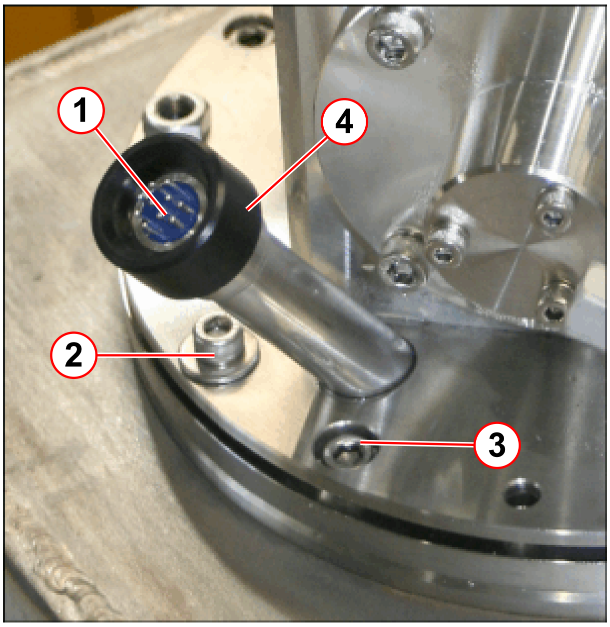
Fig. 72: Leak checking the cold head seals



(1) Cold head seals leak check points
(2) Temperature sensor cable
(3) Cold head vent line
(4) Cold head bypass line connection
(5) 8 off M6 x 16 mm O-ring clamp ring screws, torque to = 8Nm
(6) 4 off M6 x 25 mm cold head flange screws, torque to = 4Nm
(7) O-ring seal

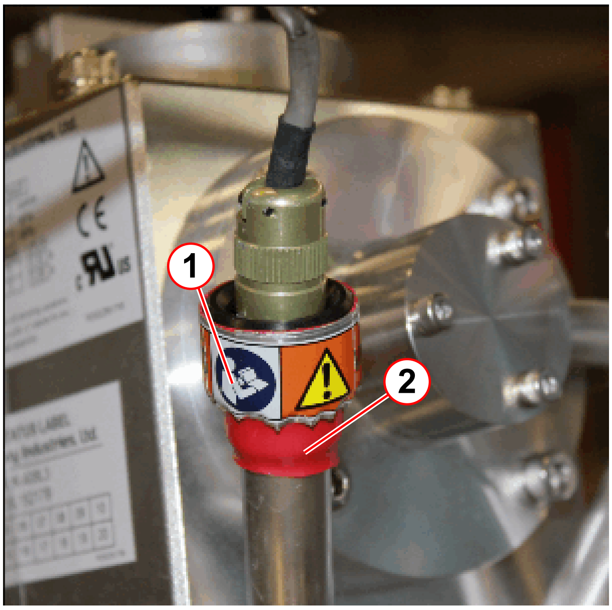
- The **cold head** is sealed in position by an **O-ring**. (→ 7/Fig. 72 Page 78)
- Check the **cold head vent line seals** at both ends. (→ 4/Fig. 72 Page 78)

Fig. 73: Cold head 10 pin seal



(1) Leak check point
(2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
(3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
(4) Black sealing cap

Fig. 74: Protective heat shrink & warning label



(1) Warning label
(2) Heat shrink

New cold heads are now supplied (from July 2015) with protective heat shrink and a warning label fitted around the temperature sensor connector.

3.4.8.3 Leak checking the temperature sensors

- **DO NOT** turn or try to remove the 'black' sealing cap (pos.4).
- Remove the temperature sensor cable with a straight pull.away from the black sealing cap.
- Leak check the 10 pin seal connector,
- Immediately refit the cable if no leak is found.

3.4.8.4 Temperature sensors

The cold head is fitted with **temperature sensors**:

- The **feed-through** for the wires is sealed with an **O-ring**.
- Check the **feed-through** and **cold head seals.**
- **Check** the **torque** of the cold head **screws**.

If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

3.4.8.5 Turret and venting

After any service procedures are performed, the venting must be warmed to room temperature, and dried.

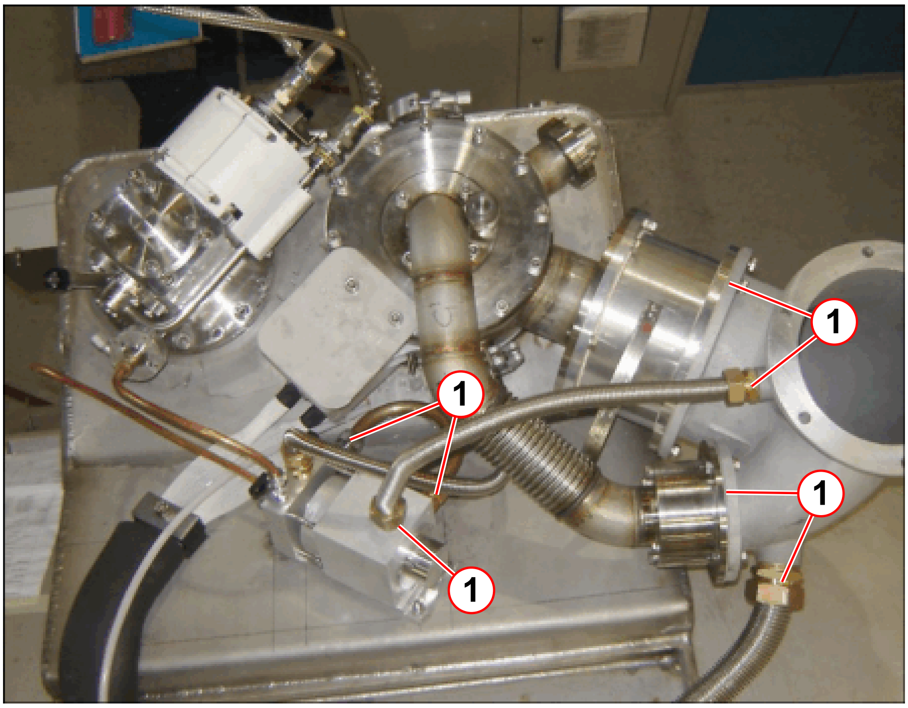
This must be completed BEFORE performing a leak check.

Low pressure leak checks

- Check for **cold gas escaping** around **low pressure joints** during:
  - depressurization.
  - helium filling the magnet.
- Check that all **clamps** on the venting **low pressure** side are **tight**.
  - **Replace gasket** seals if found damaged.

The figure below shows the **low** pressure **leak check positions.**

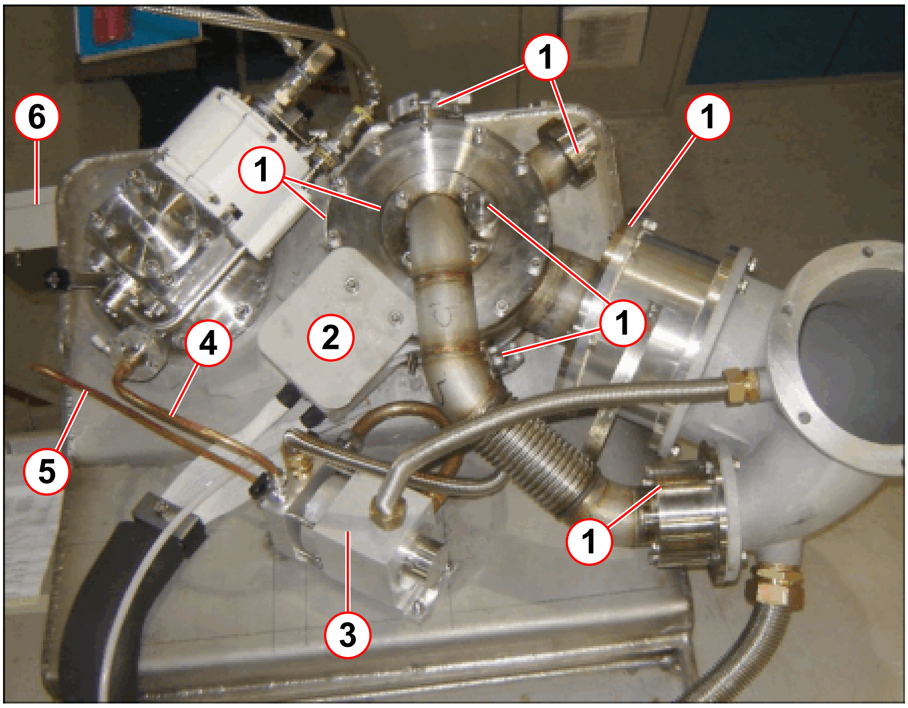
Fig. 75: Turret and venting leak check points - low pressure



(1) Low pressure joints

The figure below shows the **high** pressure **leak check positions**.

Fig. 76: Turret and venting leak check points - high pressure



(1) Turret top plate, syphon port and quench valve O-rings (high pressure)
(2) Positive FCL connector (high pressure)
(3) Venting valve assembly
(4) Cold head vent line
(5) Pressure gauge vent line
(6) Pressure gauge

High pressure leak checks

- **Check** that all **clamps** on the **high-pressure side** of the venting are tight. **Do not** try to **over-tighten** them - it could cause a leak, or make an existing leak worse.

> Replace a high load seal only with another identical high load seal. High load seals must only be used ONCE for a guaranteed leak tight seal.

- Check the **syphon port** and other seals on the **turret top plate**. (→ 1/Fig. 76 Page 81)
  - Use a **heat gun** to warm up and dry the **turret** and **syphon port**.
  - leak check the **syphon port**.
- After all **helium fills**:
  - Ensure the **internal** port surface is clean and free of ice.
  - Fit a new, undamaged **copper sealing washer**. **DO NOT** over- tighten the syphon plug.
  - Leak check the **syphon port**.
- **Positive FCL connector**. (→ 3/Fig. 76 Page 81)
- **Connections** to the **vent valve assembly**. (→ 3/Fig. 76 Page 81), (→ 1/Fig. 75 Page 80).

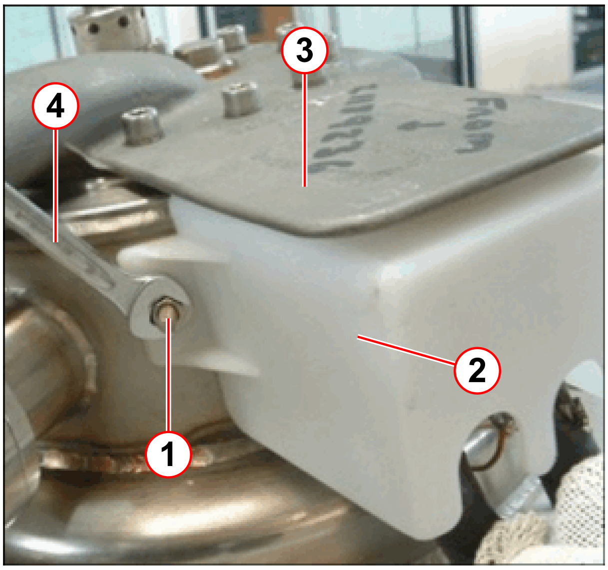
- **Cold head** line. (→ 4/Fig. 76 Page 81), see also (→ Cold head checks / Page 77)
- **Pressure gauge** line. (→ 5/Fig. 76 Page 81), see also (→ Fig. 80 Page 84)

Positive FCL connector

The FCL connector is now a replaceable component. The connector passes through the turret via a **ceramic interface** made leak tight by a high load seal. The FCL connector seal should only be checked if no other leaks have been found.

If a leak check is required, **DO NOT remove** the metal **cover plate**.

Fig. 77: Fitting/removing the FCL plastic cover

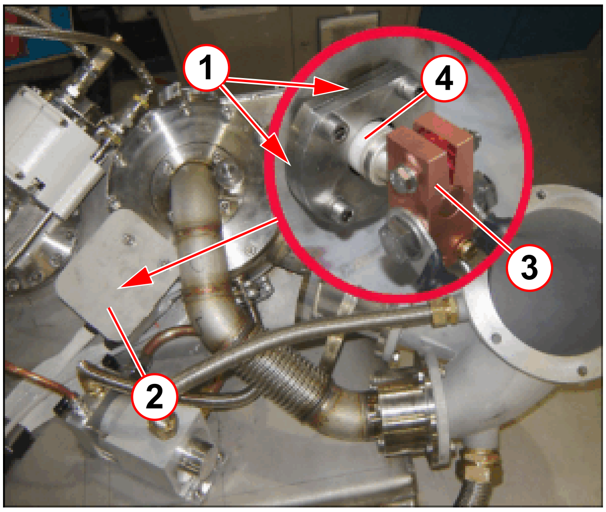


(1) M6 nuts and washers (2 off)
(2) FCL plastic cover
(3) Metal cover plate
(4) 10mm wrench/spanner (non-magnetic)

To access the **positive FCL connector**:

- **Remove** the plastic FCL cover.
  - The plastic cover is mounted on 2 **turret studs**.
  - Use a 10 mm spanner.
  - Remove the M5 nuts and washers (2 off).
  - Remove the **plastic cover**.

Fig. 78: FCL leak check points



(1) FCL leak check points (high pressure)
(2) Metal cover plate
(3) Positive bus bar

- **Leak check** all around the:
  - **metal seal**.
  - **ceramic seal**.
- **Refit** the plastic FCL cover. (→ 2/Fig. 77 Page 82)
  - Ensure that the FCL cover is **secure**
- If a leak is detected on the **ceramic seal**, contact **Magnet HSC immediately**.

CAUTION

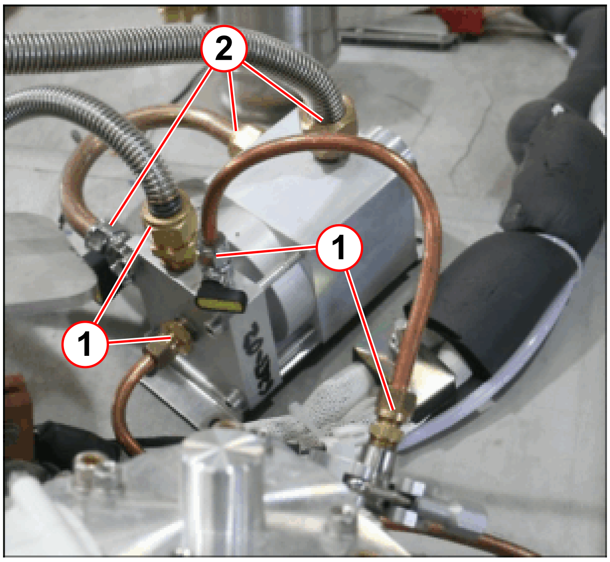
Fragile ceramic component!

Take care when working near to the FCL connector. The ceramic seal can fracture easily.

» **Do not overtighten, or make contact with the ceramic seal.**

Vent valve assembly

Fig. 79: Vent valve assembly leak check points

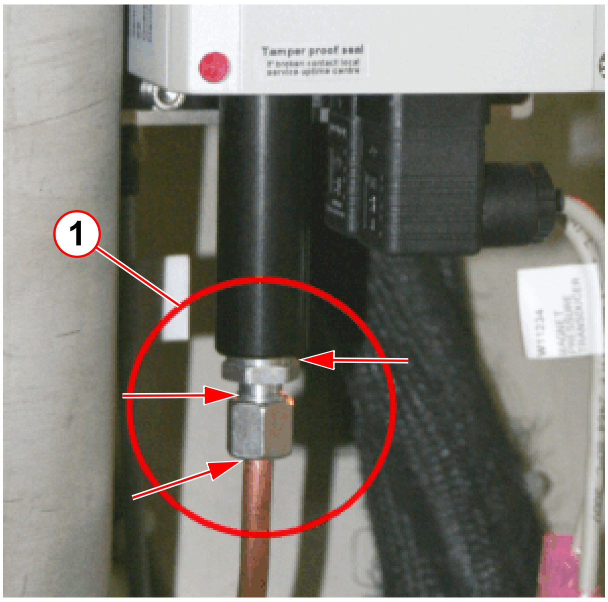


(1) High-pressure joints
(2) Low-pressure joints

- Leak check the vent valve **high pressure** assembly.
- For leaks found at pipe couplings, first try **tightening** the clamp or securing screws. **DO NOT OVERTIGHTEN!**
- If tightening does not solve the leak, the **complete** vent valve assembly must be replaced.

Pressure gauge

Fig. 80: Leak check the pressure line



(1) Leak check points

Leak check the **high pressure points**.

- For **leaks** found at pipe couplings, first try tightening the clamp or securing screws. **BEWARE OF OVER-TIGHTENING!**
- If tightening does not solve the leak, the **metal seal** must be **replaced**.
- Contact your local **Uptime Service Centre** if further leaks are suspected

3.4.9 Internal leak checks

The auxiliary and quench valve burst disks (including the quench valve O-ring) must be checked after a magnet quench.

The magnet must be at ZERO FIELD before leak checking inside the quench valve or auxiliary vent.

3.4.9.1 Internal leaks

- Auxiliary **1.38 bar** metal **bursting disk**. (→ Auxiliary vent 1.38 bar metal bursting disk / Page 85)
- **Quench valve**. (→ Quench valve and vent valve assembly / Page 86)
- Vent **valve assembly.** (→ Quench valve and vent valve assembly / Page 86)

CAUTION

Internal leak checks.

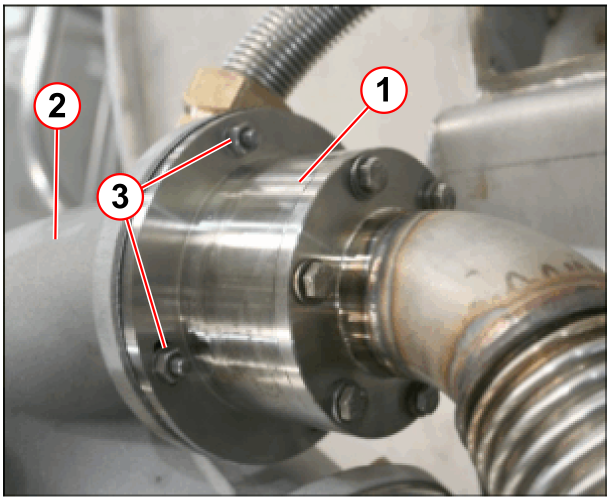
Risk of air ingress causing internal icing of turret.

» **Replace burst disk or valve immediately if found to be broken or leaking.**

3.4.9.2 Auxiliary vent 1.38 bar metal bursting disk

To check for any leak path through the **auxiliary bursting disk**:

Fig. 81: Auxiliary burst disk assembly

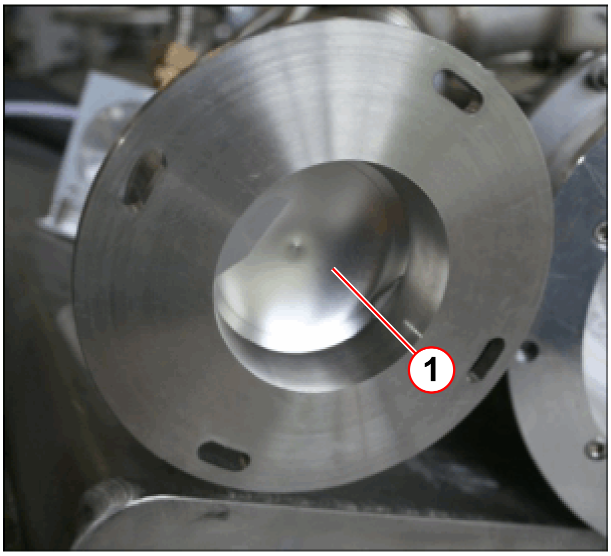


(1) Auxiliary burst disk
(2) Elbow
(3) M6 screws, washers and nuts (4 off)

- **Disconnect** the auxilliary burst disk from the **elbow,** on the low pressure side.
  - **Remove** the **M6** screws, washers and nuts (4 off).
- **Remove** the vent pipe from the bypass assembly at the **quench elbow**.
- **Immediately** place a ball of paper towelling into the **open port**.

The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 82: Auxiliary burst disk



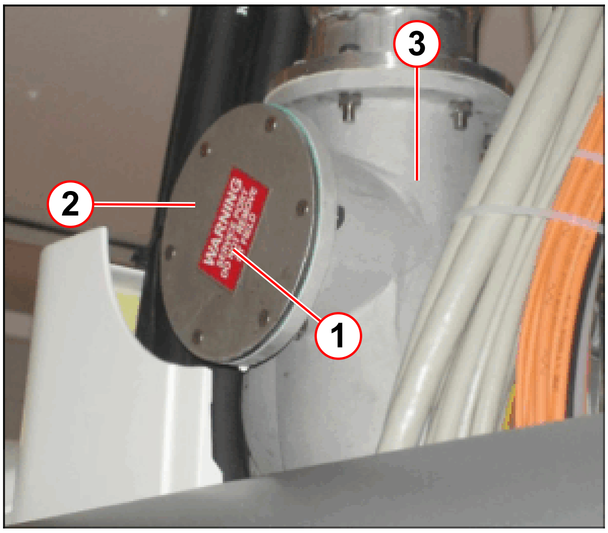
(1) Burst disk

- Visually inspect the **auxiliary burst disk membrane** to ensure that it is not damaged.

- Leak check the **membrane**.
- Replace the auxiliary metal burst disk if a leak is detected.

3.4.9.3 Quench valve and vent valve assembly

Fig. 83: Horizontal service port



(1) Warning label
(2) Port (blanked)
(3) Quench elbow

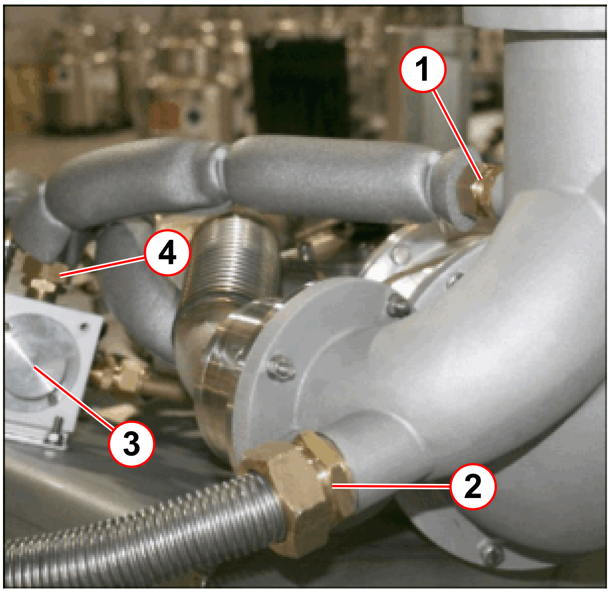
OR98 only - Horizontal service port

- **This port cover must NOT be removed.**
- **Do not** use as an inspection cover to check the quench valve burst disk.
- This port allows horizontal fitment of the quench pipe.
- The port is **blanked** during normal vertical quench line configuration.
- If used as a quench outlet, the **vertical port must be blanked.**

If the magnet is **not** at **normal operating pressure**,(**15.5 psi (A)**), it is possible that the **quench valve** or **vent valve assembly** has a leak.

- Check for an **internal** leak through the **quench valve.**

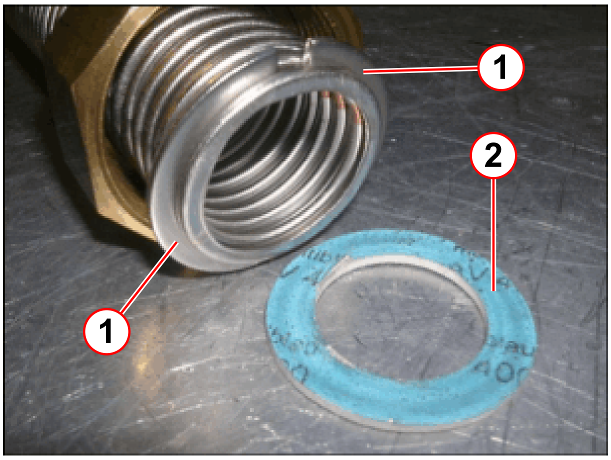
Fig. 84: Vent line connections



(1) Elbow outlet line
(2) OVC outlet line
(3) Vent valve assembly
(4) Vent valve outlet line

- **Disconnect** the **vent valve outlet line** (pos 4).

Fig. 85: Vent valve connection

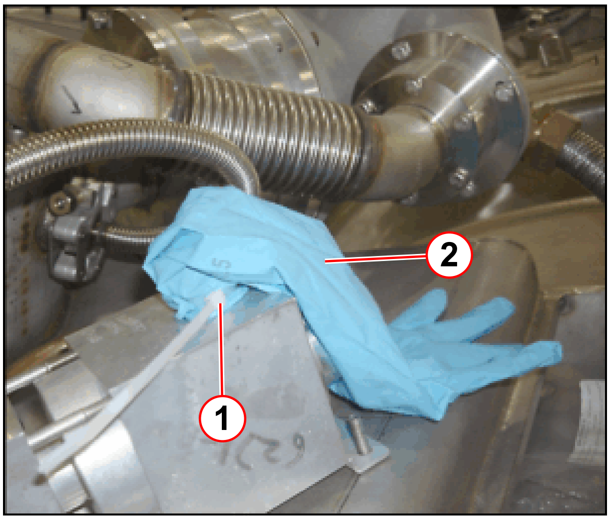


(1) Split rings
(2) Gasket

Take care not to lose the **split rings** during this process!

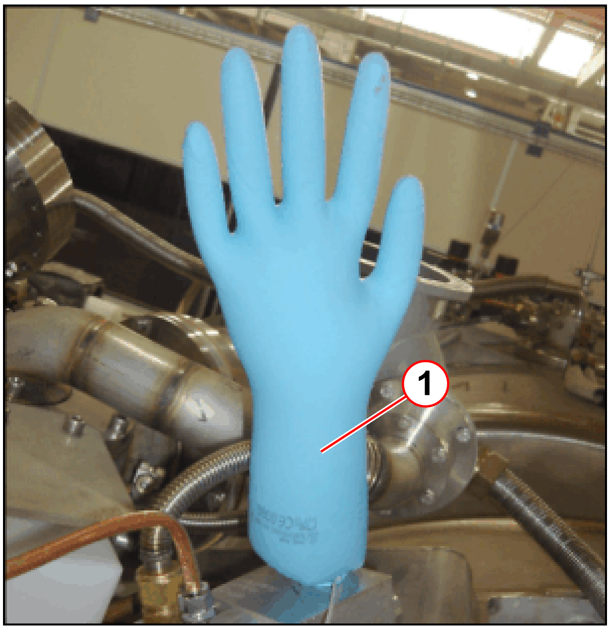
- Ensure the **split rings** do not fall out of position.
- **Be careful** to retain the gasket when disconnecting the vent line.
  - The **gasket** is not retained by the split rings and nut.

Fig. 86: Glove fitment



(1) Surgical glove
(2) Cable tie

Fig. 87: Gross leak through the vent valve assembly



(1) Glove inflated

Leak check the vent valve assembly

- Fit a surgical glove (or small airtight plastic bag) over the end of the open vent line.
- Seal the glove with a tie wrap.

  » If the glove expands rapidly with the magnet pressure **lower** than **15.6 psi (A)** (i.e.0.3 psi (G)), this indicates a **gross leak**.

  » The valve is not sealing and the **complete assembly will need to be replaced.**

- Leak check the **quench valve**: (➔ 1/Fig. 84 Page 86)
  - Check the room **background** helium level is **less than** 10^{-5}.

- If the background level is **above** 10^{-5}, **wait one hour** with the port **open**. This will allow the helium gas to disperse.
- Blowing cold air (ie. from a heat gun set to cold), will accelerate the process.
- There should be **no helium gas** inside the **quench valve**.
- When the background level is **less than** 10^{-5}, **insert the probe** of the leak checker **inside the quench valve port**.
- If a leak is detected, the **burst disk** and **quench valve O-ring** must be re-inspected. The **quench valve** will need to be **disassembled**.

- **Re-connect** the flexible **vent line**, ensuring the split rings and gasket are correctly fitted.

3.4.10 Final work steps

- At the end of leak checking, ensure again that the leak detector is working correctly by re-checking its function with a known helium gas source.
- Ensure that all parts of the venting system have been re-fitted correctly.

4.1 OR98/99/103 Ramping

Only Siemens CSEs trained in ramping magnets are allowed to perform the magnet ramping procedure.

Before ramping check that the quench line is correctly installed from the magnet to the outlet.

WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- **Ramping is not allowed if the visually inspection of the quench line was not performed successfully.**
- **Before initially ramping up the magnet, the 'Testing the ERDU circuit' has to be performed.**
- **Procedure described in the safety related test can be used if ERDU test load is available.**

WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- **Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.**
- **Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.**
- **The sign must be posted in a way that it is visible even when the door to the controlled area is opened.**

CAUTION

Risk of asphyxiation and hypothermia, if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness in personnel.

- **Quench valve must be connected to the Quench pipe.**

Initial installation ramp helium level must be:

- ≥ 60 % for the OR98
- ≥ 65 % for the OR99 and the OR103.

Ensure a minimum of:

- OR99/OR103 = 2000 litres of helium on standby.

During normal Service ramping the minimum helium level for ramping up must be:

- > 40 % for the OR98 and the OR103
- > 60 % for the OR99.

The full functionality of the MRC (MRAWP) Host computer is not available during ramping procedure.

Do not re-boot the system during ramping.

Wait at least one hour after completing a previous ramp before starting the next ramp, (including lead checks and aborted ramps).

Wait at least two hours after helium filling before ramping the magnet.

Check that the venting bypass valve is in the CLOSED position.

Always

- Always restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- Always inspect all equipment and tools for defects prior to use.
- Always inspect the Anderson ramp lead connections at the filter plate.
- Clean the area of the magnet room, dispose off empty cartons or other packing materials
- Always inspect magnet room for the presence of non identified objects. Before ramping, the nature of those objects must be clarified.

Prepare a minimum of 4 pictures of the room **prior to first ramping.**

These pictures must be colored and should have 2 to 4 megapixel resolution. They must show the complete room, all 4 corners and walls. In case there are un-removable cupboards installed, their doors must be open to prove that they are empty.

Later those pictures are to be attached to the installation report.

Never

- Never use defective or damaged equipment.
- Never allow unauthorized people to stand too close to the magnet during ramping.
- Never block emergency exits.
- Never smoke or use open flames in the magnet room.
- Never ramp a magnet when you are unsure about magnetic parts inside the magnet room. Those parts could be hidden in e.g. supposed empty boxes or in closed cupboards or sliding of a desk.

4.1.1 Tools required

WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

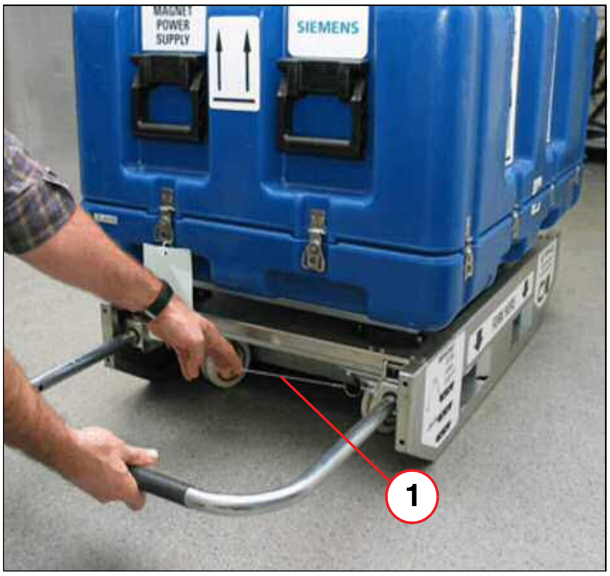
» **Only non-magnetic equipment may be taken into the magnet room.**

- Magnet Power Supply Unit (MPSU 3600); (order number: **8395613**). This includes:-
  - 3600 MPSU Power lead.
  - CAN Open lead.
  - RS232 lead.
- Heat gun.
- MRC (MRAWP) Host or Service PC.

4.1.2 Preparation

- Move the 3600 MPSU into the equipment room close to the RF filter plate.

Fig. 88: MPSU



(1) Wire locking strap

- Lower the trolley handle into the horizontal position. This will raise the wheels and allow the trolley to rest on the two skids.
- Pull the wire strap to unlock the handle, and slide the handle into the recess under the trolley.
- Remove the shipping case lid by un-clipping the 12 spring clips.

Fig. 89: MPSU

(1) MPSU LID

- Open the hinged lid of the 3600 MPSU by pulling the two silver knobs and turning them through a 1/4 turn.
- Raise the lid.

4.1.3 Installation

Fig. 90: 3600 MPSU Power connections

(1) X1 (MPSU power)
(2) CB1 (Mains circuit breaker)

- Ensure the mains circuit breaker **CB1** is in the **OFF** position on the 3600 MPSU.

Fig. 91: Adapter plate (view from Equipment Room)

- Unscrew the securing screws and remove the **adapter plate** from the **RF filter plate.**
- This will allow access to the magnet ramp leads in the **magnet** room.

> Do NOT connect the 3600 MPSU ramp leads to the MPSU until AFTER the Transport test has been completed. See (→ Transport Test Procedure / Page 113).

> The mandatory transport test was introduced from 3600 MPSU software revision level 17.
>
> This test is only required the FIRST TIME time the 3600 MPSU is used on site.

Tab. 8 MPSU lead connections

Lead	From	To	Length
Magnet Ramp leads	MPSU X3 MPSU X4	Filter plate (grey) Filter plate (blue)	5m
CAN OPEN lead	MPSU X7	EPC X125	5m
RS232 lead	MPSU X6	MRC (MRAWP) Host, or Laptop **W10220 / X12 / MPSU** Cable W10220 to **X128**, disconnect cable at **X12** on top of the EPC.	5m
MPSU Power lead	MPSU X1	EPC **X130**	10m

Fig. 92: OR98, OR99 and OR103 Ramping configuration

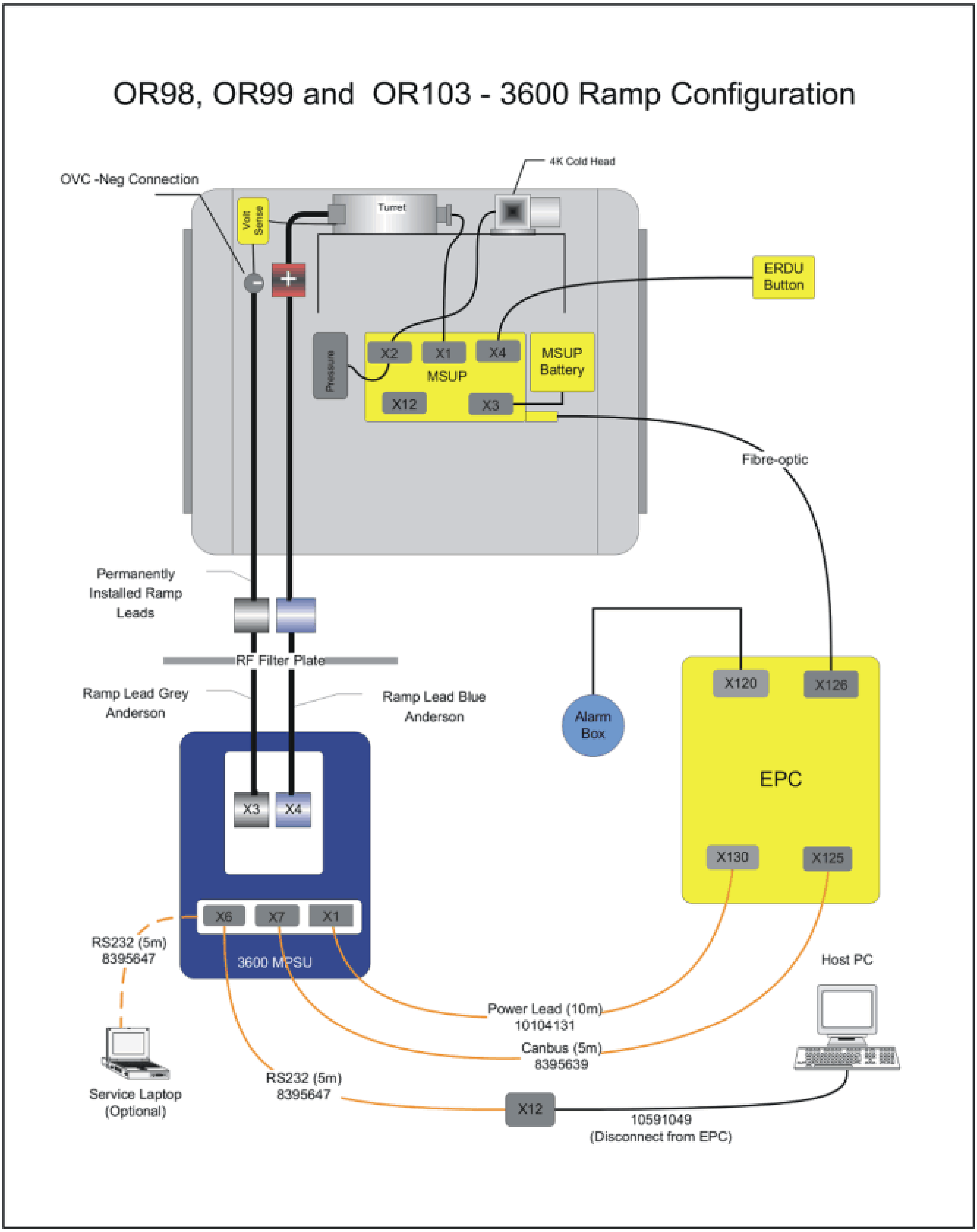
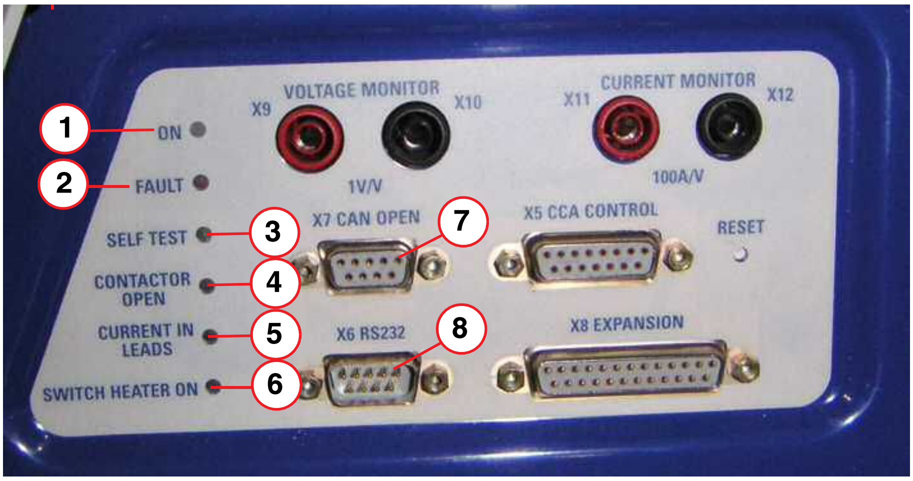


Fig. 93: 3600 MPSU Connector and indicator panel

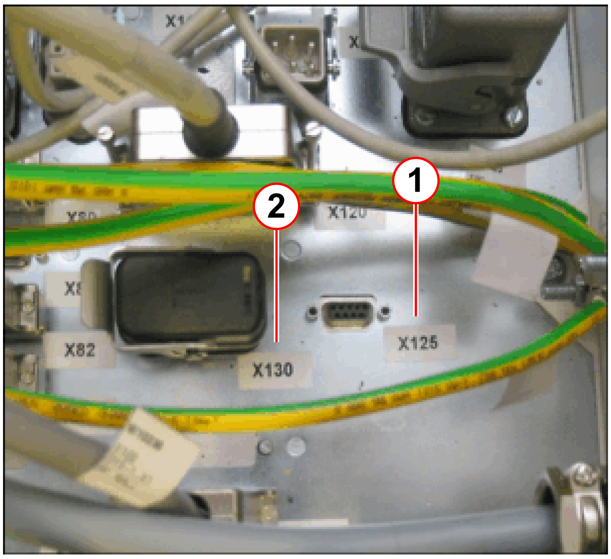


(1) ON indicator
(2) FAULT indicator
(3) SELF TEST indicator
(4) CONTACTOR OPEN indicator
(5) CURRENT IN LEADS indicator
(6) SWITCH HEATER ON indicator
(7) X7 CAN OPEN connector
(8) X6 RS232 connector

Fig. 94: 3600 MPSU leads

(1) CAN OPEN lead
(2) RS232 lead
(3) MPSU power lead

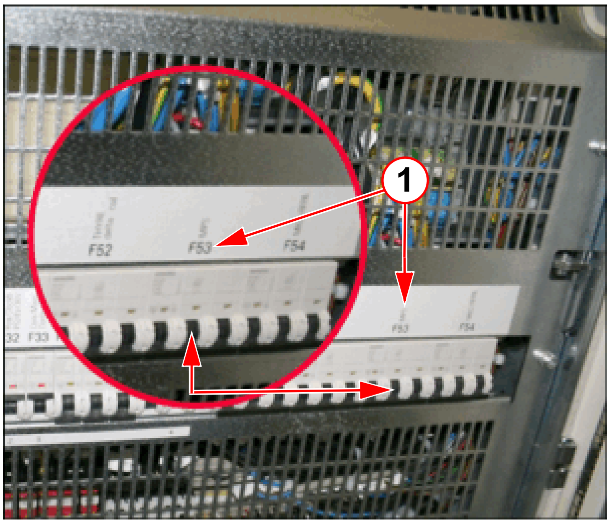
Fig. 95: EPC connections



(1) X125
(2) X130

- Connect the **CAN OPEN**, **RS232** and power leads. (➔ Fig. 94 Page 95)
- Ensure the system is switched **ON** on the **alarm box**.

Fig. 96: MPSU Circuit breaker F53



(1) Circuit breaker F53

- Turn ON **F53** in the **EPC.**

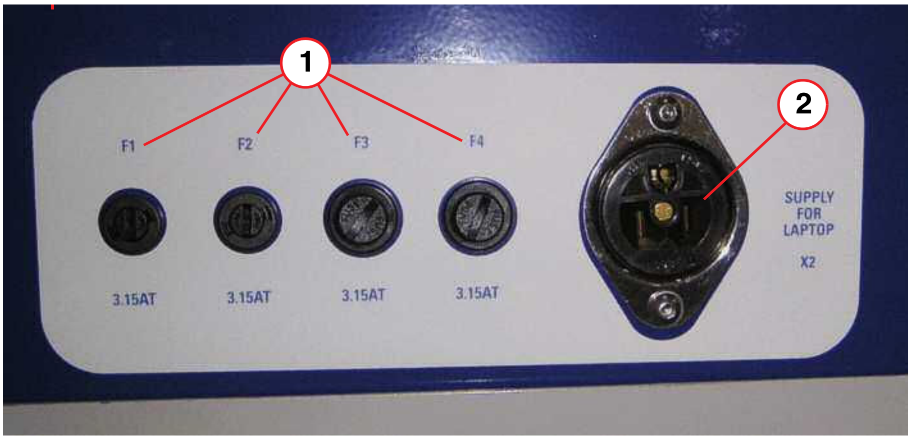
4.1.4 Set up of the MPSU

> To start ramping, the **MRC Host** or **Service laptop** must be connected to **X6** of the 3600 MPSU (RS232 interface) (→ 8/Fig. 93 Page 95).

4.1.4.1 Service Laptop connection

Power to a Service laptop is supplied at **X2** on the 3600 Power Supply.

Fig. 97: 3600 MPSU Fuses and laptop power supply



(1) Fuses F1 - F4
(2) X2 Supply for Laptop computer

Establish communication using a**Terminal emulation software** (e.g. PuTTY) (→ Using Putty / Page 98)

4.1.4.2 MRC Host connection

> The following work step is only necessary with SW N4_VD11A.
>
> With SW N4_VD11D, the ttyrecorder stops automatically before the terminal emulator starts, and re-starts it when the terminal emulator is closed.

- Hold down CTRL/ESC together
  » select **programs** - **advanced user** - **OK**
- Type in password
  » meduser1
- Select OK
- Select Start - Run
  » The menu appears
  » Type in CMD
- Select OK

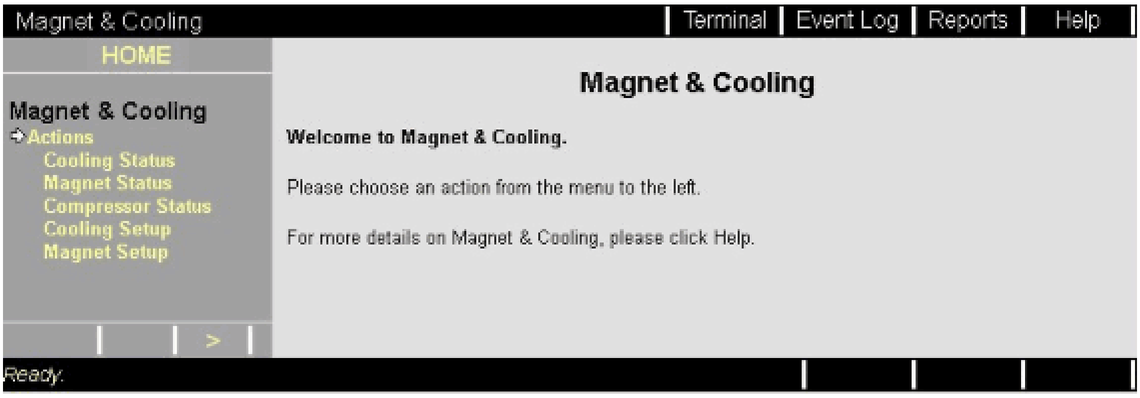
Before running **Terminal emulation software**, the Host process **stop_ttyrecorder.bat** must be run. This will release the serial port to enable the host to communicate with the MPSU. This process is automatically re-started after a reboot.

- In the DOS window type **stop_ttyrecorder.bat** <enter>

4.1.4.3 Configuring the Host

- Open the **Magnet & Cooling** service platform.
- Select **Terminal** from the **toolbar.**

Fig. 98: Open the Terminal via Magnet&Cooling



- **Terminal** option: the program will **open**. No configuration is required.
- **Terminal** emulator option: the program will open, configuration is required

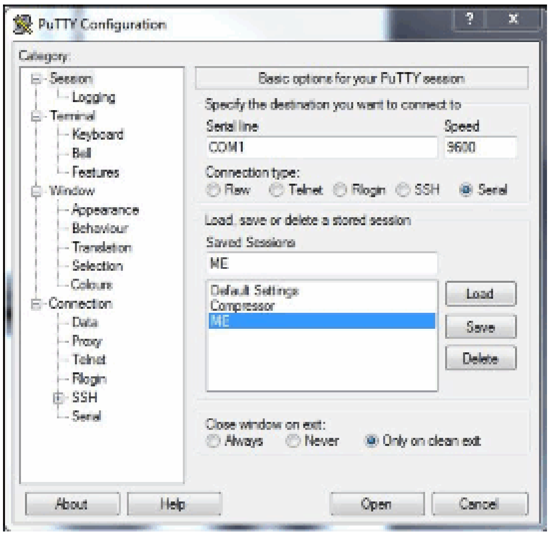
4.1.4.4 Using Putty

Setting up PuTTY

You can use software called PuTTY to connect to MSUP’s (Magnet Supervisorys) and MPSU’s (Magnet Power Supplies). This software has to be setup with the following settings in order to work.

Ensure you have PuTTY version 0.60 or newer software installed on your PC.

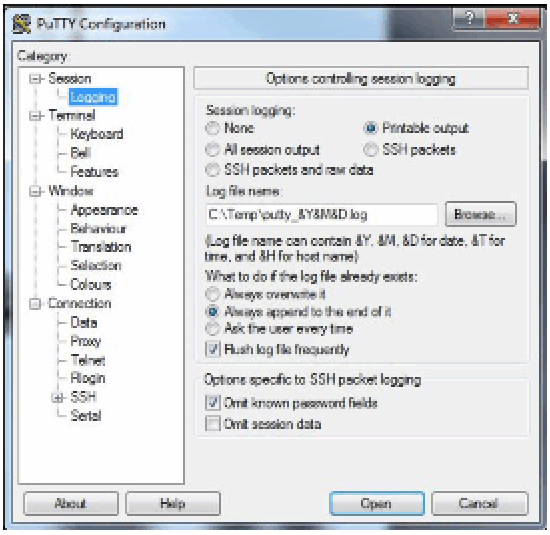
Fig. 99: PuTTY Configuration - Session



- Under Session tab.
- Set the ‘Connection type:’

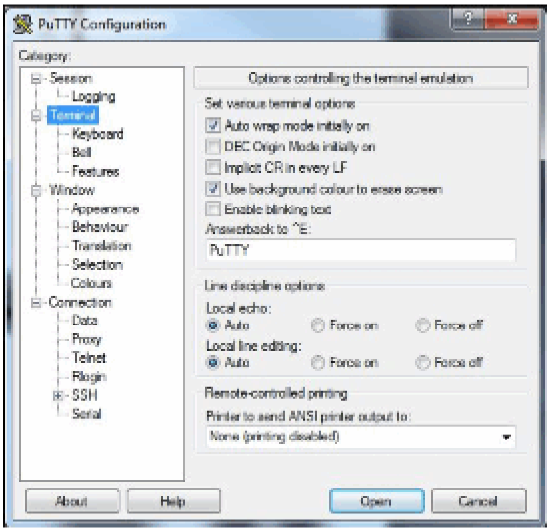
  to Serial.
- Set the ‘Serial line’ to the correct COM port being used. In this case COM1. Only state this if the com port that is being used. Change as required.
- Set the ‘Speed’ to 9600.
- Set the ‘Close window on exit’ to Only on clean exit.

Fig. 100: PuTTY Configuration - Logging



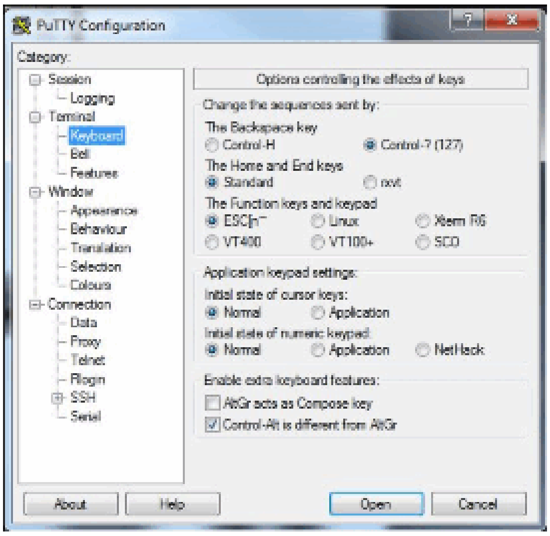
- Under Logging tab.
- Set the ‘Session logging’ to Printable output.
- Set the ‘Log file name:’ to C:\Temp \putty_&Y&M&D.log.
- Set the ‘What to do if the log file already exists:’ to Always append to the end of it and tick Flush log file frequently.
- Tick the ‘Options specific to SSH packet logging’ to Omit known password fields.

Fig. 101: PuTTY Configuration - Terminal



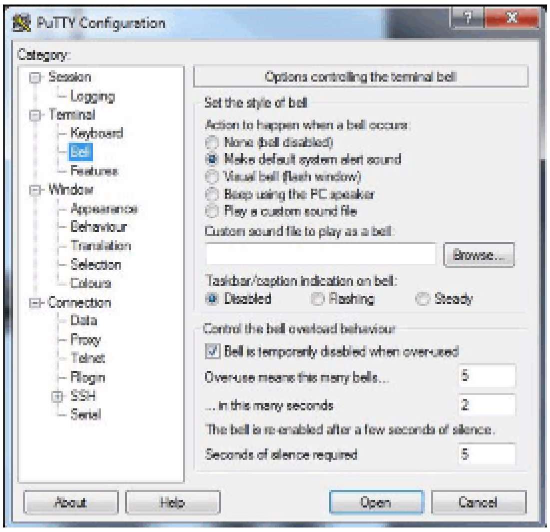
- Under Terminal tab.
- Under 'Set various terminal options' tick the following 'Auto wrap mode initially on' and 'Use background colour to erase screen'.
- Under the 'Answerback to ^E:' ensure it states 'PuTTY'.
- Under 'Local echo:' and 'Local line editing:' ensure 'Auto' is selected
- Under 'Printer to send ANSI printer output to:' select 'None (printing disabled).

Fig. 102: PuTTY Configuration - Keyboard



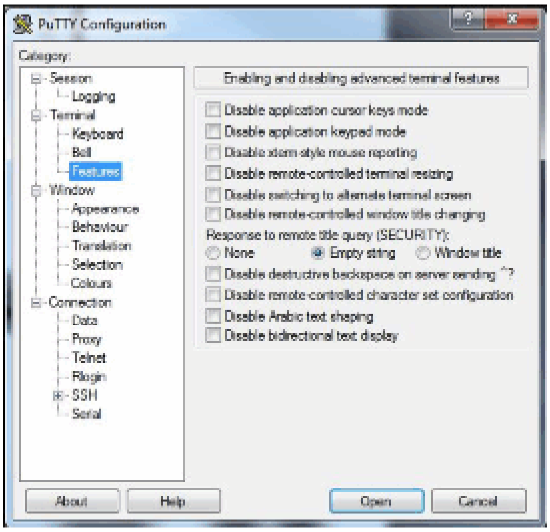
- Under the 'Keyboard' tab.
- Set 'The Backspace key' to 'Control-? (127)'.
- Set 'The Home and End keys' to 'Standard'.
- Set 'The Function keys and keyboard' to 'ESC{n~'.
- Set 'Initial state of cursor keys:' and 'Initial state of numeric keypad:' to 'Normal'.
- Tick 'Enable extra keyboard features' to 'Control-Alt is different is different from AltGr'.

Fig. 103: PuTTY Configuration - Bell



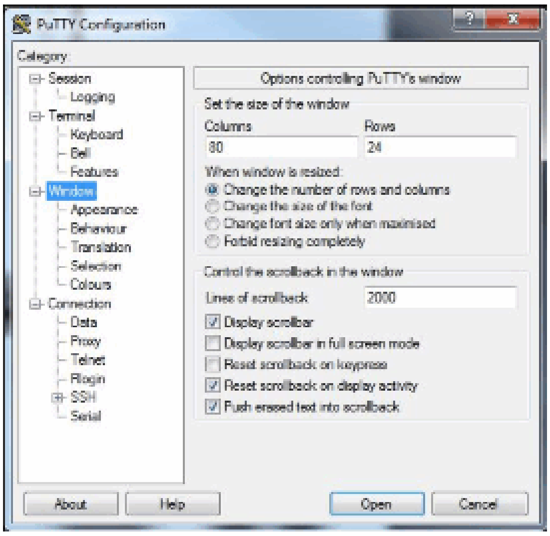
- Under the ‘Bell’ tab.
- Set ‘Action to happen when a bell occurs’ to ‘Make default system alert sound’.
- Set ‘Taskbar/caption indication on bell:’ to 'Disabled’,
- Tick ‘Bell is temporarily disabled when over-used’.
- Set ‘Over-use means this many bells...’ to ‘5’.
- Set ‘...in this many seconds’ to ‘2’.
- Set ‘Seconds of silence required’ to ‘5’.

Fig. 104: PuTTY Configuration - Features



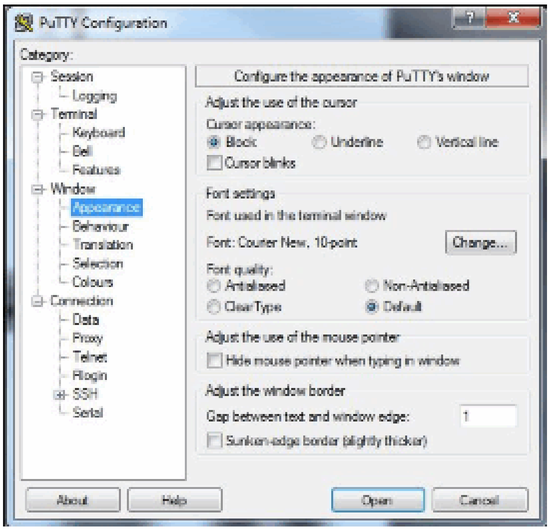
- Under the ‘Features’ Tab.
- Set ‘Response to remote title query (SECURITY):’ to 'Empty string’.
- Ensure nothing else is selected.

Fig. 105: PuTTY Configuration - Window



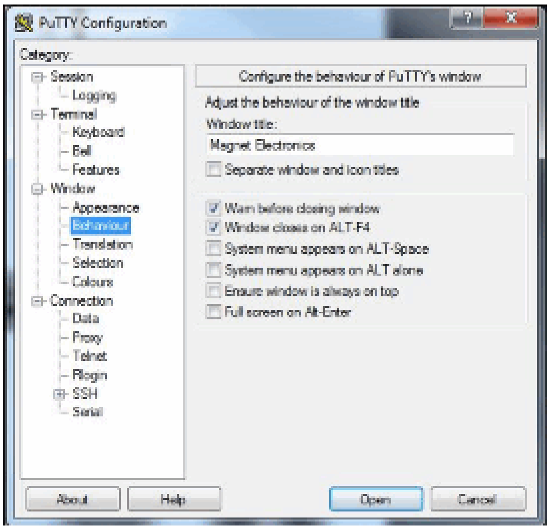
- Under the 'Window' Tab.
- Set the 'Columns' to '80' and the 'Rows' to '24'.
- Set 'When window is resized:' to 'Change the number of rows and columns'.
- Set the 'Lines of scrollback' to '2000'.
- Ensure 'Display scrollbar', 'Reset scrollback on display activity' and 'Push erased text into scrollback' are all ticked.

Fig. 106: PuTTY Configuration - Appearance



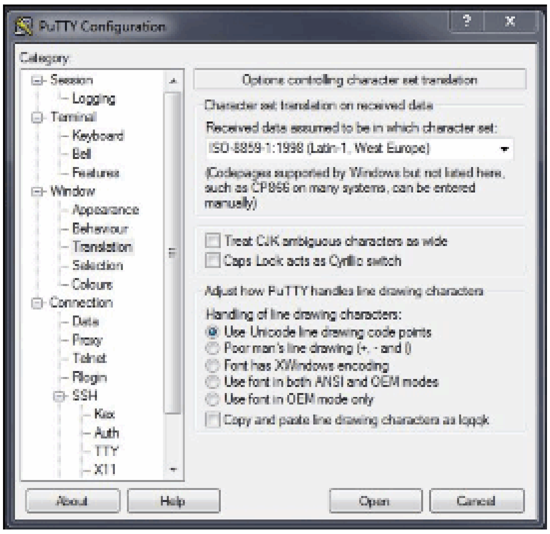
- Under 'Appearance' tab.
- Set 'Cursor appearance' to 'Block'.
- Set 'Font quality:' to 'Default'.
- Set 'Gap between text and window edge:' to '1'.

Fig. 107: PuTTY Configuration - Behaviour



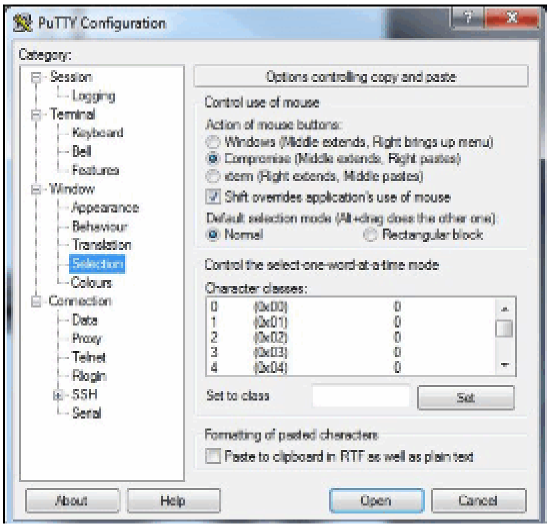
- Under 'Behavior' tab.
- Under 'Window title:' type 'Magnet Electronics'
- Ensure 'Warn before closing window', and 'Window closes on ALT-F4' are ticked.

Fig. 108: PuTTY Configuration - Translation



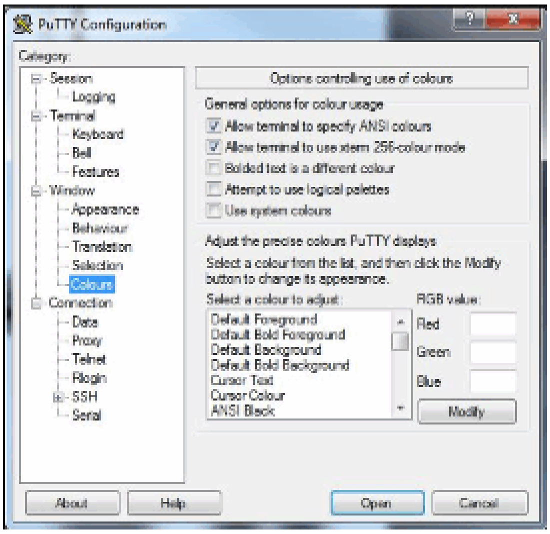
- Under 'Translation' tab.
- Set 'Received data assumed to be in which character set:' to 'ISO-8859-1:1998 (Latin-1, West Europe)'.
- Set 'Handing of line drawing characters:' to 'Use Unicode line drawing code points'.

Fig. 109: PuTTY Configuration - Selection



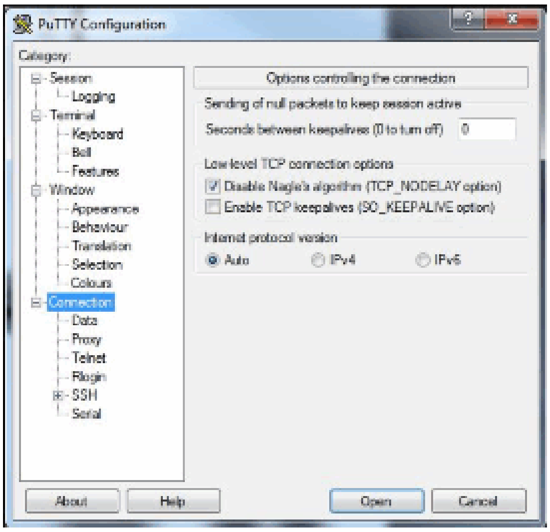
- Under 'Selection' tab.
- Set 'Action of mouse buttons:' to 'Compromise (Middle extends, Right pastes)'.
- Tick 'Shift overrides application's use of mouse'.
- Set 'Default selection mode (Alt + drag does the other one):' to 'Normal'.

Fig. 110: PuTTY Configuration - Colours



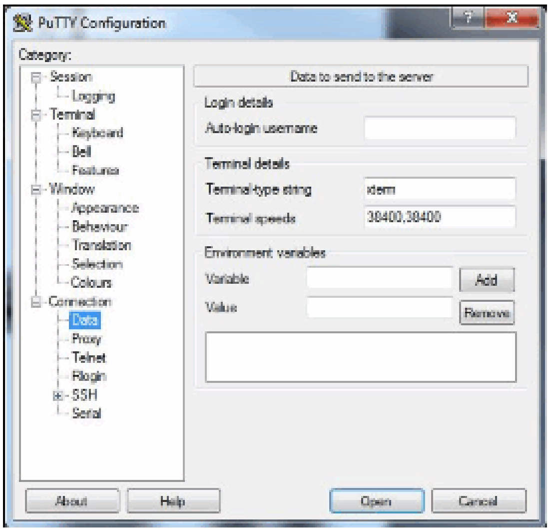
- Under 'Colours' tab.
- Tick 'Allow terminal to specify ANSI colours' and 'Allow terminal to use xterm 256-colour mode'.

Fig. 111: PuTTY Configuration - Connection



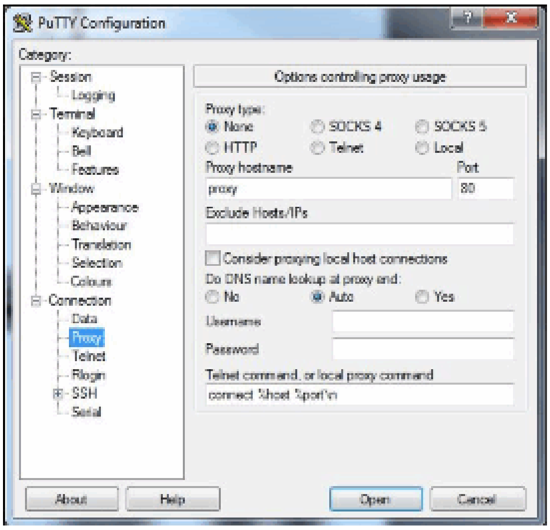
- Under ‘Connection’ tab.
- Set ‘Low-level TCP connection options’ to ‘Disable Nagle's algorithm (TCP_NODELAY option)’.
- Set ‘Internet protocol version’ to ‘Auto’.

Fig. 112: PuTTY Configuration - Data



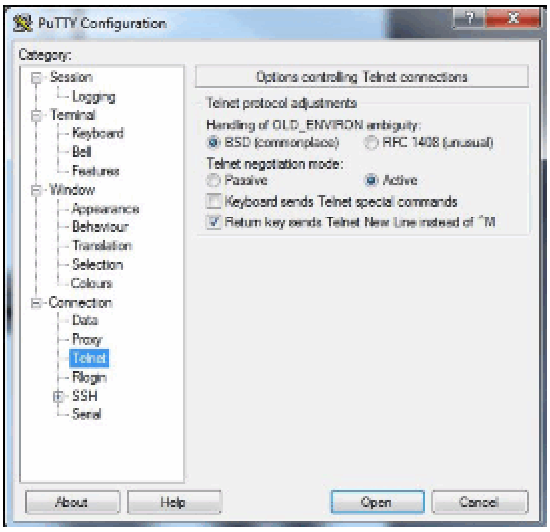
- Under ‘Data’ tab.
- Under the ‘Terminal-type string’ ensure it states ‘xterm’.
- Under the ‘Terminal speeds’ ensure it states ‘38400.38400’.

Fig. 113: PuTTY Configuration - Proxy



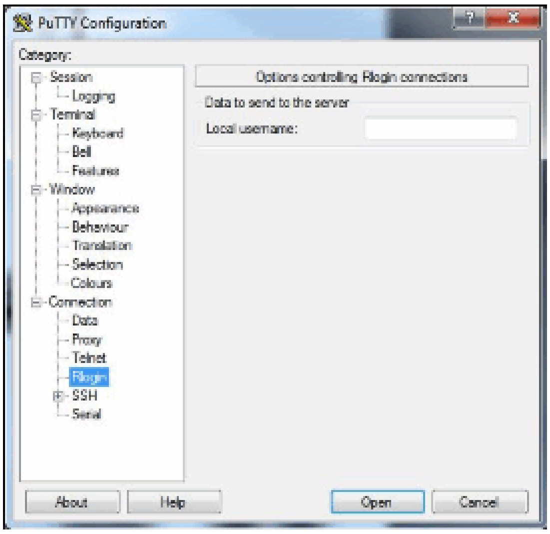
- Under ‘Proxy’ tab.
- Set ‘Proxy type:’ to ‘None’.
- Under the ‘Proxy hostname’ ensure it states ‘proxy’.
- Under the ‘Port’ ensure it states ‘80’.
- Set ‘Do DNS name lookup at proxy end:’ to ‘Auto’.
- Under the ‘Telnet command, or local proxy command’ ensure it states ‘connect %host %port\n’.

Fig. 114: PuTTY Configuration - Telnet



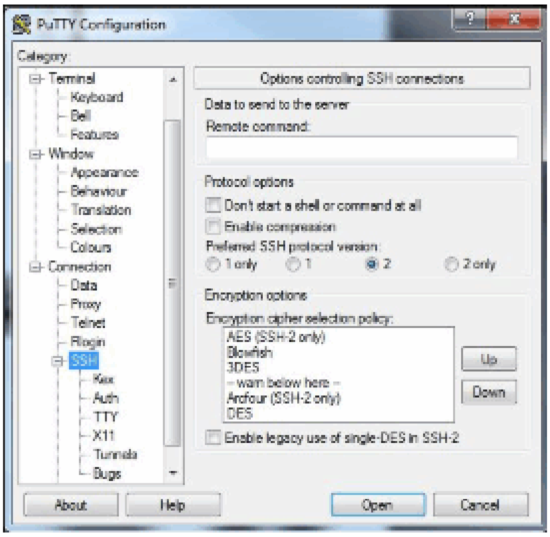
- Under ‘Telnet’ tab.
- Set ‘Handling of OLD_ENVIRON ambiguity:’ to ‘BSD’.
- Set ‘Telnet negotiation mode:’ to ‘Active’.
- Tick ‘Return key sends Telnet New Line instead of ^M’.

Fig. 115: PuTTY Configuration - Rlogin



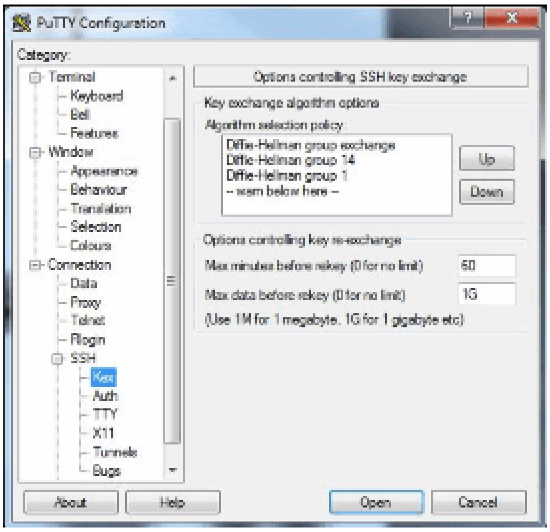
- Under 'Rlogin' tab.
- Nothing is selected.

Fig. 116: PuTTY Configuration - SSH



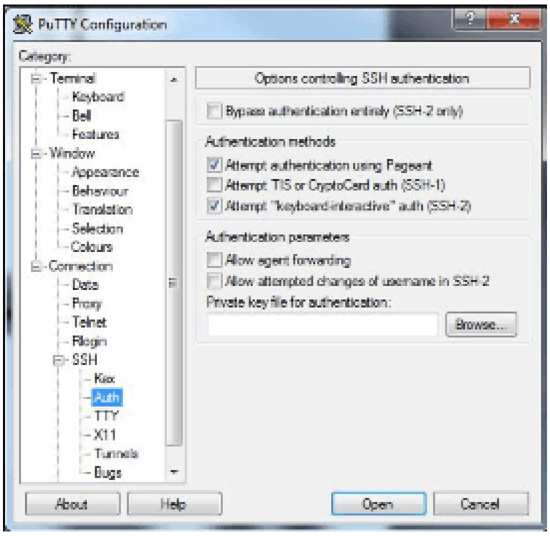
- Under 'SSH' tab.
- Set 'preferred SSH protocol version' to '2'.

Fig. 117: PuTTY Configuration - Kex



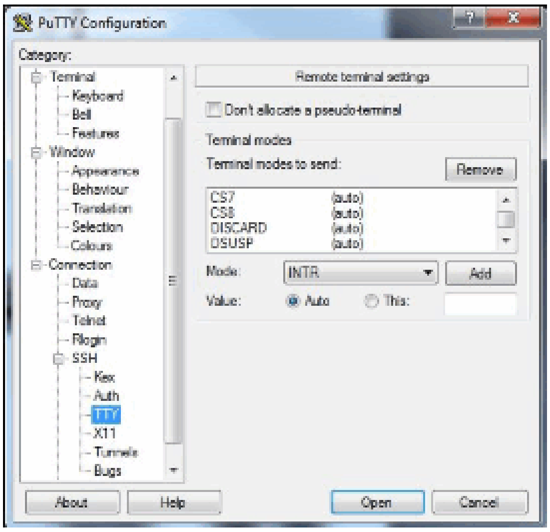
- Under 'Kex' tab.
- Under the 'Max minutes before re-key (0 for no limit)' ensure it states '60'.
- Under the 'Max data before rekey (0 for no limit)' ensure it states '1G'.

Fig. 118: PuTTY Configuration



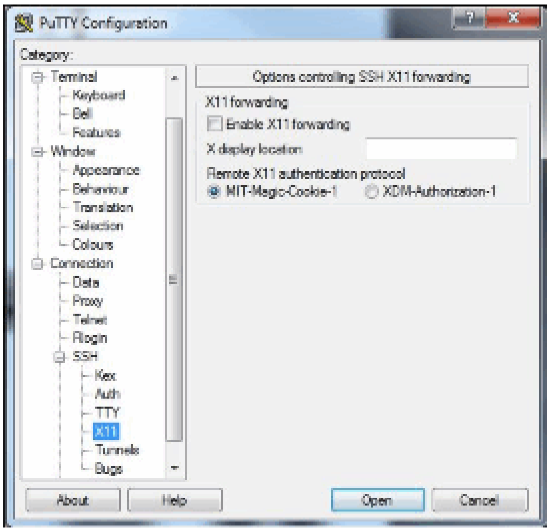
- Under 'Auth' tab.
- Tick 'Attempt authentication using Pageant' and 'Attempt "keyboard-interactive" auth (SSH-2).

Fig. 119: PuTTY Configuration - TTY



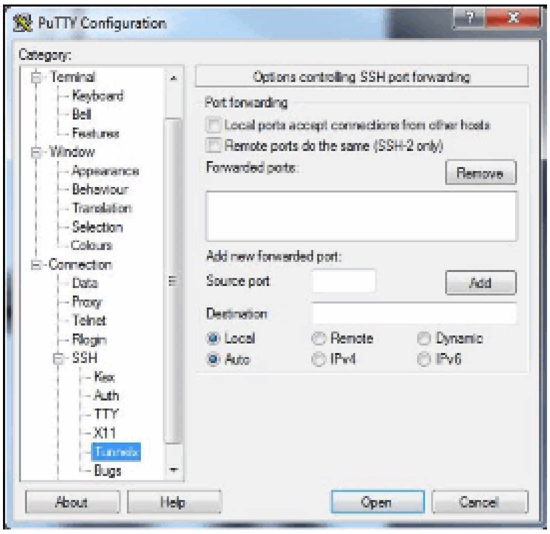
- Under 'TTY' tab.
- Set 'Value' to 'Auto'.

Fig. 120: PuTTY Configuration



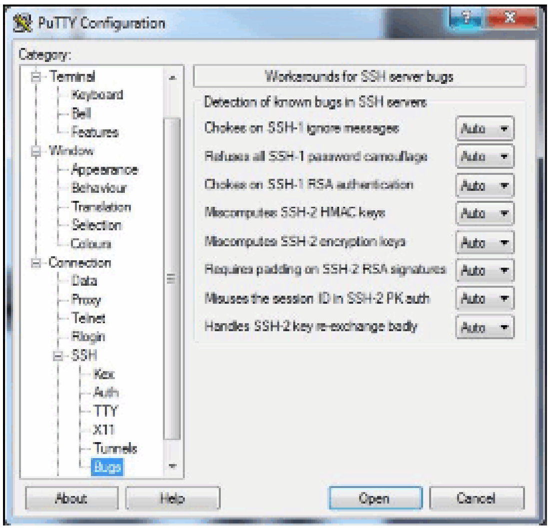
- Under 'X11' tab.
- Set 'Remote X11 authentication protocol' to 'MIT-Magic-Cookie-1'.

Fig. 121: PuTTY Configuration - Tunnels



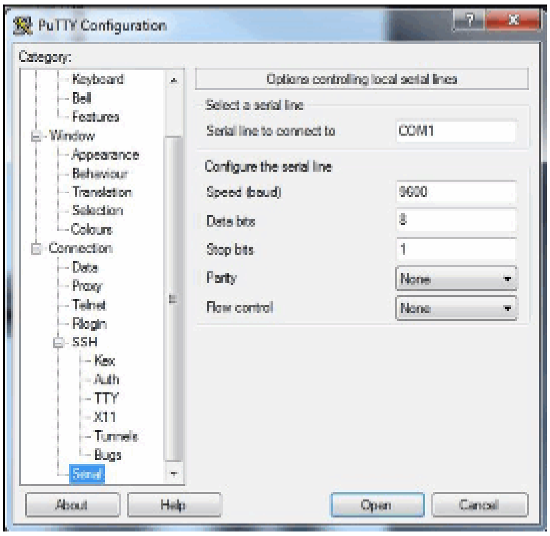
- Under 'Tunnels' tab.
- Set 'Destination' to 'Local' and 'Auto'.

Fig. 122: PuTTY Configuration - Bugs



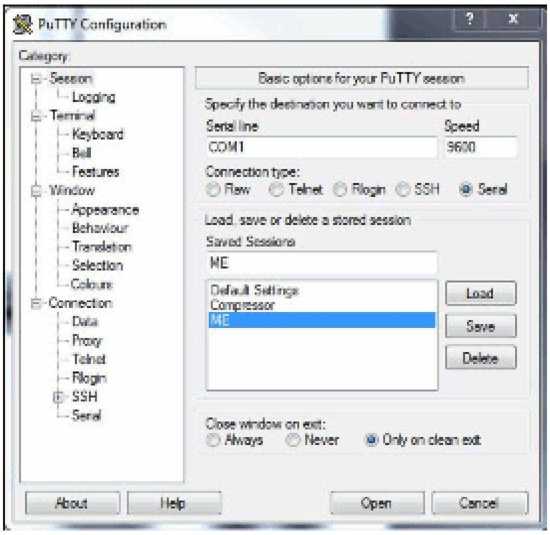
- Under 'Bugs' tab.
- Ensure all settings are set to 'Auto'.

Fig. 123: *PuTTY Configuration - Serial*



- Under ‘Serial’ tab.
- Under the ‘Serial line to connect to’ ensure it states ‘COM1’. Only state this if the com port that is being used. Change as required.
- Under the ‘Speed (baud)’ ensure it states ‘9600’.
- Under the ‘Data bits’ ensure it states ‘8’.
- Under the ‘Stop bits’ ensure it states ‘1’.
- Under the ‘Parity’ ensure it states ‘None’.
- Under the ‘Flow control’ ensure it states ‘NONE’.

Fig. 124: *PuTTY Configuration - Session*



- Under ‘Session’ tab.
- Under ‘Saved Sessions’ type in ‘ME’ or any name which you want to save these settings under.
- Click the ‘Save’ button.
- Click the ‘Open’ button.
- PuTTY software will now open and run.

In future requirements to use PuTTY follow this short sequence.

- Open the PuTTY software.
- Under the Session tab select the session required. In the case above the session is called ME.
- Click the load button.
- Click the ‘Open’ button.
- PuTTY will now open and run.

4.1.4.5 Ramp Cables

- Ramp cables **must not be fitted** until the **Transport test** has been **sucessfully** completed.

4.1.4.6 Initialising the 3600 MPSU

Ensure that there is a valid helium level reading before switching on the 3600 MPSU.

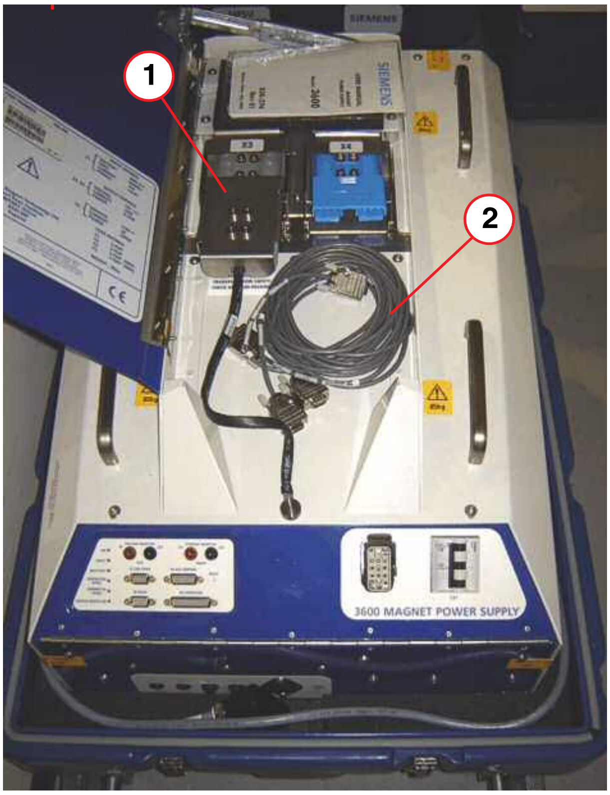
CAUTION

Risk of quench.

Failure to observe the following process may cause the magnet to quench.

» **Do not perform any other software sequence or downloads on the Host or Service laptop during ramping.**

Fig. 125: Transport test tool



(1) Transport test plug (fitted)
(2) RS232 cable

The **transport test tool** consists of 2 x 470 ohm resistors housed in a grey Anderson plug.

The center tap of the resistors is connected to a flying lead which is permanently bolted to the 3600 MPSU chassis.

The MPSU is supplied **with the test plug (X3)** already connected. (Pos 1)

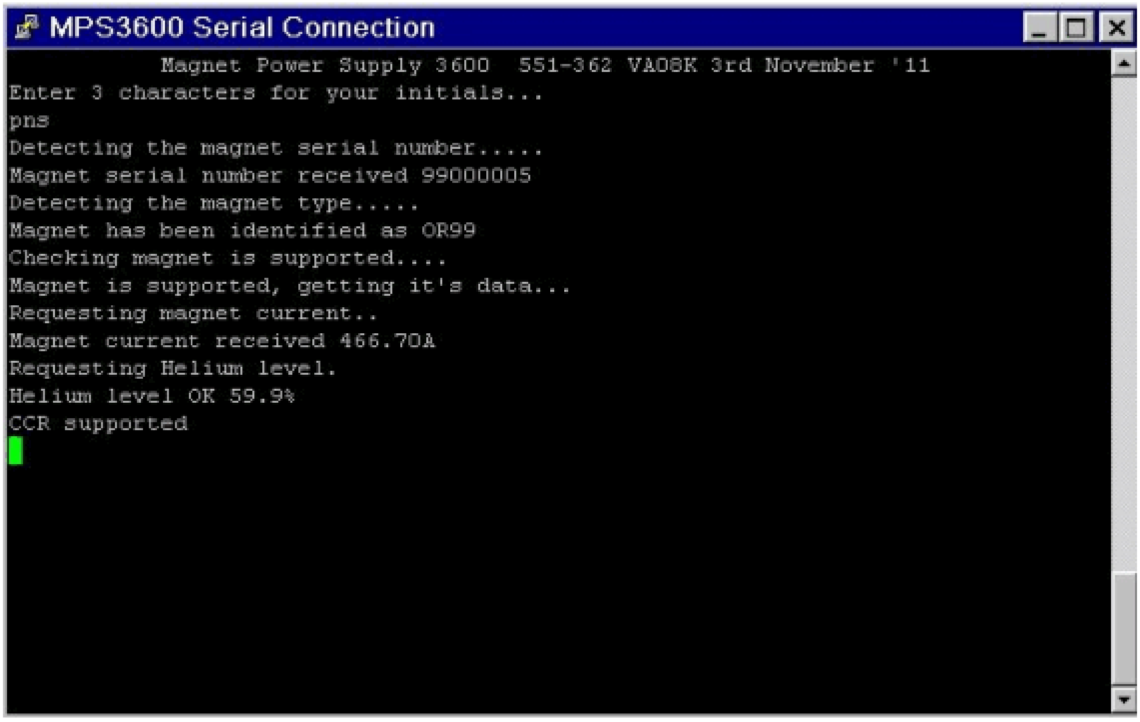
When the power supply arrives at site, the **transport test plug X3** should already be fitted. (→ 1/Fig. 125 Page 111). If not, connect it into **X3**. The test must be **passed** before it is removed and the **ramp cables** connected.

- Ensure the **RS232 cable** is connected between the **X6 connector** on the 3600 MPSU and the **laptop**.
- Ensure the **Terminal emulation software** is set up and running.

> The Transport test is only required for the first time the MPSU is used on a new installation.

- Ensure that **EPC circuit breaker** (**F53**) is set to the **ON position**.
- **At the MPSU**
  - set circuit breaker **CB1** to the **ON** position **'I'**
  - check the **green LED** indicator is **'ON'** and **lit**.
- The **'boot' screen** is displayed.

Fig. 126: Terminal boot-up screen



> If the screen does not appear, make sure that the correct COM port and settings have been selected. If they have, turn the MPSU off and then back on, using the mains circuit breaker CB1. (→ 1/Fig. 90 Page 93)

Reboot the scanner with the MPSU switched on.

A Reboot is necessary to update the loadware and to download the ramp logfile to the HOST.

The logfile is saved to C/MedCom/log directory.

Always reboot the scanner after the MPSU is switched off and on again. Rename the logfile, e.g. "MPSRampLog1.dat" after the ramp is finished.

If not renamed, the file will be overwritten with the next ramp logfile.

Repeat booting of the scanner if following warning message still persists: “MPS not operational”

Mandatory Transport test

During transport, it is possible that the 3600 MPSU may have suffered internal damage due to rough handling.

The mandatory Transport test was introduced from 3600 MPSU software revision level 17. (→ Transport Test Procedure / Page 113)

For units prior to this revision, continue with (→ Ramp lead connection / Page 117).

This test only needs to be completed the first time the 3600 MPSU is used on site.

This test must be performed PRIOR to connecting the ramp leads and using the 3600 MPSU on the magnet.

4.1.4.7 Transport Test Procedure

When prompted, enter your **initials.**

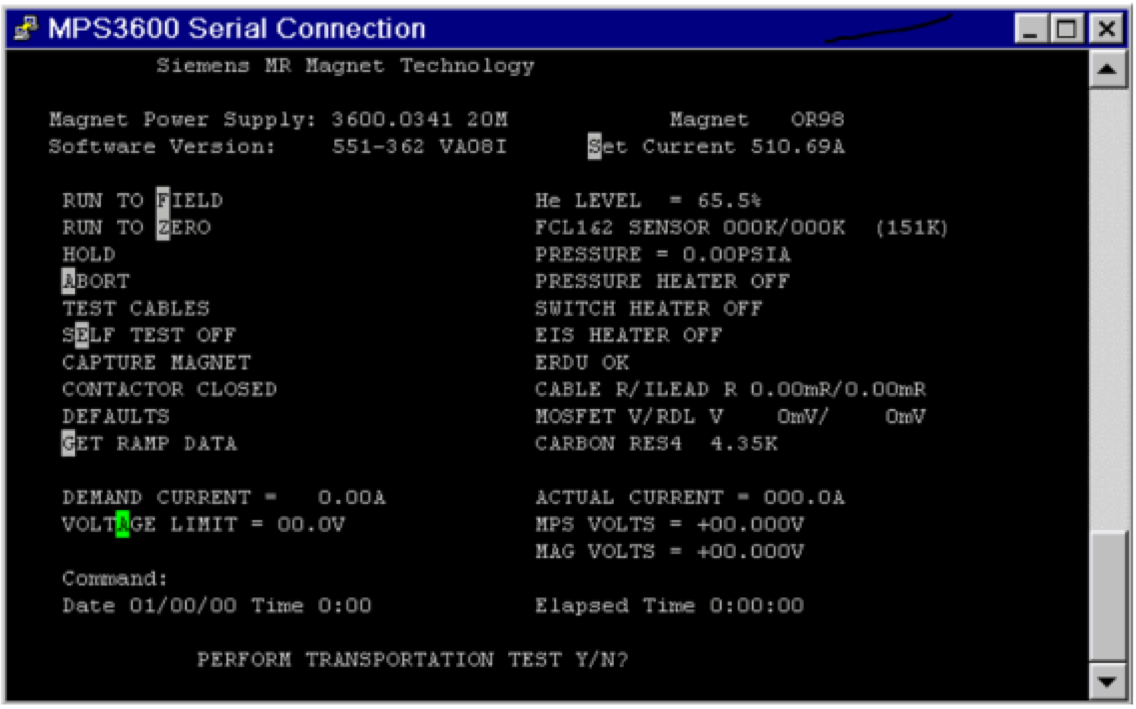
If transport test is NOT required

- Turn off the 3600 MPSU via **CB1.**
- **Connect** the **ramp leads** from the **Filter panel** to **X3** and **X4**:(→ Ramp lead connection / Page 117)
- Then refer to **Run to field**.

Transport test required

- The 3600 MPSU will now retrieve the **magnet type**, **magnet serial number, Last current set**, **He-level**, **date** and **time** from the **MSUP**.
- The following screen will prompt **‘Perform Transport Test’.**
    - » Type **Y** and **<ENTER>**.

Fig. 127: Transport Test - prompt



- The 3600 now performs a **transit test**.
- If the test is **successful**, the power supply will perform an **internal self-test**.
- The self test **LED** indicator will **light**.
- Turn off the 3600 MPSU via **CB1.**
- **Connect** the **ramp leads** from the **Filter panel** to **X3** and **X4**:(➔ Ramp lead connection / Page 117)
- Then refer to **Run to field**.

Failure of the Transport test

Failure of the **Transport test** is displayed briefly in the event of the test detecting an **internal fault** . (➔ Fig. 128 Page 115).

This is followed by a request to **check the connections** and **repeat** the test.
(➔ Fig. 129 Page 115)

> After 3 successive Transport tests have FAILED, further operation of the 3600 MPSU is prevented. Return the MPSU for repair and order a replacement unit.
>
> Line power to the 3600 MPSU must be switched OFF then ON to enable further tests to be performed.

> If all attempts fail, the 3600 MPSU must be returned for repair, and a replacement unit obtained.

Fig. 128: Transport Test failure

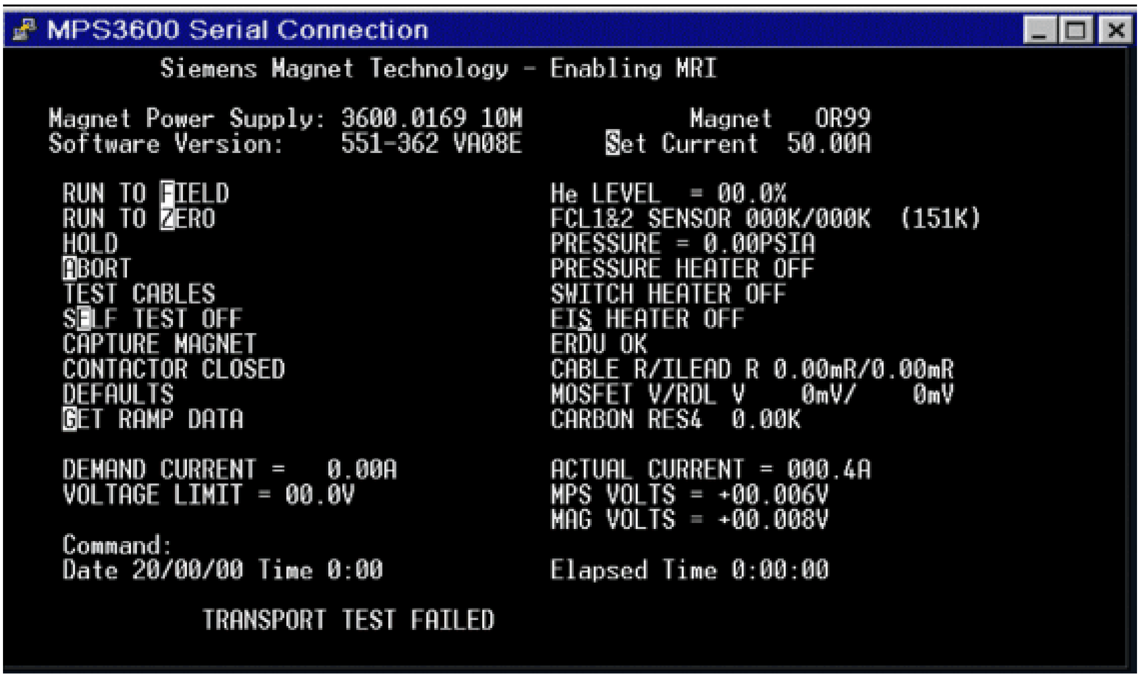
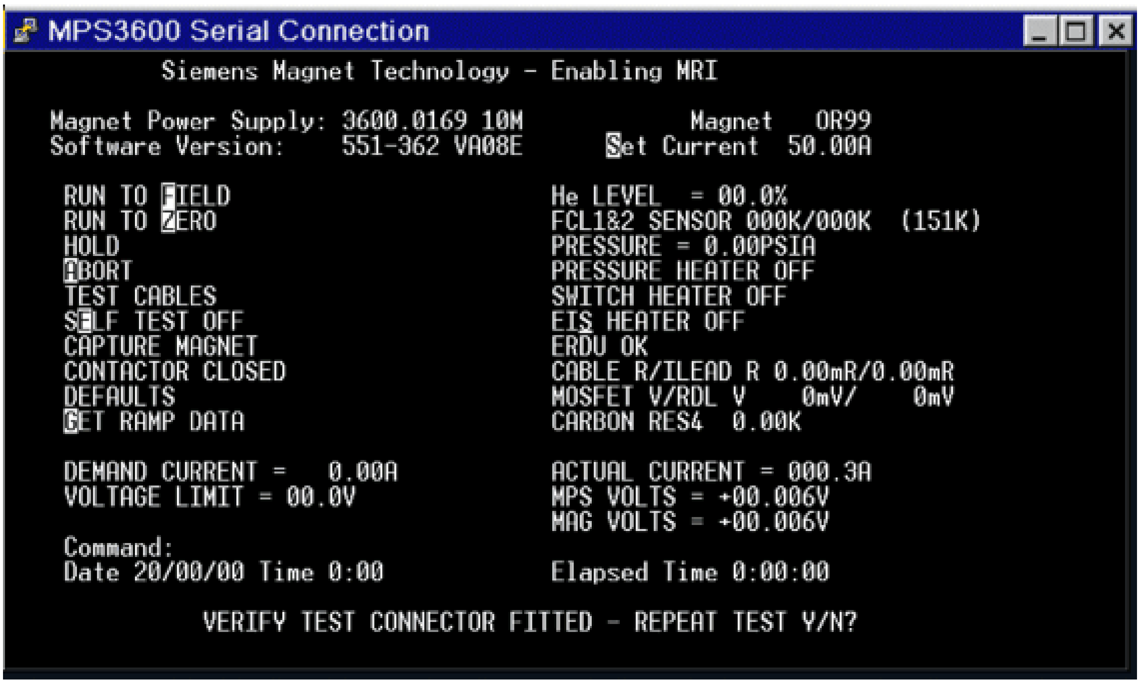


Fig. 129: Transport Test - verify test connector

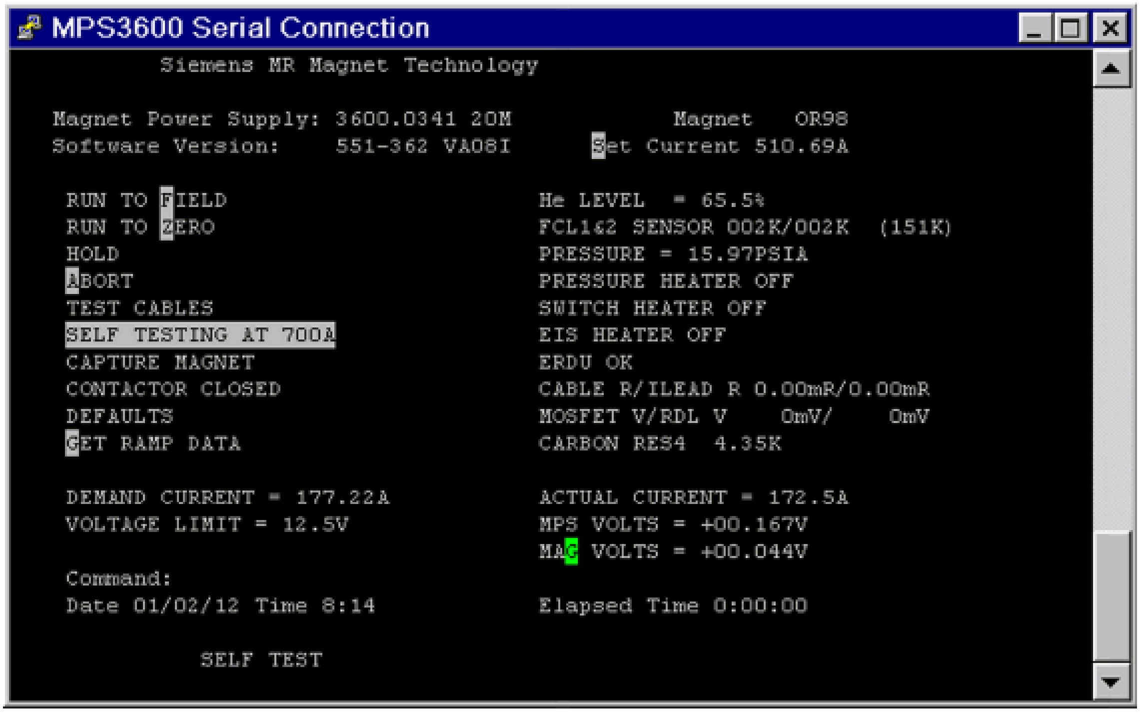


Internal self-test

During the self-test, the **LED indicator** indicator will light during the test. The self-test runs up to 700A with the output shorting contactor closed.

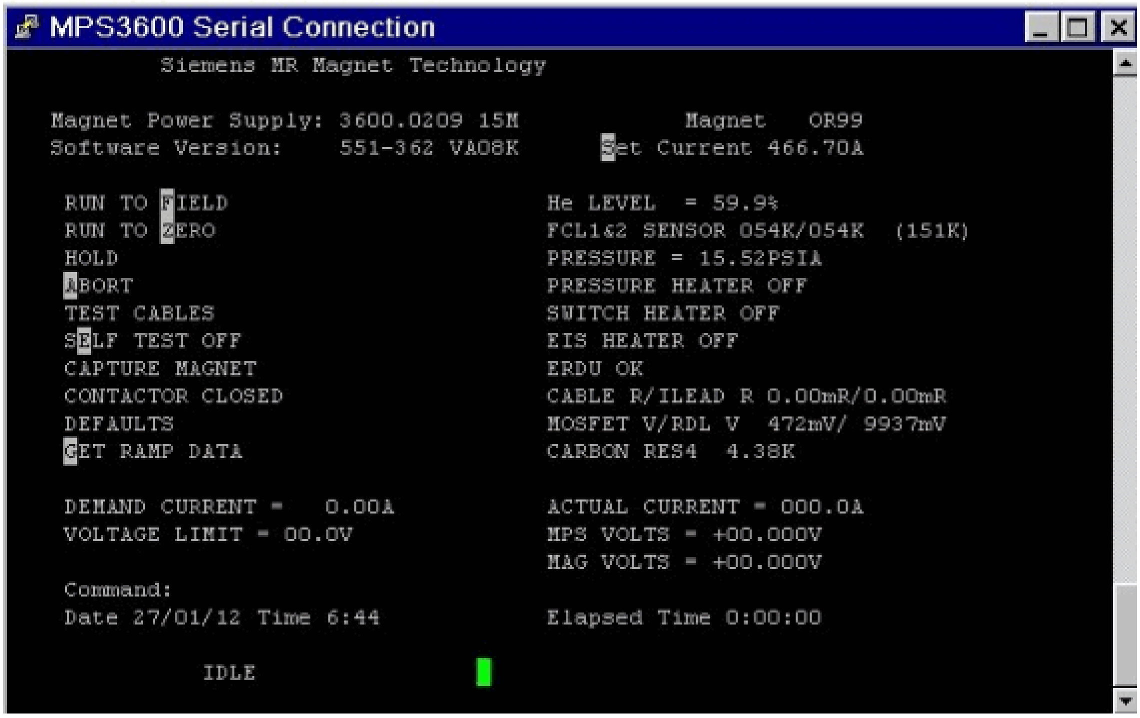
This enables the voltage across the internal Run-down Load and MOFSET switch to be measured and displayed on the screen.

Fig. 130: Self test



- When the self-test is complete, the **Operational screen** is displayed.

Fig. 131: Operational screen



- When packing the MSPU after use, ensure that the **Transport test plug** is refitted, ready for the next site.

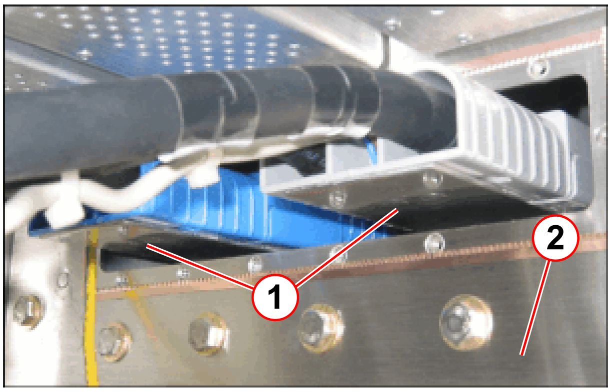
4.1.4.8 Ramp lead connection

After the Transport test has been successfully completed, (if applicable), the following connections must be made/changed **prior to ramping the magnet.**

There is no need to switch off the 3600 MPSU at this point.

- **Remove** the **Transport test plug**.

Fig. 132: Equipment Room - locking plate



(1) Locking plate
(2) RF filter plate

Equipment room

- **Connect** the **3600 MPSU ramp leads** to the **magnet leads** at the **RF filter plate**.

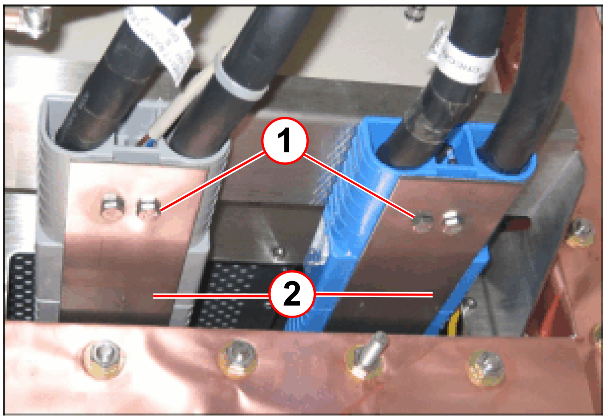
Magnet room

- Prevent the **ramp leads** from being accidentally **disconnected.**

  » **Fit** locking **bolts** and **wing nuts**.

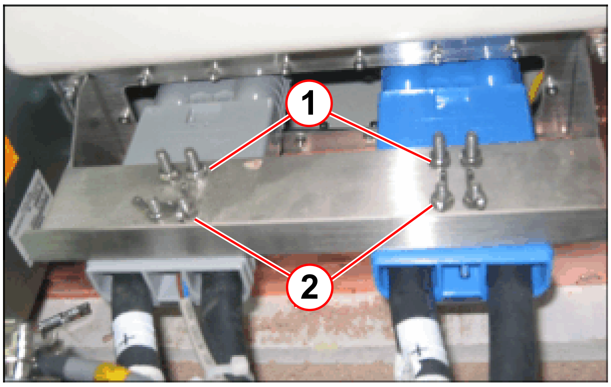
Ramp leads are heavy - to prevent stress on the connection, use tie wraps to support the weight if locking bolts are missing.

Fig. 133: Magnet room - locking plate (viewed from below)



(1) Locking bolts
(2) Locking plate

Fig. 134: Magnet room - locking plate (viewed from above)



(1) Anderson fixing bolts
(2) Wing nuts

Fig. 135: MPSU - ramp cables fitted

- **Connect** the two **3600 MPSU ramp leads** to Anderson connectors **X3** and **X4** on the top of the **3600 MPSU**.
- The screen will prompt **'ARE RAMP CABLES FITTED'** Y <**ENTER**>
- The 3600 MPSU now performs an **internal self-test.** (→ Internal self-test / Page 115)

CAUTION

Incorrect locking of the ramp leads.

If these instructions are not observed, the magnet and the 3600 MPSU can be damaged.

» **Connect and secure the leads.**

4.1.5 Run to Field

4.1.5.1 Co-Siting of Magnets

Co-siting describes the ramping requirements where two MR systems with the same operating frequency are sited next to each other in adjacent rooms.

If two systems are installed next to each other in adjacent rooms:
- One magnet must be ramped to the upper frequency limit.
- The other magnet must be ramped to approx.100 kHz less at the lower frequency limit.

Shimming is only optimized with both magnets ramped up.

Neither magnet is ramping, while the other is running applications. If one magnet is switched off, the other magnet has to be re-shimmed.

4.1.5.2 Run To Field Prerequisites:

- **Wait at least 1 hour** after completing any previous ramp, or aborted ramp (including lead checks).

> Following a helium fill, wait at least 2 hours before ramping.

- **Helium** level must be **> 40 %** for the **OR98** and the **OR103**. Or **> 60 %** for the **OR99** for all ramps after the initial ramp.

> Initial installation ramp helium level must be **≥ 60 %** for the **OR98** and **≥ 65 %** for the **OR99** and the **OR103**.

- **At the MPSU**
  - set circuit breaker **CB1** to the **ON** position **'I'**
  - check the **green LED** indicator is **'ON'** and **lit**.
- The **'boot' screen** is displayed.
- When prompted, enter your **initials.**

4.1.5.3 Setting the magnet current

The last ramp current will be stored in the MSUP/.iButton and will be displayed on the screen when the 3600 MPSU is switched on. Check the **actual magnet current** required. This is recorded on the **Magnet Acceptance Document.** To change the current:-

- **Enter** the command: **S<space>magnet current<ENTER>.**

4.1.5.4 Run to Field

- **Enter** the command **F <ENTER>** for ramping up the magnet.
  - » **Checks** will be carried out and the 3600 MPSU will then perform an **automatic ramp** to field.

The sequence of screen displays viewed during the ramp up is as follows:

- **Cable test**.
- The MPSU prompts the question **'Has magnet been filled in last 2 hrs Y / N?**
  - » Enter Y / N **<ENTER>**
- Magnet filled **Yes**
  - » The power supply prompts - **The magnet been filled in the last 2 HRS**.
  - » Answer **Y.**

- Cable test.

Fig. 136: Helium fill check

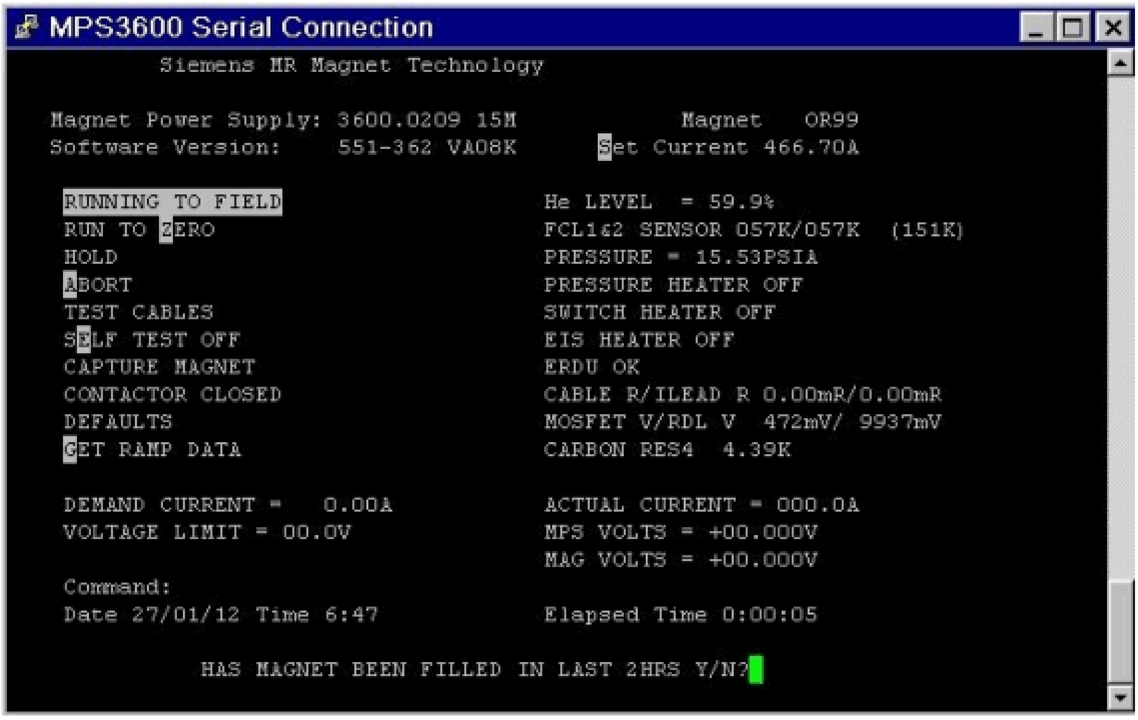
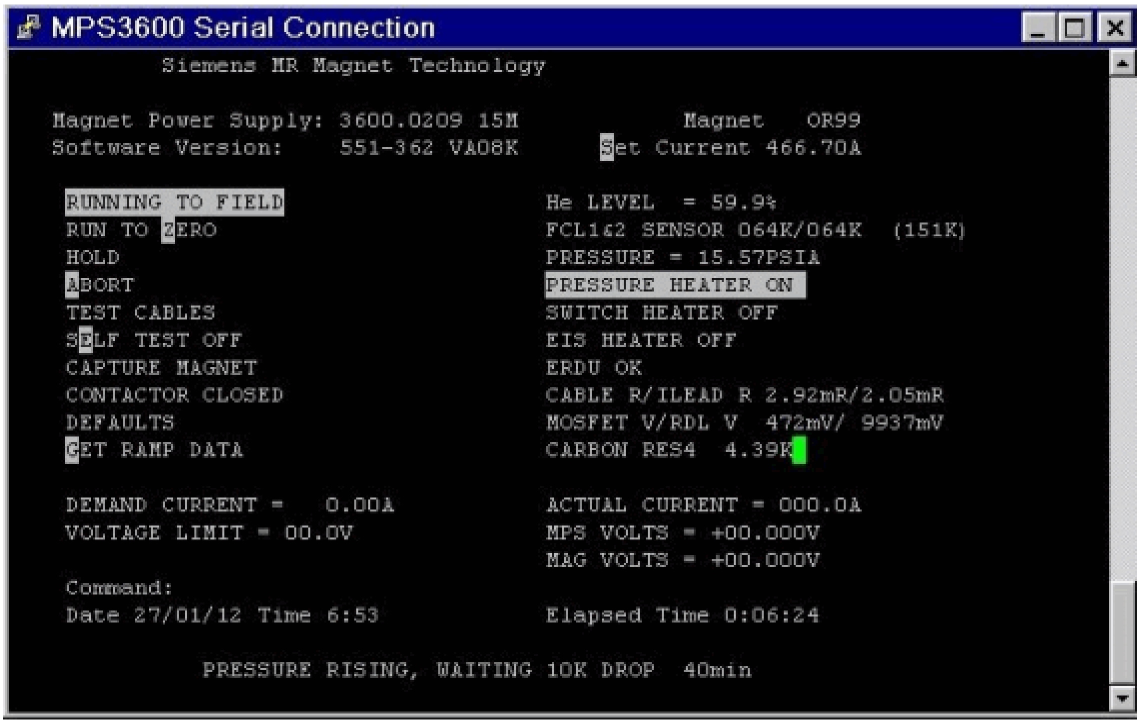


Fig. 137: Pressure heater ON (pressure rising)



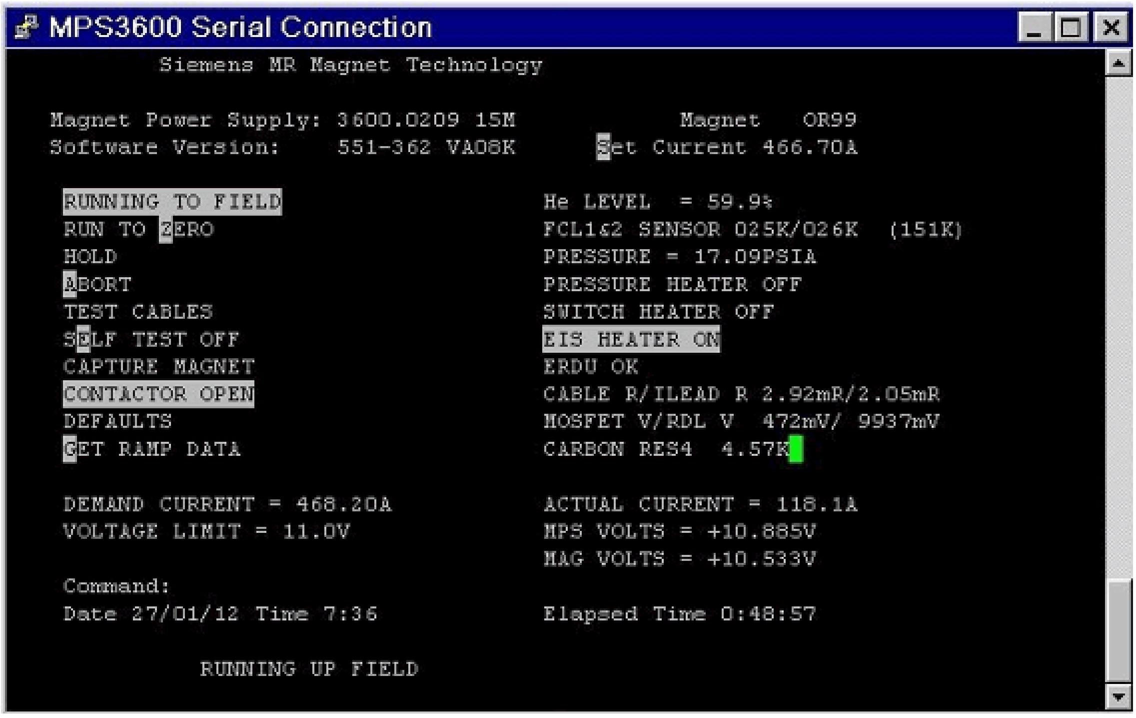
- Pressure rising - waiting for **10K drop**.

> Pressure raising time will vary, depending on magnet pressures and temperatures.
>
> Maximum 10K drop delay time is:
>
> OR98 = 30 mins
>
> OR99/OR103 = 40 mins

> If the 3600 MPSU software times out during the 10K drop the ramp has failed. Abort ramp and wait one hour before restarting.

- Waiting vent delay (5 mins).
- Ramp to field (set current) after performing the overfleld algorithm.

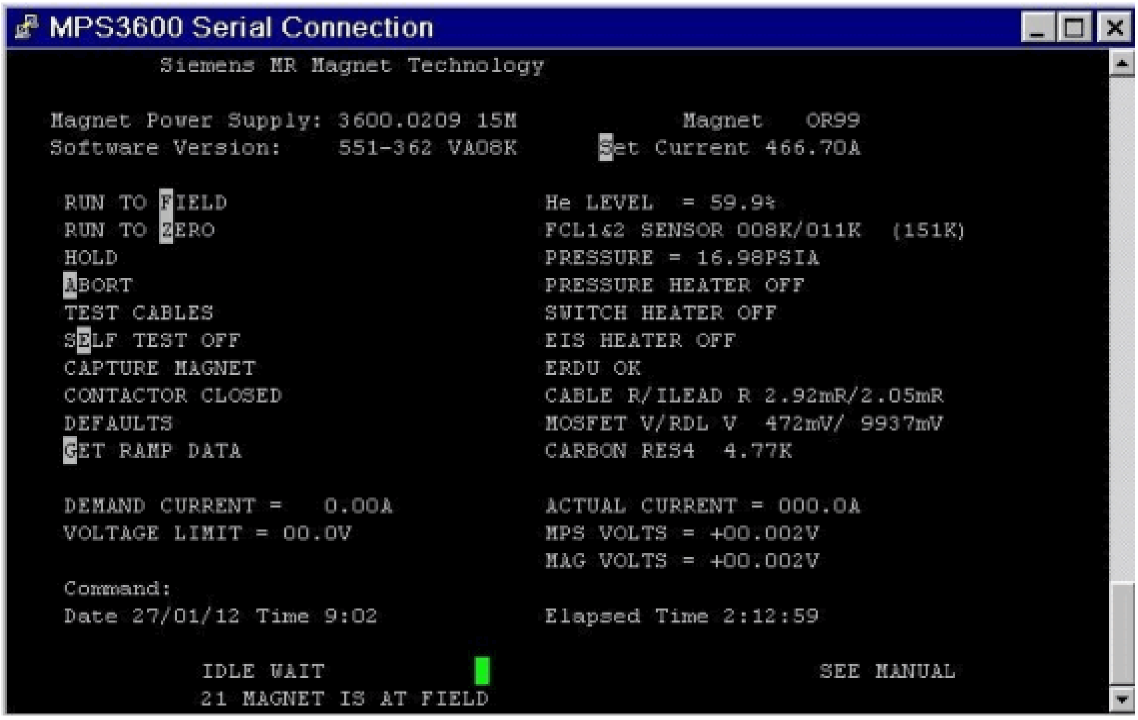
Fig. 138: Ramping to field



- Field delay (15 seconds).
- Waiting switch delay 2 mins (switch heater off).
- Running lead to zero - EIS delay wait.

- Idle wait - magnet is **at field**.

Fig. 139: Magnet at field (leads down)



- Ensure the **actual current** is equal to zero.
  » Turn 3600 MPSU **OFF** at **CB1**.

> If performing a shim plot:
> - Reconnect cable W10220 / X12 / MPSU to W10220 / X12 on top of EPC.
> - Restart Software

4.1.6 Run to Zero

4.1.6.1 Run to zero prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).

> Following a helium level fill, wait at least 2 hours before ramping.

4.1.6.2 Run to Zero

- Ensure the ramp leads are connected to the **3600 MPSU.**

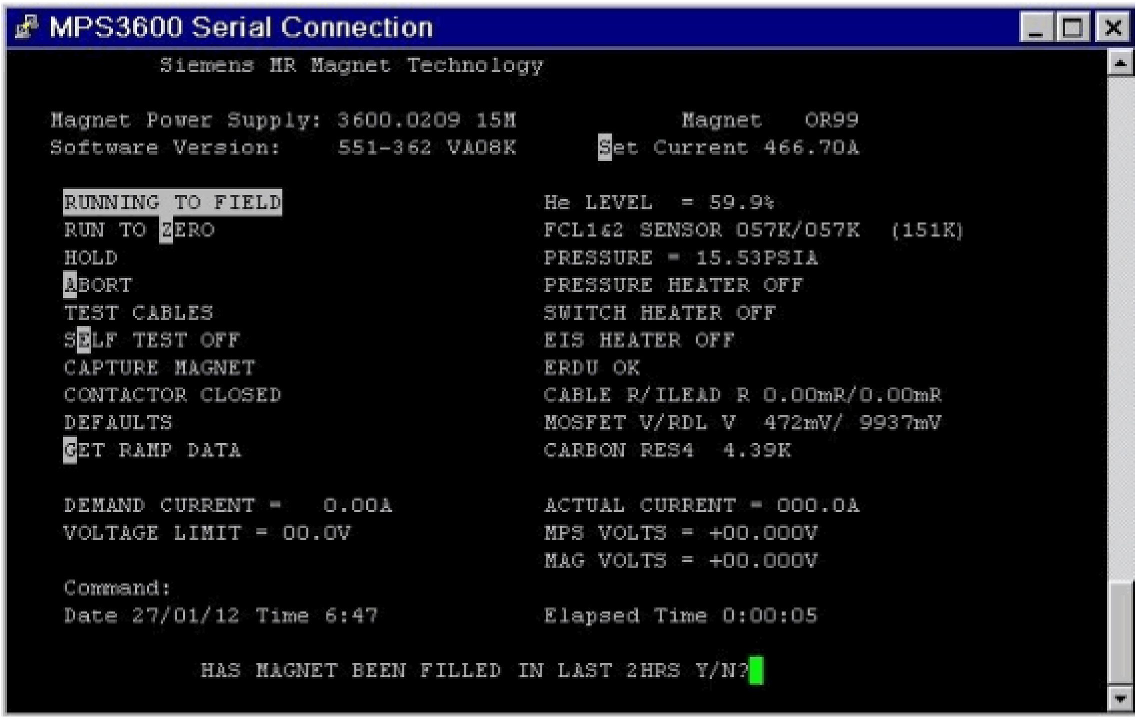
> It is not necessary to set the ramp current.

- **At the MPSU**
  - set circuit breaker **CB1** to the **ON** position **'I'**
  - check the **green LED** indicator is **'ON'** and **lit**.
- The **'boot' screen** is displayed.
- When prompted, enter your **initials.**
- Ensure the 3600 MPSU is in **Idle mode**. .
- Enter the command **Z<ENTER>** for ramping down the magnet.
  » Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to **zero**.

The sequence during the ramp to zero is as follows:-

- Cable test.

Fig. 140: Helium fill check



Magnet filled **Yes/No?**

» The power supply prompts - **Has the magnet been filled in the last 2HRS**

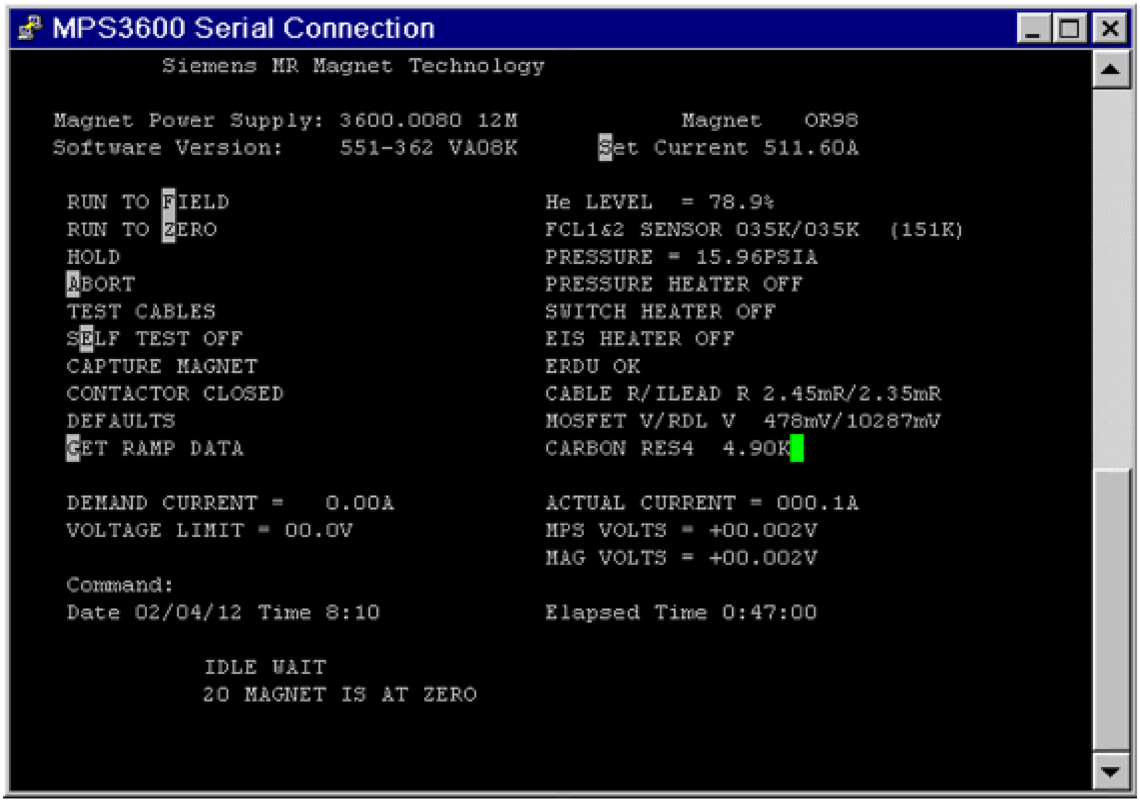
» Answer Y / N

The MPSU unit now controls the run to zero sequence with the following steps:

- 10K drop, 5 minute cooling time.
- Switch heater on signal to MSUP.

- Magnet capture.
- Magnet current running to zero.
- EIS delay wait (after contactor closes).

Fig. 141: Idle wait - Magnet is at Zero



- Idle wait - 20 magnet is **at zero.**
- When the **ramp is complete**,
  - **Download** the ramp logs (→ Downloading the 3600 MPSU log file / Page 126)
  - **Turn off** the 3600 MPSU at circuit breaker switch (**CB1**) - refer to (→ Switching off the MPSU / Page 126).
  - **Disconnect the ramp leads** from the 3600 MPSU.

4.1.7 Ramping the magnet to the required frequency

The magnet must be ramped to the upper limit of the frequency limit. This will require making small changes to the magnetic field current either up or down. The procedure is the same as ramping the magnet.

4.1.7.1 Magnet current setting OR98

Refer to the **Magnet Acceptance Document/MSUP** to obtain the operating **current value**.

> OR98 range of frequency:
> Maximum frequency: 63,700 000 MHz + 0 kHz
> Minimum frequency: 63,500 000 MHz

Use the mathematical formula below to calculate the current to determine the **centre frequency.**

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

OR98	**I final = (63,680 000 MHz x I actual) / Freq actual**

4.1.7.2 Magnet current setting OR99 / OR103

Refer to the **Magnet Acceptance Document/MSUP** to obtain the operating **current value**.

> OR99 / OR103 range of frequency:
> Maximum frequency: 123,265 000 MHz + 0 kHz
> Minimum frequency: 123,100 000 MHz

Use the mathematical formula below to calculate the current to determine the **centre frequency.**

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

OR99 / OR103	**I final = (123,260 000 MHz x I actual) / Freq actual**

Magnetom Skyra with Multinuclear Spectroskopy Option (MNO)

> OR99 / OR103 range of frequency:
> Center frequency: 123,200 000 MHz
> Maximum frequency: 123,265 000 MHz +0 kHz
> Minimum frequency: 123,125 000 MHz -0 kHz

Use the mathematical formula below to calculate the current to determine the **centre frequency.**

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

OR99 / OR103	**I final = (123,200 000 MHz x I actual) / Freq actual**

4.1.8 Work after ramping the magnet

4.1.8.1 Downloading the 3600 MPSU log file

The last ramp data is stored in a log file within the 3600 MPSU, even if the 3600 MPSU is switched off. Data is overwritten when the next ramp is started. It is recommended that after each ramp, the log file is downloaded and stored for future analysis.

The Magnet Peak Current is recorded. If the magnet quenches, the recording will tell you at what current level the magnet quenched.

> The ramp log files may be saved locally if the host is operating
>
> However it is still recommended to recover these logs as described manually to ensure data consistency.

To download the log file:

- **G** <Enter>, this will ensure that you are in the **error screen**.
- From the Terminal menu, select:-
  - Transfer.
  - Capture Text.
  - Browse.
  - Select the target directory and enter a filename e.g.**C:\Medcom\log** \MR25123_51099log.txt.
  - Start.
- **L** <Enter>, the data screen appears.
- A scrolling data screen will appear. Wait until the data scrolling has stopped.

  The data screen contains seven columns. Each line displays the data stored at a specific period of time during magnet ramping.
- From the Terminal menu, select:
  - Transfer.
  - Capture Text.
  - Stop.

4.1.8.2 Switching off the MPSU

> Verify that the yellow “CURRENT IN LEADS” LED is OFF when the 3600 MPSU is switched on. (→ 5/Fig. 93 Page 95)

NEVER attempt to remove the magnet ramp leads when the “CURENT IN LEADS” LED is flashing.

- Set the mains circuit breaker **CB1** to the **OFF** position. (→ 1/Fig. 90 Page 93)
- Set the circuit breaker **F53** in the **EPC** to the **OFF** position. (→ 1/Fig. 96 Page 96)
- **Disconnect** the 3600 MPSU ramp leads from the magnet leads at the RF filter panel.
- The **magnet ramp leads** remain fitted and stored on the lead tray in the magnet room.
- **Re-fit** the equipment room **adapter plate** using the securing screws.

Connect the Anderson connectors (blue/blue and grey/grey).

Fig. 142: RF filter plate

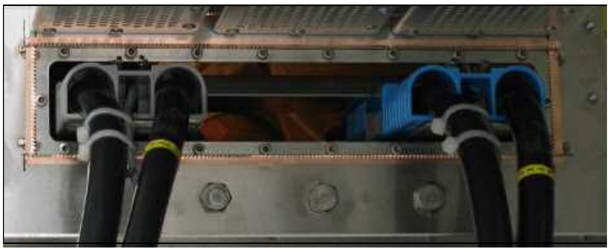


Fig. 143: Adapter plate (view from Equipment Room)

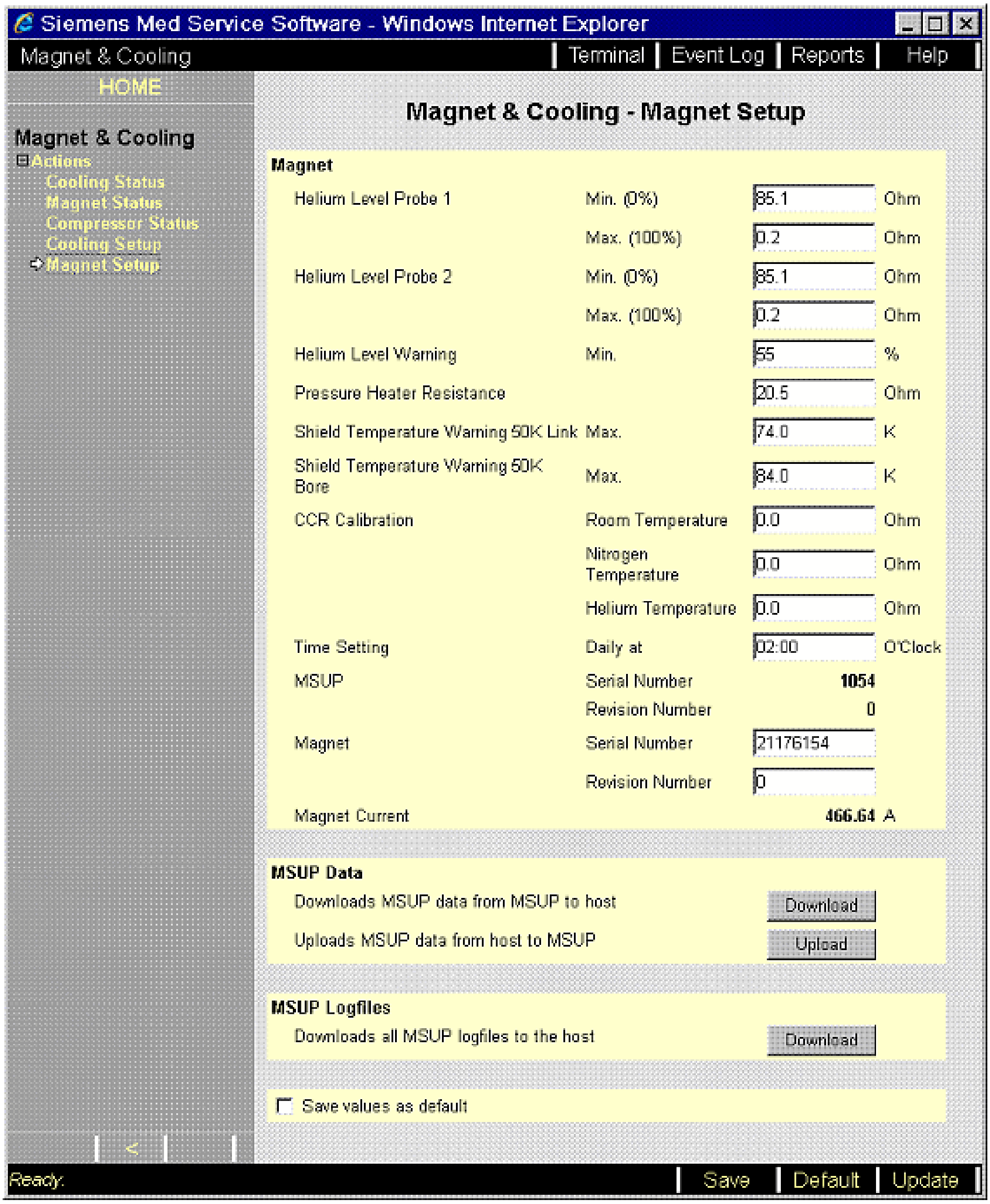
- Reconnect cable W10220 / X12 / MPSU to W10220 / X12 on top of EPC.
- Restart the Software.

4.1.9 Final work steps

4.1.9.1 Save the magnet current to the HOST

- Access the service menu in the **SeSo at MRC (MRAWP) Host**.
- Select menu item **“Magnet & Cooling”.**
- Select **“Magnet Setup”** in the **“Action”** menu.
- Click **“Download”** to save the magnet current value on the **HOST**.

Fig. 144: Magnet setup



4.1.9.2 Warm Turret & Venting

Fig. 145: Warm the turret and venting with a heat gun.

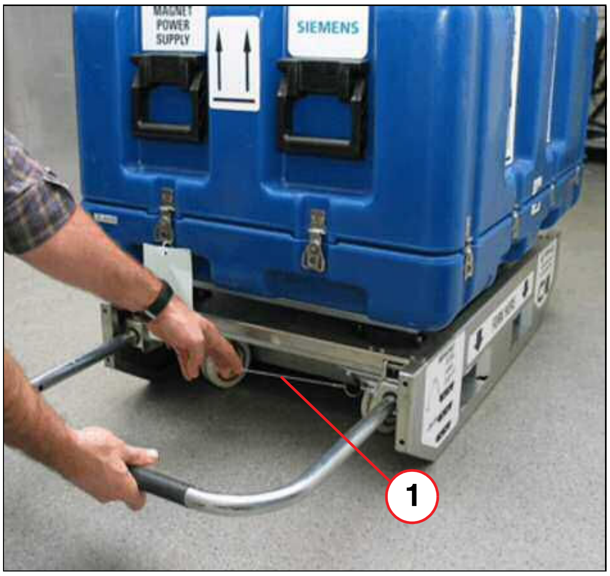
- Ensure that the **turret and venting** remains leak tight and the valves continue to function correctly.
- **Warm** up the **turret and venting** with a heat gun.

4.1.9.3 Pack the 3600 MPSU

Fig. 146: MPSU - ramp cables fitted

- Disconnect the two **3600 extension ramp leads** from the Anderson connectors **X3** and **X4.**
- Re-fit the **Transport Test Plug** into **X3**.
- Disconnect the remaining leads and pack inside the transport case.
- Close the lid and secure by pulling the two silver knobs and turning them through a 1/4 turn.

Fig. 147: MPSU



(1) Wire locking strap

- Replace the shipping case lid and secure using the 12 spring clips.
- Pull the wire strap to unlock the handle, and slide the handle out of the recess under the trolley.
- Release the wire strap to lock the handle in the extracted position.
- Raise the handle to lower the wheels and lift the trolly off of the two skids.
- The 3600 MPSU and trolley can now be safely moved.

4.1.10 Mains failure during ramping

CAUTION

If the supply to the 3600 MPSU fails while the magnet is ramping through the ramp leads up or down, the magnet current will continue to flow.

Failure to observe the following safety warning may result in physical injury and equipment damage.

» **Never remove the ramp leads from the magnet while current is in the magnet ramp leads.**

Current is in the ramp cable whenever:

» The yellow “CURRENT IN LEADS” LED is flashing.

Failure of the 3600 MPSU may prevent this LED from lighting up. So, in the event of a failure (e.g. mains failure), it is essential to measure the voltage across the terminals on the turret.

Any voltage on these terminals other than zero is an indication that there is current in the ramp leads.

» Magnet was ramping up or down and there was a mains failure.

» Magnet was ramping up or down and the 3600 MPSU detected a fault such as over temperature, which would shut down the 3600 MPSU.

» Normal ramp up or down of the magnet.

CAUTION

If the magnet is suspected of ramping down through the ramp leads, owing to one of the reasons above.

Failure to observe the following safety warning may result in physical injury and equipment damage.

» **Do not remove the ramp leads. The magnet must be run to zero or persist at field before removing the ramp leads.**

If the cause of the shutdown was failure of the mains, nothing can be done until both the 3600 MPSU and the Magnet Supervisory (MSUP) are working again.

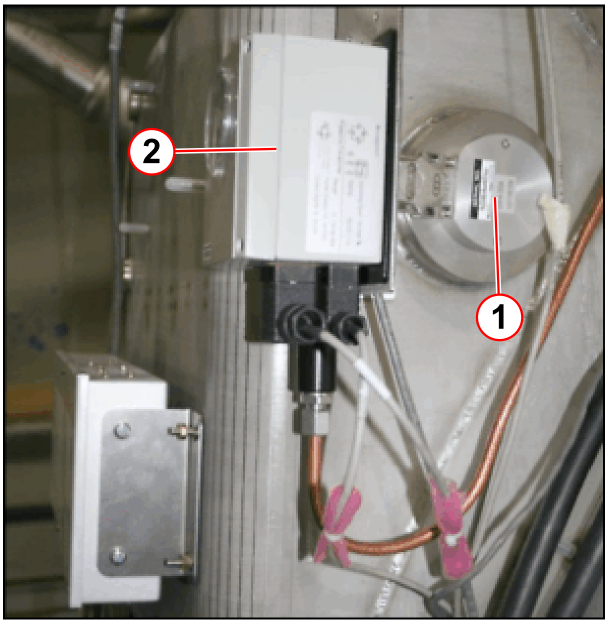
When the 3600 MPSU and the MSUP are operational, ramp the magnet to zero.

Do not delay. Ramp down as soon as mains power returns.

Do not wait one hour to ramp down or capture.

4.1.11 iButton function

Fig. 148: i-Button location



(1) iButton
(2) Pressure gauge

The **iButton** is a non-volatile memory device which stores vital magnet data.

It is connected to the **MSUP** at **X2** (CH/DIAG/PRESS/ID/VALVE).

4.2 What to do after a quench

4.2.1 Post quench procedure

4.2.1.1 General

Should the magnet quench, a large volume of helium gas will be generated within the helium vessel. The **quench valve** will **open**, and stay open until the flow of gas subsides. The quench valve will then re-seal preventing any air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP will automatically maintain the system pressure above atmospheric. **If this is not the case, contact the Magnet HSC immediately**.(→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

> When a quench is detected, Magnet HSC must be notified.
>
> Fill out the **Quench Report database** (→ Product Information MR, https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.

CAUTION

If the quench valve or auxiliary vent burst disk breaks:

Failure to observe the following will result in ICE FORMING within the magnet.

- » **Replace the graphite, or auxiliary vent bursting disk IMMEDIATLEY.**
- » **If an auxiliary vent burst disk is not available, fit the red blanking plate immediately.**
- » **Inform Magnet HSC.**
- » **DO NOT RAMP THE MAGNET until the burst disks have been replaced.**

The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature. (➔ Fig. 152 Page 139)

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms. Compare to the values on the Magnet Acceptance Document.

Check the helium level at least 1 hour after the quench. If there is still helium in the magnet the system can be top filled.

The magnet must be re-filled immediately with helium.

To re-fill the OR99/OR103 magnet from zero helium, or warm ceramic carbon resistor (CCR) temperatures between LN2 and He, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for liquid helium.

OR98 - requires approximately 1000 to 1500 litres of liquid helium.

OR99/OR103 - requires approximately 2000 liters of liquid helium.

CAUTION

IF the magnet is allowed to warm up above 77K, a full nitrogen pre-cool is required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

A full nitrogen pre-cool takes about 10 days.

» **Contact Magnet HSC IMMEDIATELY before proceeding. (➔ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)**

Fig. 149: OR98 Magnet Acceptance Document (example)

SIEMENS

Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21236477 **SMT Reference** 21245226 05.10

System Description
1.495T Actively Shielded Superconducting Twin Turret Magnet System OR98

Measured Magnet Centre (Offset)	+1 (P)	mm
Gradient Coil Position	39	mm
Overall System Length	1409	mm

This system requires 12 Z2 rings and 2.38 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Measured value	
Operating Current @ 1.495T	510.83	AMPS
Field/Current Ratio	12.46	kHz/0.1 Amp

Magnet Type: OR98--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC Outer (D-E	926		2951
CCR3 Top Inner (C-E	1060		2812
CCR2 Middle Inner (B-E)	1035		2977
CCR1 Bottom Inner (A-E	1093		3138
TDC Calibration Values:	882.4	1181.8	3075.3

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2	
0% Resistance	107.0	107.0	Ohms
0 % Calibration	448	448	
100 % Resistance	17.9	17.9	Ohms
100 % Calibration	75	75	
Diode Table No	-2		
Pressure Heater Res	22.1 Ohms		

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
56	55	41	K	782 Milliwatts

Checked by K Suraina Signature

Date 28/05/2010

Fig. 150: OR99 Magnet Acceptance Document (example)

SIEMENS | Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21240447 **SMT Reference** 21243544 05.10

System Description

2.894T Actively Shielded Superconducting Twin Turret Magnet System OR99

Measured Magnet Centre (Offset)	-3 (S)	mm
Gradient Coil Position	43	mm
Overall System Length	1669	mm

This system requires 12 Z2 rings and 1.44 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Specification value	Measured value	
Operating Current @ 2.894T	< 500 AMPS	465.97	AMPS
Field/Current Ratio	N/A	26.44	kHz/0.1 Amp

Magnet Type: OR99--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC (D-E)	1072		2863
CCR3 Top (C-E)	872		3114
CCR2 Middle (B-E)	1027		3182
CCR1 Bottom (A-E)	1036		2963
TDC Calibration Values:	1022.6	1322.1	2991.4

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2	
0% Resistance	80.3	80.3	Ohms
0 % Calibration	336	336	
100 % Resistance	0.5	0.5	Ohms
100 % Calibration	002	002	
Diode Table No	-2		
Pressure Heater Res	20.5	Ohms	

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
54	50	40	K	874 Milliwatts

Checked by K Suraina Signature

Date 10/05/2010

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Fig. 151: OR103 Magnet Acceptance Document

SIEMENS

Magnet Acceptance Documentation
Installation data
MR Magnet Technology Quality Regulations

Cryostat number 33002284 **SMT Shipped System** 33002281 07.11

System description 2.894T Actively Shielded Superconducting Twin Turret Magnet System OR103

	SKYRA	MONET	
Measured magnet centre (Offset)	-4 (S)	-4 (S)	mm
Gradient Coil Position	174	174	mm
Overall Magnet Length	1670	1670	mm
This system requires	0	0	Z2 Rings
This system requires	2.63	2.63	kg of iron to shim

(This is a theoretical calculation that may change due to site conditions)

Operating parameters

Description	**Specification value**	**Measured value**
Operating current @ 2.894T	< 500 Amps	487.35 Amps
Field/Current Ratio	N/A	25.28 kHz/0.1 Amp

Magnet type: OR103-00

Carbon resistor values

Position	**Room temperature**	**Nitrogen temperature**	**Helium temperature**
CCR4 d - e top	1071 Ohms		2573 Ohms
CCR3 c - e	991 Ohms		2612 Ohms
CCR2 b - e	929 Ohms		2797 Ohms
CCR1 a - e bottom	1007 Ohms		2841 Ohms
CCR4 calibration values	1031 Ohms	1304 Ohms	2621 Ohms

Helium level probe resistance (when using a 250mA supply)

	Probe 1	**Probe 2**
0% resistance	107 Ohms	107 Ohms
0% calibration	448	448
100% resistance	10.8 Ohms	10.8 Ohms
100% calibration	45	45

Diode table no. 2

Pressure heater resistance 20.5 Ohms

Final temperature sensor recon values

Shield bore	Shield Link	Coldhead	Recon margin
55K	48K	39K	853mW

Checked by K. Baker **Signed**

Date 11/07/2011

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Compare the CCR temperatures with the values on the Magnet Acceptance Document.

4.2.1.2 Safety

Switch off and isolate the magnet power supply from the mains before further action.

WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogens.

- **Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.**
- **Ensure the room is ventilated.**
- **Ensure that the turret venting kit is connected to the quench pipe.**
- **O_2 monitors must be used in the imaging suite.**

Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

CAUTION

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- **Replacement of burst disk(s) must be performed by a qualified CSE.**
- **Replace broken or damaged burst disk/O-rings IMMEDIATELY.**
- **Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.**
- **If ice is suspected contact Magnet HSC immediately.**

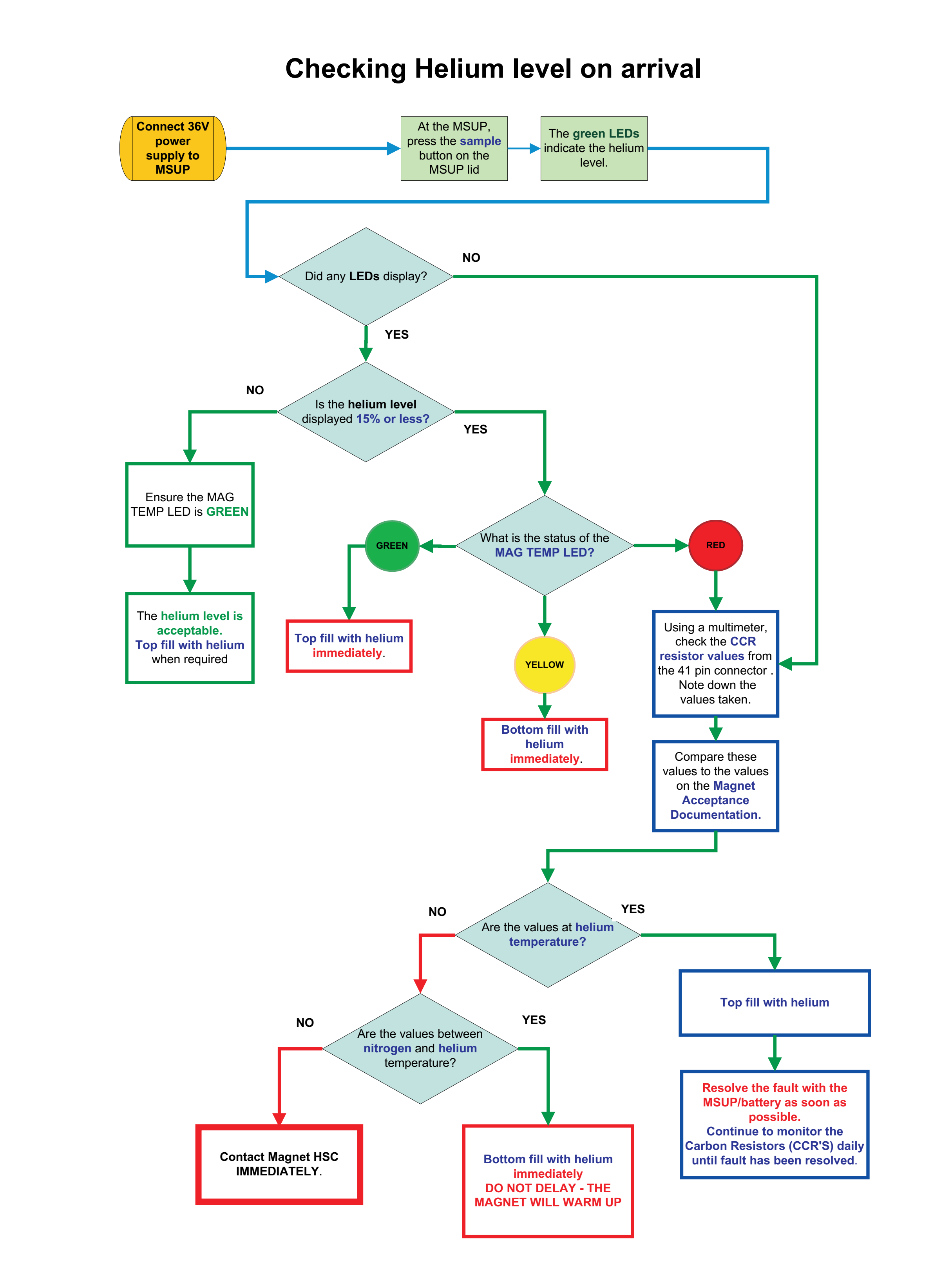
4.2.2 Post quench steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	■ Isolate the **MPSU** from the magnet. ■ Download the **3600 MPSU ramp log files**.	■ Stop filling.	■ **Check** that no ERDU buttons are pressed. ■ **Re-set** ERDU button.
Step 2	■ **Check** the **liquid helium level**. at the Host, MRC - Magnet & Cooling. ■ **Check** the ceramic **carbon resistors** at the MRC. (→ Check the Ceramic Carbon Resistor (CCR) values / Page 141)		
Step 3	■ Order **1000 - 2000** liters of liquid helium. [1]		
Step 4	■ **Warm up** and **check** the **quench valve** and **auxiliary vent burst disk are intact.** Refer to **Checking the quench valve burst disk and venting**. (→ Checking the quench valve burst disk and aux vent burst disk / Page 143)		
Step 5	After **1 hour**: ■ **Download** the **Magnet Supervisory log files.** (→ Log file capture / Page 140) ■ **Helium fill**. Refer to **Start up**.		
Step 6	■ **Leak check** the quench valve and venting. Refer to **Turret leak check**.		
Step 7	■ Fill out the **Quench report**. Refer to **Quench report.**		
Step 8	■ **Ramp** the magnet. Refer to **Start up.**		
Step 9	■ Perform the **shim check**. » If the **shim check** exceeds the specifications, a re-shim is necessary.		
Step 10	■ **Contact Magnet HSC**: » Email Magnet 3600 Power Supply and Magnet Supervisory logs to: **Magnet HSC** with magnet serial no. and all log data. Refer to **Start up.**		

1. The quantity of the order depends on the helium level remaining in the system after the quench, and volume already on site.

Fig. 152: Checking Helium level on arrival



4.2.3 Procedure

4.2.3.1 Required tools

Heat gun

Standard tool kit

Helium leak detector (part no **10614850**)

Laptop and RS232 cable

O_2 monitor

Eye protection

Gloves

4.2.3.2 Unattended quench

Unattended quench indicators are:

- If the **Magnet STOP (ERDU)** has been activated:
  - » the yellow **WARNING LED** lights on the alarm box.
  - » the alarm **buzzer** is activated.
- If the **helium level** is low:
  - » the yellow **WARNING LED** lights on the alarm box.
  - » the alarm **buzzer** is activated.
- If the **magnet pressure** is very low, the
  - » **cold head** switches off.
  - » the yellow **WARNING LED** lights on the alarm box.
  - » the alarm **buzzer** is activated.
- Imaging is not be possible if there is no magnetic field.

Press the **AUDIO ALARM OFF** button on the alarm box to silence the audible alarm.

4.2.3.3 Log file capture

Magnet Supervisory

- The **quench data is lost**, if:

  a) The system is **re-booted** within 2 hours of a quench
  b) The system is **turned off** within 2 hours of a quench
  c) The data is downloaded within 2 hours of a quench

- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the **Magnet Power Supply** log file.

> The MPSU log files MUST be downloaded BEFORE the next ramp is performed, otherwise previous data will be overwritten!

- **Switch off and disconnect** the MPSU from the magnet.

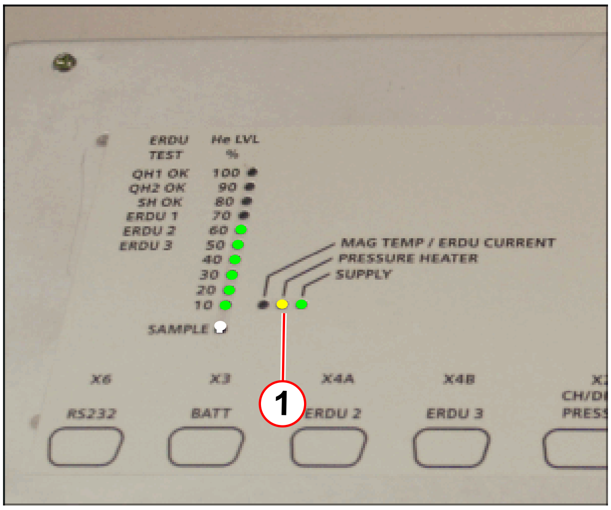
If the magnet quenches during helium fill:

- **Stop** the transfer.

Do not remove or disconnect the syphon as the magnet will be at high pressure.

For all types of quench, allow up to 1 hour for the quench pressure to subside.

Fig. 153: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

Check the **pressure heater status**

- Check the MSUP **PRESSURE HEATER LED** is lit, **or**
- Access the **Service Software** at the **MRC** (MRAWP) Host. (→ 1/Fig. 157 Page 144)

If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. Ensure the compressor **switches off**.

4.2.4 Check the Ceramic Carbon Resistor (CCR) values

Compare the temperatures with the values on the Magnet Acceptance Document.

- Access the **Service Software** at the **MRC (MRAWP) Host.**
- Select the menu item **Magnet & Cooling - Magnet Status**.

Fig. 154: Checking the Ceramic Carbon Resistors

Tab. 9 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resis-tors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

The above table is a guide only. Always compare the readings with those on the Magnet Acceptance Document.

The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms.

If temperatures are between N2 and helium temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

If the host (or laptop), is not available, check CCR values directly from the pins (a-e) on the 41 pin connector on the turret.

4.2.5 Checking the quench valve burst disk and aux vent burst disk

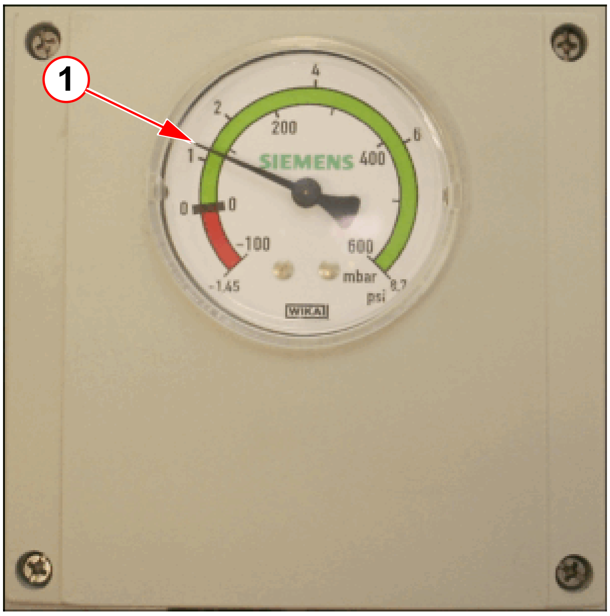
After helium filling, the quench valve and venting must be leak checked.

Ensure the **helium vessel** has a **positive** pressure:

- at the **pressure gauge.** (→ Fig. 155 Page 143), (→ Fig. 156 Page 143)
- at the **MRC (MRAWP) Host.**(→ 1/Fig. 171 Page 153)
- Check for **positive pressure** on the pressure gauge.

OR98

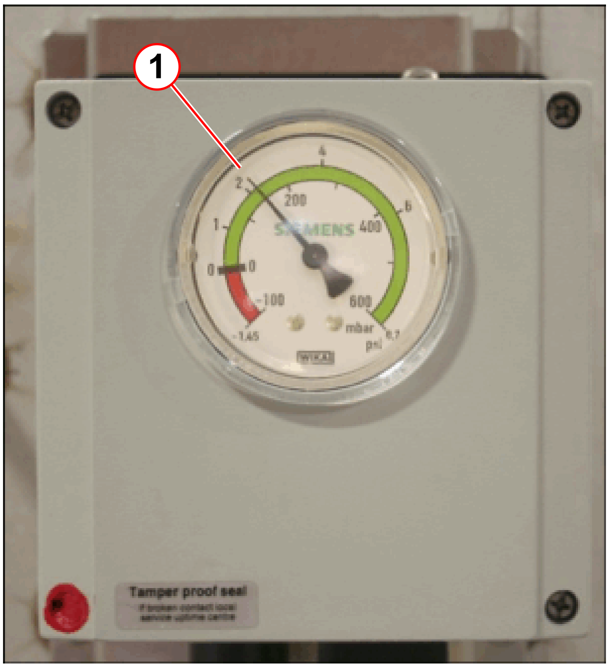
Fig. 155: OR98 - Pressure gauge



(1) Expected gauge pressure 0.7 - 2.3 psi (G)

OR99/OR103

Fig. 156: OR99/OR103 - Pressure gauge



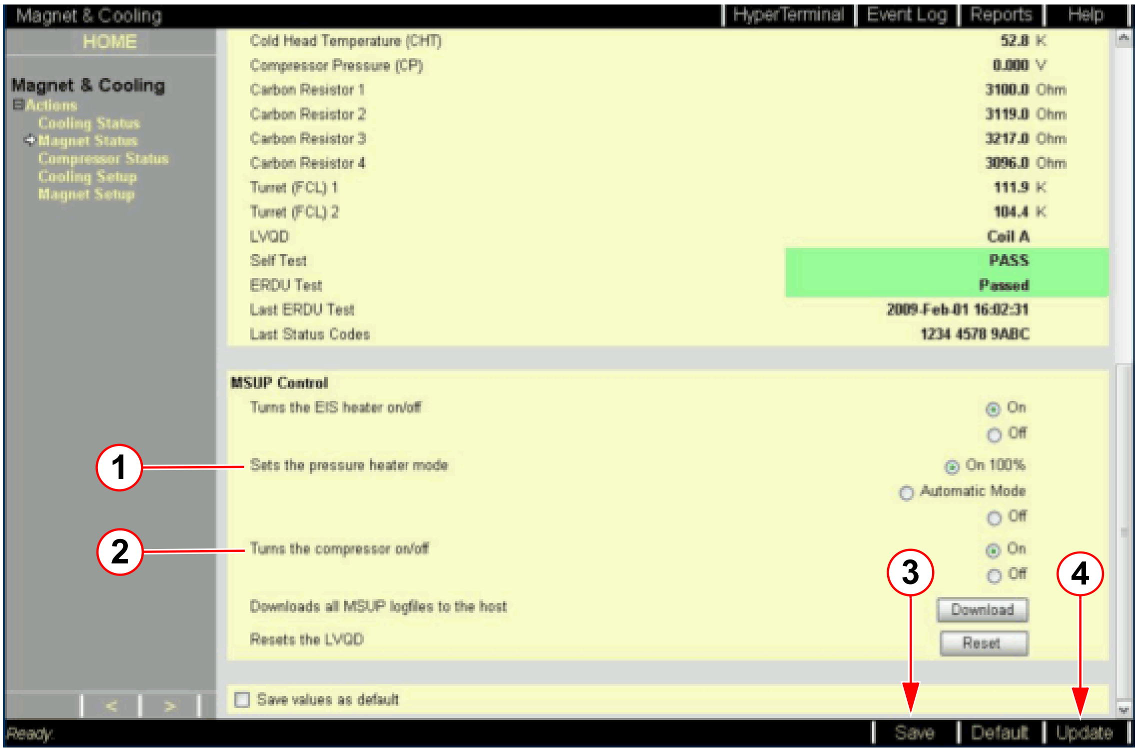
(1) Expected gauge pressure 1.7 - 3.3 psi (G)

- If the **pressure gauge** is within the **RED zone**, the magnet is **sub-atmospheric**.
  » **DO NOT** open the magnet **bypass valve**, or disturb any high pressure fitting.

4.2.5.1 Deactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **OFF** button (→ 1/Fig. 157 Page 144).
- Select **SAVE** (→ 3/Fig. 157 Page 144).

Fig. 157: Service screen Magnet & Cooling - Magnet status - MSUP Control



(1) Sets the pressure heater mode
(2) Sets the the compressor on/off
(3) Save button
(4) Update button

> The pressure heater will reactivate within 60 min.

4.2.5.2 Quench valve burst disk and O-ring seal check

> The auxiliary and OVC burst disk vent lines do not need to be removed from the quench elbow.
>
> The quench valve body and burst disk assembly can be removed separately from between the elbow and turret flanges.

- Remove the **quench valve assembly**. (The high load seal is broken when the quench valve is removed).
  - » A new **high load seal must** be fitted. (→ 1/Fig. 165 Page 148)

Depressurize the helium vessel

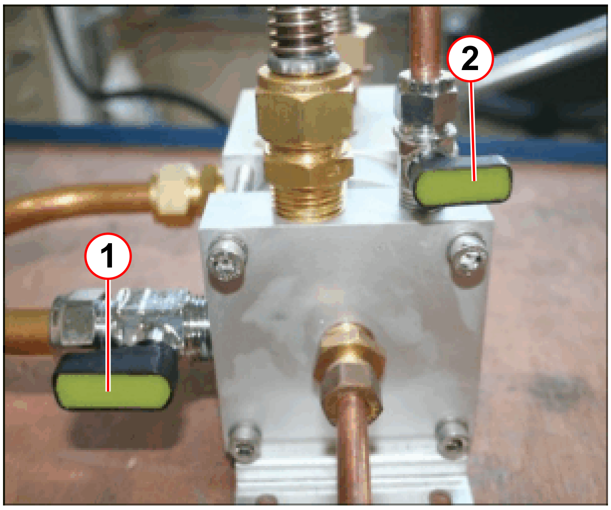
⚠ CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

» **To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.**

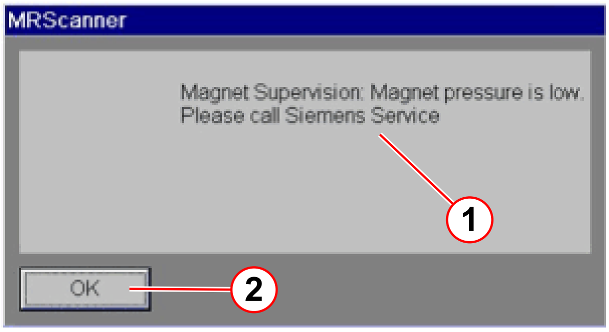
Fig. 158: Bypass valve positions for depressurization



(1) Magnet bypass valve OPEN
(2) Cold head bypass valve CLOSED

- **Fully open** the **magnet bypass valve** to **depressurize** the helium vessel.
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch** will activate.
  - » the **cold head** will turn **off**.
- **Close** the magnet bypass valve to prevent **air ingress**.
- **Warm up** the vent valve assembly.
  - » **Take care** not to damage any components when using a **heat gun**.

Fig. 159: MSUP - Magnet pressure is low



(1) Pop-up message
(2) Click OK to clear

On the **MR Scanner:**

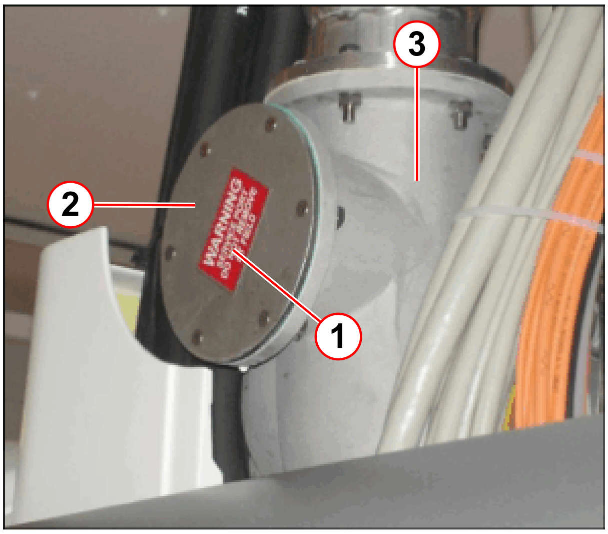
- A pop-up **alarm message** is displayed.
  - » **Magnet pressure is low**.
  - » Click **OK** to clear the alarm.
- The **alarm box** will sound when **pressure falls** to **alarm limits.**
  - » The yellow **WARNING LED** lights on the alarm box.
  - » The alarm **buzzer** is **activated**
  - » Press **AUDIO ALARM OFF** to silence the alarm.

4.2.5.3 Quench valve removal and burst disk inspection

- When the magnet has reached **zero pressure**, warm up the **quench valve and venting** to room temperature, using a **heat gun**.

- The magnet must be at **zero field** before checking the quench valve burst disk.
- The quench valve assembly must be **removed completely** for inspection.

Fig. 160: Horizontal service port

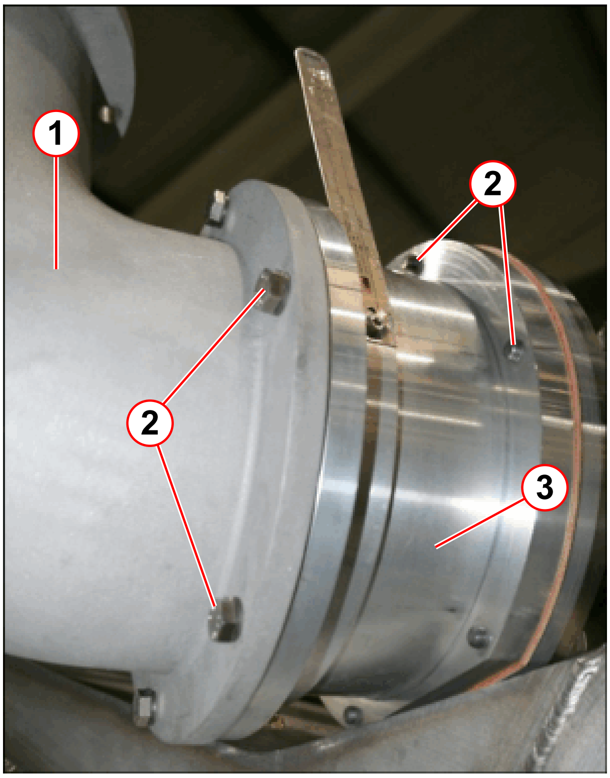


(1) Warning label
(2) Port (blanked)
(3) Quench elbow

OR98 only - Horizontal service port

- **This port cover must NOT be removed.**
- **Do not** use as an inspection cover to check the quench valve burst disk.
- This port allows horizontal fitment of the quench pipe.
- The port is **blanked** during normal vertical quench line configuration.
- If used as a quench outlet, the **vertical port must be blanked.**

Fig. 161: Quench valve removal



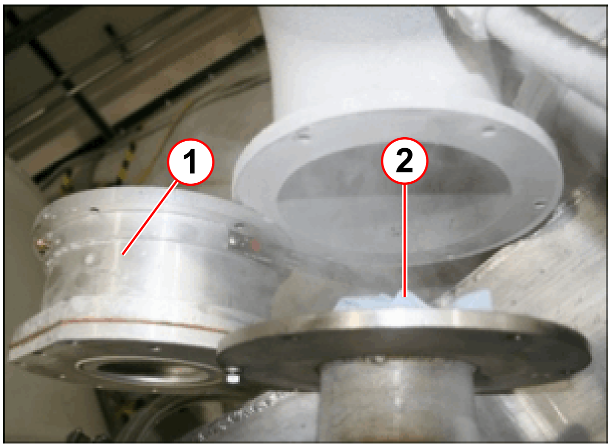
(1) Elbow
(2) M8 screws and nordlock washers (6 off) each end
(3) Quench valve assembly

The **quench valve assembly** is secured between the **elbow** and **turret**. There are 6 off M8 screws and nordlock washers fitted on each end of the quench valve.

- **Disconnect** the **quench valve** from the elbow:
  - **Remove** the (6 off) M8 screws and washers from the elbow.
- **Disconnect the quench valve** from the turret:
  - **Remove** the (6 off) M8 screws and washers from the **turret flange.**

The quench valve turret flange is on the high pressure side of the turret. When the M8 screws are removed, the magnet will be OPEN TO ATMOSPHERE.

Fig. 162: Quench valve rotated into postion

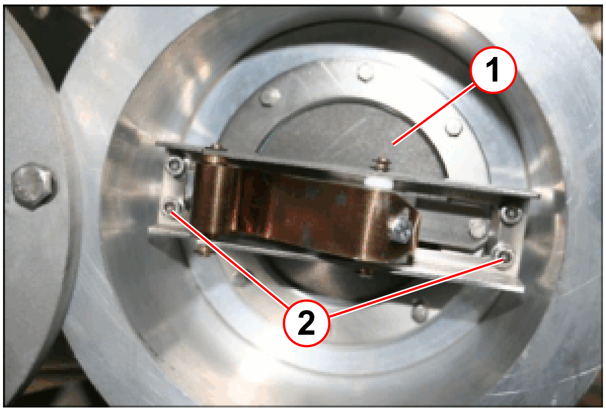


(1) Quench valve
(2) Paper towelling

- Carefully **lift out** the quench valve assembly.
- **Immediately** place a ball of paper towelling into the **open turret port**.

4.2.5.4 Quench valve inspection

Fig. 163: Check the quench valve burst disk



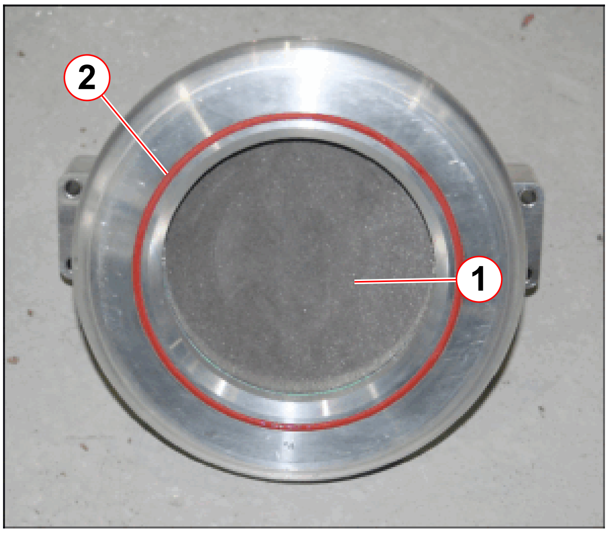
(1) Graphite burst disk
(2) Spring plate retaining screws M5 x 4

- **Inspect** the **graphite burst disk** for breakage, or cracks.
- **Evenly** unscrew and remove the (4 off) M5 cap head screws.
- **Carefully** lift out the **burst disk assembly** from the **quench valve body**.

The spring exerts significant force on these screws.

4.2.5.5 Burst disk inspection

Fig. 164: Quench valve burst disk O-ring



(1) Burst disk
(2) O-ring

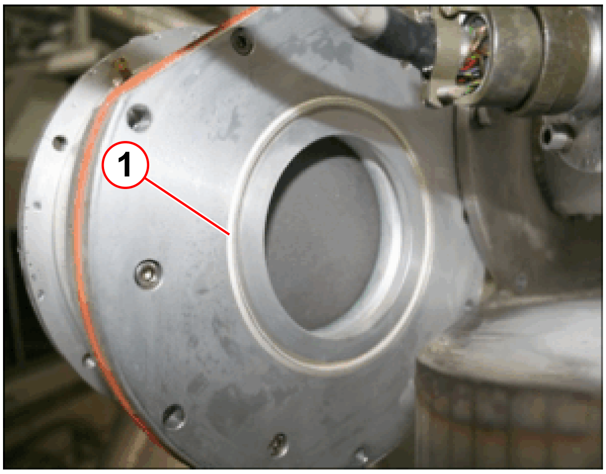
- **Inspect** the burst **disk**:
  » If the **burst disk** has ruptured, or is cracked, it must be replaced **immediately**.
- Inspect the **quench valve O-ring**. Replace if damaged.
- **Clean** the O-ring groove and O-ring before re-fitting.
- **Ensure** the O-ring is **seated evenly** all around the groove.

> The O-ring must be checked each time the quench valve is removed, and changed if damaged.

- Check the **aux burst disk.** (→ Check the auxiliary burst disk / Page 149)
- Re-fit the **burst disk assembly** into the **quench valve body**.
- Refit the **elbow**.

Refit the quench valve

Fig. 165: Quench valve seal



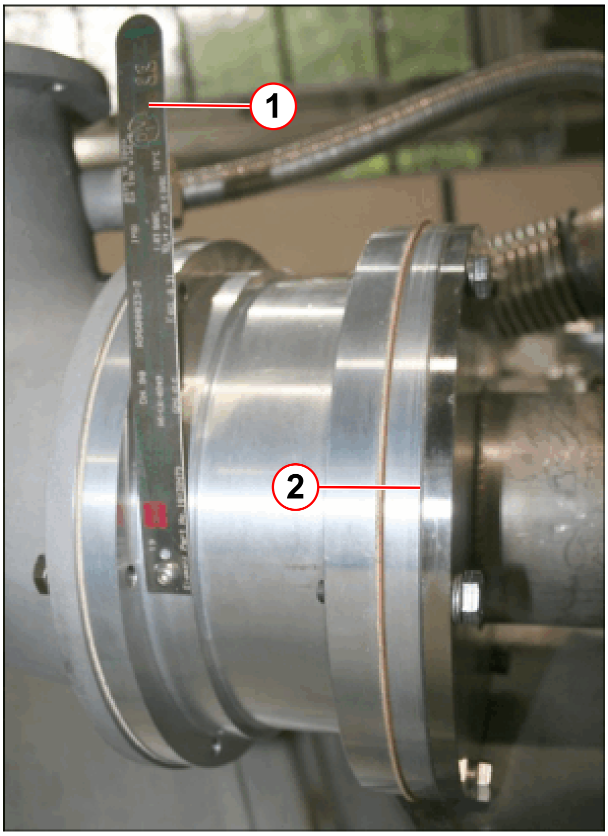
(1) High load seal

- Clean and dry the **sealing face** on the quench valve.
- Fit a **NEW high load seal**.
- **Remove** the **paper towelling** from the turret port. Ensure the surfaces are **clean** and **dry**.
- **Immediately** re-fit the **quench valve assembly** to the **turret flange**.

A new high load seal MUST be fitted every time the quench valve is removed.

Take care not to damage / scratch the flange face.

Fig. 166: Quench valve burst disk identification tag



(1) Burst disk tag
(2) Leak check turret flange

- Warm up the **sealing face** on the quench valve flange.
- **Close** the magnet **bypass valve**.
- **Leak check** the burst disk and O-ring assembly.
  - **inside** the quench valve body.
- **Leak check** the turret quench valve flange.
  - **outside** the quench valve body.

4.2.5.6 Check the auxiliary burst disk

CAUTION

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

» **Replacement of burst disk(s) must be performed by a qualified CSE.**

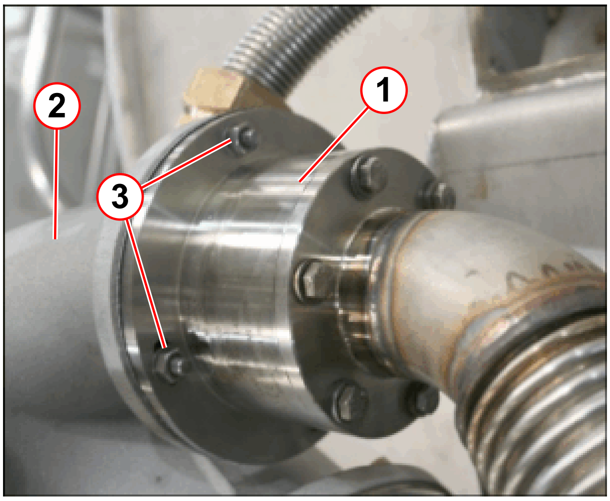
» **Replace broken or damaged burst disk/O-rings IMMEDIATELY.**

» **Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.**

» **If ice is suspected contact Magnet HSC immediately.**

- Check the magnet is still at **zero pressure.**
- If the elbow **has not** been removed for the quench valve inspection:
  - Warm up the **auxiliary burst disk assembly** to room temperature, using a **heat gun.**

Fig. 167: Auxiliary burst disk assembly

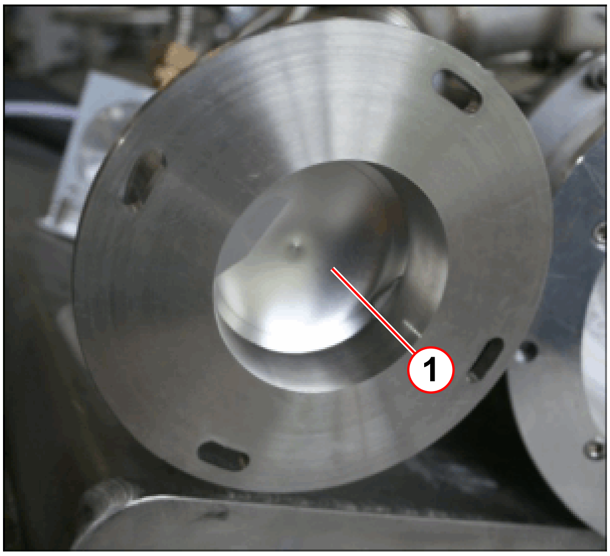


(1) Auxiliary burst disk
(2) Elbow
(3) M6 screws, washers and nuts (4 off)

- **Disconnect** the auxilliary burst disk from the **elbow,** on the low pressure side.
  - **Remove** the **M6** screws, washers and nuts (4 off).
- **Remove** the vent pipe from the by-pass assembly at the **quench elbow**.
- **Immediately** place a ball of paper towelling into the **open port**.

Do not touch the metal membrane; it is very fragile.

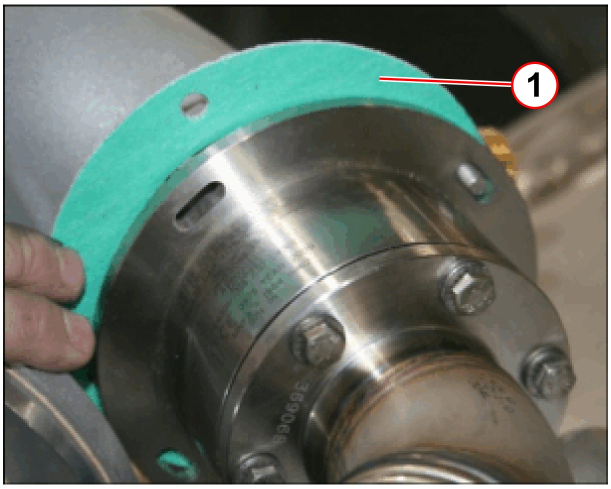
Fig. 168: Auxiliary burst disk



(1) Burst disk

- Visually inspect the **auxiliary burst disk membrane** to ensure that it is not damaged.

Fig. 169: Auxiliary burst disk gasket



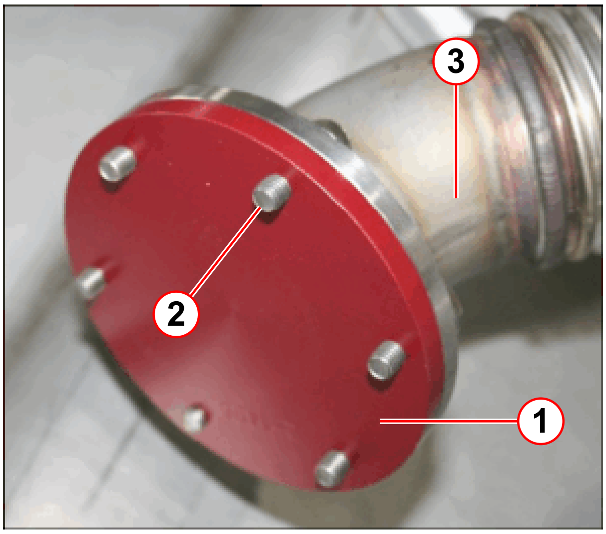
(1) Gasket

- Ensure the **gasket seal** is not damaged, and that the surface is clean.
- **Re-fit** the auxiliary burst disk assembly to the elbow.
  - use the M6 screws, washers and nuts.
  - **torque** the M6 screws to **6 Nm**.

- **Warm up** the turret venting to room temperature, using a **heat gun**.

Fitting the auxiliary burst disk blanking plate

Fig. 170: Auxiliary burst disk blanking plate



(1) Blanking plate (red anodized)
(2) M5 screws and washers (6 off)
(3) Vent line

- Two red annodized **blanking plates** and O-rings are provided in the magnet site box.
- If the burst disk is damaged **in any way, the red burst disk blanking plates MUST** be fitted immediately to
  - aux vent line.
  - elbow.
- A replacement burst disk must be obtained urgently.
- When fitted, **leak check** the red blanking plate.

⚠ WARNING

Safety device removed!!!!!

The magnet must not be ramped with the auxiliary burst disk removed or broken.

» **Replace the auxiliary vent burst disk before ramping.**

WARNING

Risk of air ingress

If the auxiliary burst disk is broken, IMMEDIATELY fit the red anodized blank from the customer spare parts box.

» **A blanking plate for (a) elbow auxiliary port (b) auxiliary vent line is supplied in the customer site box.**

» **A replacement auxiliary burst disk must be fitted BEFORE ramping the magnet**

4.2.5.7 OVC burst disk

A quench **will not** cause the OVC metal burst disk to rupture. The OVC burst disk only requires checking during **routine safety checks.**

4.2.5.8 Reactivate the pressure heater

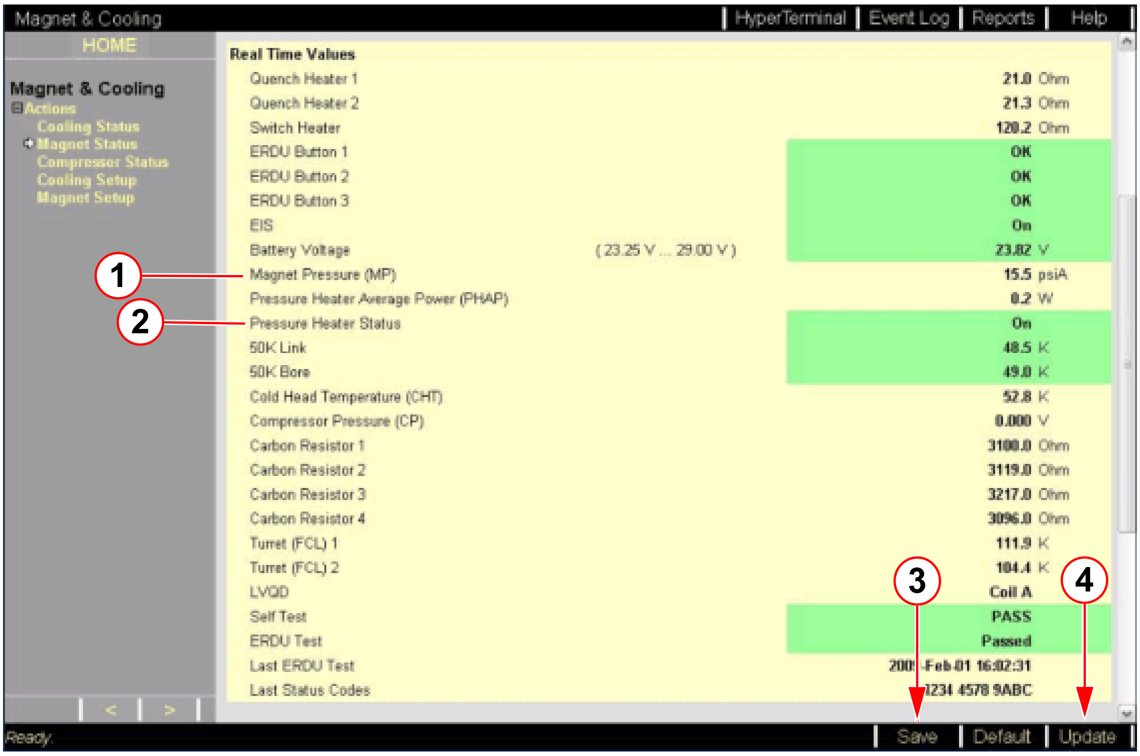
- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status.**
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button. (→ 1/Fig. 157 Page 144)
- Select **SAVE.** (→ 3/Fig. 157 Page 144)
- **At Real-Time Values** - Monitor the **magnet pressure.**
  - Select **Update** to refresh data until pressure reaches ≥ **15.5 psi (A).**

Data is only accurate from the last refresh of the page, or from when the page was opened last.

During pressurization, the cold head will switch on automatically.

If the cold head was turned off at MRC, select (→ 2/Fig. 157 Page 144)

Fig. 171: Check magnet pressure level



(1) Magnet pressure (MP)
(2) Pressure heater status
(3) Save button
(4) Update button

- At **MSUP Control - Sets the pressure heater mode** select the **Automatic mode** button. (→ 1/Fig. 157 Page 144)
- Select **SAVE.** (→ 3/Fig. 171 Page 153)

4.2.6 Final steps

- **Helium fill**.
- Perform a **system leak check**.
- **Ramp the magnet** to field.

> Allow a minimum of 2 hours after filling before ramping the magnet.

4.2.7 Quench report

- Fill out the **Quench Report report.** (→ Product Information MR, https://bps-healthcare.siemens.com/productinformation/ProductInformation.aspx?

uuid=7abf6bb6-5daf-440c-9920-fab2d4640084) - Links - MR - Quality Database (and Quench Report) and send to Magnet HSC.

5.1 Prerequisites for Shimming

5.1.1 Only for Systems with a Tim Dockable Patient Table

> If the system has only one dockable patient table, no further action is necessary.
>
> If the system has more than one dockable patient table, you have to check which patient table is registered.
>
> Only one dockable patient table is registered in the factory.
>
> A correctly registered patient table is necessary for shimming.

1. **Check which dockable patient table is already registered**.
   - Open a command tool and enter the command:
     `C:\Users\meduser>`**`MrVerifyPTAB`**

     Example:

Fig. 172: MrVerifyPTAB

```
86); Numaris 4 VD13A ; 25-Sep-2011 00:51; .

-------------------------------------------------------------------
current system settings:
    mobile.................:1
    frequency..............:123200000Hz
    field strength.........:2.9T

current PTAB settings read from HW:
    material number........:10433522
    serial number..........:1002

PTAB devices registered on host:
 device 0:
    material number........:10433522
    serial number..........:1002
-------------------------------------------------------------------

finished successfully.
```

2. Use the registered dockable patient table for shimming.

> Check the correct adjustment of the dockable patient table.
>
> Refer to the Installation Instructions (→ Tim Dockable Table (Mobile Patient Table) / M7-010.812.01)if an adjustment is necessary.

If the system has a 2nd or more dockable patient tables, these additional patient tables need to be registered in the system software.

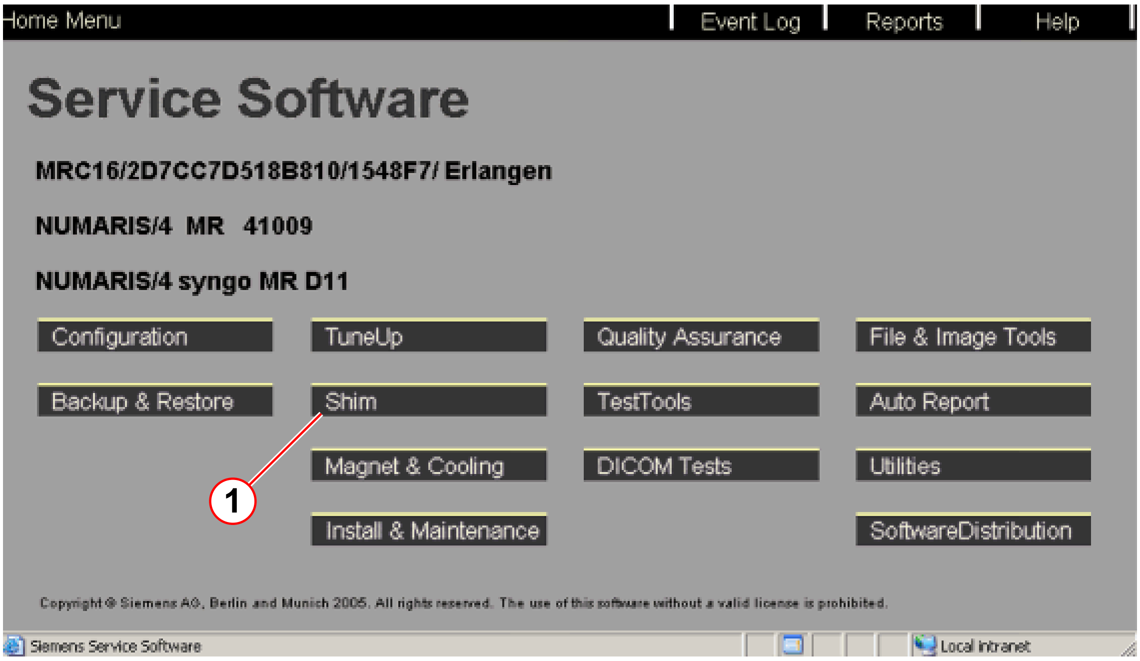
For detailed information, please refer to document (→ Additional Tim Dockable Table / MR07-000.814.05)

5.2 Passive Shim

Information regarding passive Shim is not published as hardcopy documentation. For more information, refer to the Online Help

- Start the Shim workflow from the SeSo platform.

Fig. 173: Service Software



(1) Shim workflow

6.1 Changes to previous version

Version 02

- Chapter 03
  - Renaming of the chapter.
  - Update of the chapter.
  - Note of the Quench line acceptance protocol added.
- Chapter 04
  - Note of the Quench line acceptance protocol added.

Version 03

- Chapter 03
  - Update of links
- Chapter 04
  - Update of links
  - Update of MRC Host connection.

Version 04

- Chapter 03
  - Venting Leak Check. Reference to use 5 Nm added.
- Chapter 04
  - What to do after Quench. Mobile reference deleted.

Version 05

- Chapter 03
  - New leak detector added.
- Chapter 05
  - Dockable patient table added.

Version 06

- Chapter 04
  - MNO Frequency range added.
  - Update of the screenshots

Version 07

- Chapter 04
  - Update of note: functionality of host computer.
  - Update of terminal emulation software.

Version 08

- All,
  - Editorial changes
- Chapter 03, Perquisites Ramping:
  - New path to the acceptance protocol and deleted the sign off by the project manager, Charm MR_00455599)

- Chapter 04, Ramping the Magnet:
  - Transport test, updated. Charm MR_00455599)
  - Warning for visual inspection of quench tube added. (→ OR98/99/103 Ramping / Page 89)

Version 09

- Chapter 04, Ramping the Magnet:
  - Co-Siting of Magnets added.

7 List of Hazard IDs

There are no Hazard IDs in this document.

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